



CHAPTER 6

Non-Euclidean Geometry*

6.1 Introduction

6.1.1 The Parallel Axiom

We have previously alluded to the remarkable discovery (made in the 19th century) that there exist geometries other than Euclid's. In this optional chapter we begin to explore the beautiful connections that exist between these so-called non-Euclidean geometries and the complex numbers. Since this Introduction summarizes many of the key ideas and results, you may wish to read it even if you cannot afford the time to read the entire chapter.

One way to approach Euclidean geometry is to begin with definitions of such things as “points” and “lines”, together with a few assumptions (*axioms*) concerning their properties. From there one goes on, using nothing but logic, to deduce further properties of these objects that are necessary consequences of the initial axioms. This is the path followed in Euclid's famous book, *The Elements*, which was published around 300 BCE.

Of course Euclidean geometry did not suddenly spring into existence as a fully formed logical system of axioms and theorems. It was instead developed gradually as an idealized description of physical measurements performed on physically constructed lines, triangles, circles, etc. Though the ancients did not think of it in this way, Euclidean geometry is thus not simply mathematics, it is a physical theory of space—a *fantastically accurate* theory of space.

Euclidean geometry is not, however, a perfect theory: modern experiments have revealed extremely small discrepancies between the predictions of Euclidean geometry and the measured geometric properties of figures constructed in physical space. These departures from Euclidean geometry are now known to be governed, in a precise mathematical way, by the distribution of matter and energy in space. This is the essence of a revolutionary theory of gravity (*General Relativity*) discovered by Einstein in 1915.

It turns out that the larger the figures examined, the larger the deviations from the predictions of Euclidean geometry. However, it's important to realize just how small these deviations typically are for figures of reasonable size. For example, suppose we measure the circumference of a circle having a radius of one meter. Even if our measuring device were capable of detecting a discrepancy the size of a single atom of matter, no deviation from Euclidean geometry would be found! Little wonder, then, that for two thousand years mathematicians were seduced into believing that Euclidean geometry was the *only logically possible* geometry.

It is a marvellous tribute to the power of human mathematical thought that non-Euclidean geometry was discovered a full century *before* Einstein found that it was needed to describe gravity. To locate the seeds of this mathematical discovery, let us return to ancient Greece.

Euclid began with just five axioms, the first four of which never aroused controversy. The first axiom, for example, merely states that there exists a unique line passing through any two given points. However, the status of the fifth axiom (the so-called *parallel axiom*) was less clear, and it became the subject of investigations that ultimately led to the discovery of non-Euclidean geometry:

Parallel Axiom. *Through any point p not on the line L there exists precisely one line L' that does not meet L .* (6.1)

Figure [6.1a] illustrates the parallel axiom, and it also explains why this axiom cannot be experimentally tested, at least as stated. As the line M rotates towards L' , the intersection point q moves further and further away along L . Our geometric intuition is based on figures drawn in a finite portion of the plane, but to verify that L' never meets L , we need an infinite plane. We can certainly try to *imagine* what an infinite plane would be like, but we have no first hand experience to back up our hunches.

These are very modern doubts we are expressing. Historically, mathematicians fervently believed in (6.1), so much so that they thought it must be a logically necessary property of straight lines. But in that case they ought to be able to prove it outright, instead of merely assuming it as Euclid had done.

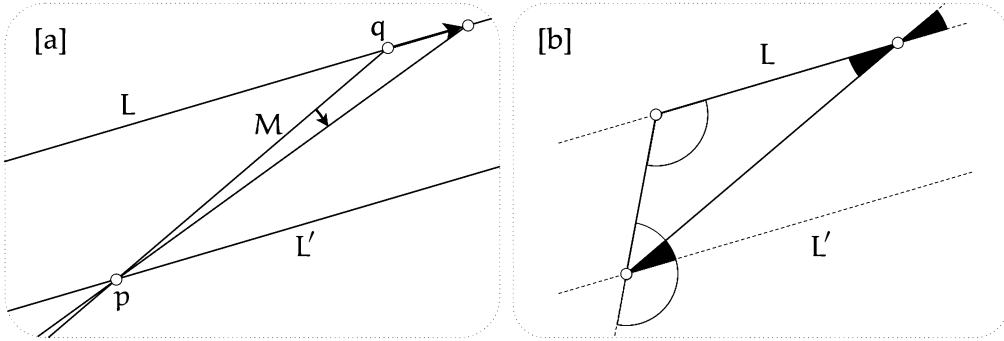
Many attempts were made to deduce (6.1) from the first four axioms, one of the most penetrating being that of Girolamo Saccheri in 1733.¹ His idea was to show that if (6.1) were not true, then a contradiction would necessarily arise. He divided the denial of (6.1) into two alternatives:

Spherical Axiom. *There is no line through p that does not meet L .* (6.2)

or

Hyperbolic Axiom. *There are at least two lines through p that do not meet L .* (6.3)

¹ See [Stillwell \(2010\)](#).



[6.1] Two forms of the Parallel Axiom. [a] *The Parallel Axiom seems beyond empirical verification: As the line M rotates towards L', the intersection point q moves further and further away along L, so to be certain that the supposedly unique parallel line L' never intersects L, we need to go all the way to infinity!* [b] *However, an alternative formulation of the Parallel Axiom is subject to empirical verification or refutation: The angles in every triangle must sum to π , so that $\mathcal{E} = 0$.*

Our naming of (6.2) will become clear shortly, but the use of “hyperbolic” in connection with (6.3) is more obscure, though standard.

In the case of (6.2), Saccheri was indeed able to obtain a contradiction, provided “lines” are assumed to have infinite length. If we drop this requirement, then we obtain a *non-Euclidean geometry* called *spherical geometry*. This is the subject of the following section.

In the case of (6.3), Saccheri and later mathematicians were able to derive very strange conclusions, but they were not able to find a contradiction. As we now know, this is because (6.3) yields another viable non-Euclidean geometry, called *hyperbolic geometry*. Of the two non-Euclidean geometries obtained from (6.2) and (6.3), hyperbolic geometry is by far the more intriguing and important: it is an essential tool in many areas of contemporary research. Furthermore, there is even a sense (to be discussed later) in which hyperbolic geometry subsumes both Euclidean and spherical geometry.

6.1.2 Some Facts from Non-Euclidean Geometry

Let’s take our first look at how these new geometries differ from Euclid’s. A very familiar theorem of Euclidean geometry states that in any triangle T,

$$(\text{Angle sum of } T) = \pi.$$

As indicated in [6.1b], this result is actually equivalent to the parallel axiom. It follows that in non-Euclidean geometry the angle sum of a triangle differs from π . To measure this difference, we introduce the so-called *angular excess* $\mathcal{E}$:

$$\mathcal{E}(T) \equiv (\text{Angle sum of } T) - \pi.$$

Euclidean geometry is thus characterized by the vanishing of $\mathcal{E}(T)$.

Note that, unlike the original formulation of the parallel axiom, this statement can be checked against experiment: construct a triangle, measure its angles, and see if they add up to π . Gauss was the first person to ever conceive of the possibility that physical space might not be Euclidean, and he even attempted the above experiment, using three mountain tops as the vertices of his triangle, and using light rays for its edges. Within the accuracy permitted by his equipment, he found $\mathcal{E} = 0$. Quite correctly, Gauss did not conclude that physical space is definitely Euclidean in structure, but rather that if it is *not* Euclidean then its deviation from Euclidean geometry is extremely small.

Let us return from physics to mathematics. Using pure logic to work out the consequences of (6.2) and (6.3), both Gauss and Johann Heinrich Lambert independently discovered that the two non-Euclidean geometries departed from Euclid's in opposite ways:

- In spherical geometry the angle sum is greater than π : $\mathcal{E} > 0$.
- In hyperbolic geometry the angle sum is less than π : $\mathcal{E} < 0$.

Furthermore, they discovered the striking fact that $\mathcal{E}(T)$ is completely determined by the *size* of the triangle. More precisely, $\mathcal{E}(T)$ is simply proportional to the *area* $A(T)$ of the triangle T :

$$\mathcal{E}(T) = \mathcal{K} A(T), \quad (6.4)$$

where $\mathcal{K}$ is a constant that is positive in spherical geometry, and negative in hyperbolic geometry.

Several interesting points can be made in connection with this result:

- Although there are no qualitative differences between them, there are nevertheless *infinitely many* different spherical geometries, depending on the value of the positive constant $\mathcal{K}$. Likewise, each negative value of $\mathcal{K}$ yields a different hyperbolic geometry.
- Since the angle sum of a triangle cannot be negative, $\mathcal{E} \geq -\pi$. Thus in hyperbolic geometry ($\mathcal{K} < 0$) no triangle can have an area greater than $|(\pi/\mathcal{K})|$.
- In non-Euclidean geometry, similar triangles do not exist! This is because (6.4) tells us that two triangles of different size cannot have the same angles.
- Closely related to the previous point, in non-Euclidean geometry there exists an *absolute unit of length*. For example, in spherical geometry we could define it to be the side of the equilateral triangle having angle sum 1.01π . Similarly, in hyperbolic geometry we could define it to be the side of the equilateral triangle having angle sum 0.99π .

- A somewhat more natural way of defining the absolute unit of length is in terms of the constant $\mathcal{K}$. Since the radian measure of angle is defined as a ratio of lengths, $\mathcal{E}$ is a pure number. On the other hand, the area $\mathcal{A}$ has units of $(\text{length})^2$. It follows that $\mathcal{K}$ has units of $1/(\text{length})^2$ and so it can be written as follows in terms of a length R : $\mathcal{K} = +(1/R^2)$ in spherical geometry; $\mathcal{K} = -(1/R^2)$ in hyperbolic geometry. Later we will see that this length R can be given a very intuitive interpretation P .
- The smaller the triangle, the harder it is to distinguish it from a Euclidean triangle: only when the linear dimensions are a significant fraction of R will the difference become obvious. This is why Gauss chose the biggest triangle he could in his experiment. Einstein's theory explains why Gauss's triangle was nevertheless much too small: the weak gravitational field in the space surrounding the earth corresponds to a microscopic value of $\mathcal{K}$ and hence to an enormous value of R . It would have been a different story if Gauss had been able to perform his experiment in the vicinity of a small black hole!

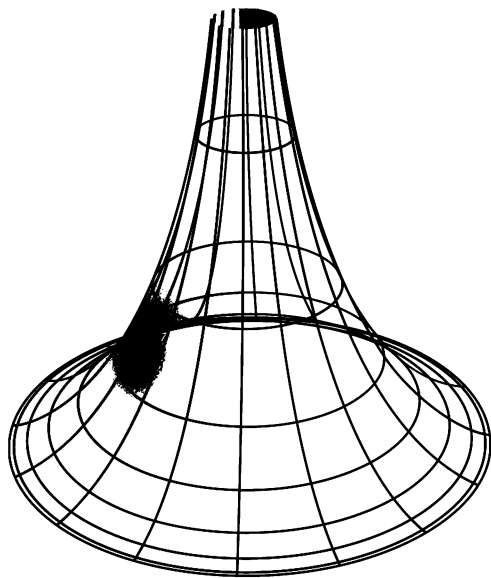
6.1.3 Geometry on a Curved Surface

We began this book by discussing how the complex numbers met with enormous initial resistance, and how they were finally accepted only after they were given a *concrete interpretation*, via the complex plane. The story of non-Euclidean geometry is remarkably similar.

Gauss never published his revolutionary ideas on non-Euclidean geometry, and the two men who are usually credited for their independent discovery of hyperbolic geometry are János Bolyai (1832) and Nikolai Lobachevsky (1829). Indeed, hyperbolic geometry is frequently also called Lobachevskian geometry, perhaps because Lobachevsky's investigations went somewhat deeper than Bolyai's. However, in the decades that followed their discoveries, Bolyai's work was completely ignored, and Lobachevsky's met only with vicious attacks.

The decisive figure in the acceptance of non-Euclidean geometry was Eugenio Beltrami. In 1868 he discovered that hyperbolic geometry could be given a *concrete interpretation*, via "differential geometry". For our purposes, differential geometry is the study of curved surfaces by means of ideas from calculus. What Beltrami discovered was that there exists a surface (the so-called *pseudosphere* shown in [6.2]) such that figures drawn on it automatically obey the rules of hyperbolic geometry². Psychologically, Beltrami's pseudosphere was to hyperbolic geometry as the complex plane had been to the theory of complex numbers.

² This oversimplification does not do justice to Beltrami's accomplishments. Later in this chapter we shall see what Beltrami *really* did!



[6.2] The Pseudosphere. In 1868 Eugenio Beltrami finally removed the mystery and scepticism that had overshadowed the strange, abstract laws of hyperbolic geometry ever since their discovery, 40 years earlier. He did so by recognizing that the hyperbolic laws simply are the intrinsic laws of geometry that hold true within a surface called the pseudosphere (shown here) that has constant negative Gaussian curvature. The pseudosphere extends upward indefinitely, but the base circle is a boundary beyond which it cannot be extended.

To explain what we mean by this, let us first discuss how we may “do geometry” on a more general surface, such as the surface of the strange looking vegetable³ shown in [6.3]. The idea of doing geometry on such a surface is essentially due to Gauss and (in greater generality) to Riemann.

The first thing we must do is to replace the concept of a straight line with that of a *geodesic*. Just as a line-segment in a flat plane may be defined as the shortest route between two points, so a geodesic segment connecting two points on a curved surface may be defined (provisionally) as the shortest connecting route *within the surface*. For example, if you were an ant living on the surface in [6.3], and you wanted to travel from a to b as quickly as possible, then you would follow the illustrated geodesic segment. The figure also shows the geodesic segment connecting another pair of points, c and d.

Here is a simple way you can actually construct such geodesic segments: take a thread and stretch it tightly over the surface to connect the points a and b. Provided

³ European readers may think this an imaginary vegetable, but Americans can buy it in the supermarket.

that the thread can slide around on the surface easily, the tension in the thread ensures that the resulting path is as short as possible. Note that in the case of cd , we must imagine that the thread runs over the inside of the surface. In order to deal with all possible pairs of points in a uniform way, it is therefore best to imagine the surface as made up of two thinly separated layers, with the thread trapped between them.

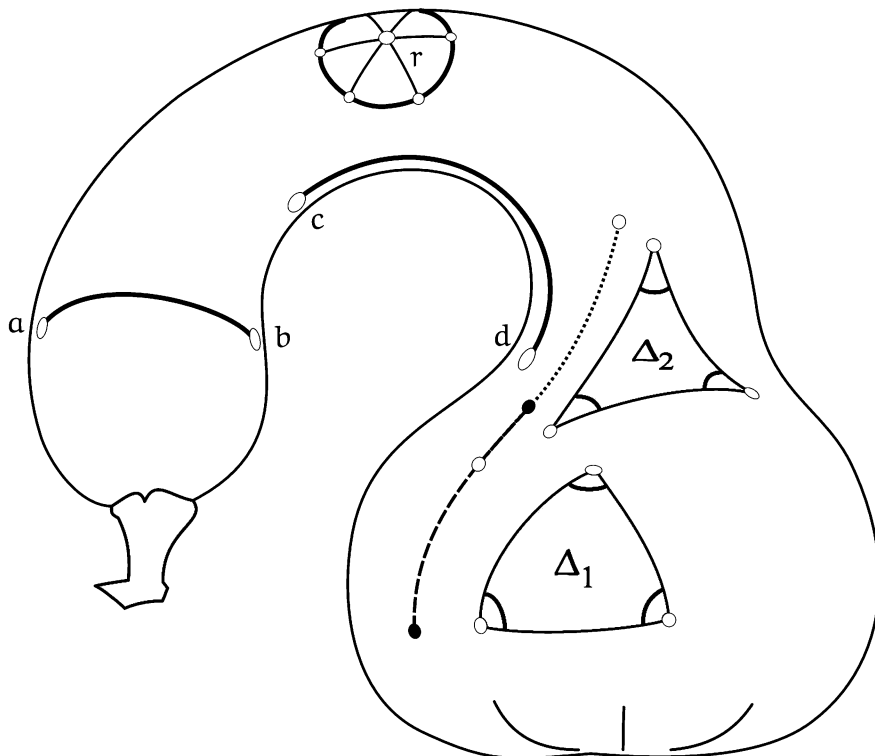
In fact there exists a wonderfully simple and *practical* (yet surprisingly little known) method of constructing geodesics on *any* part of a surface. Cut a long, straight, *narrow* strip of masking tape (aka painter's tape) and start to lay one end down on the surface in the direction you want your geodesic to go. Holding the free end up away from the surface with one hand, use your other hand to gently run your finger along the tape, allowing it to choose its own path as it sticks itself to the surface as it goes. Behold! You have created a *geodesic*, and this construction works equally well on ab and cd ! In the case of ab , you can easily check the validity of the construction by stretching a string over the surface: it will coincide with path of your sticky tape! For the explanation of *why* this works (and many applications) see [Needham \(2021\)](#).

It is now obvious how we should define distance in this geometry: the distance between a and b is the length of the geodesic segment connecting them. Figure [6.3] shows how we can then define, for example, a circle of radius r and centre p as the locus of points at distance r from p . To construct this circle we may take a piece of thread of length r , hold one end fixed at p , then (keeping the thread taut) drag the other end round on the surface.

Given three points on the surface, we may join them with geodesics to form a triangle; [6.3] shows two such triangles, Δ_1 and Δ_2 . Now look at the angles in Δ_1 . Clearly $\mathcal{E}(\Delta_1) > 0$, like a triangle in spherical geometry, while $\mathcal{E}(\Delta_2) < 0$, like a triangle in hyperbolic geometry.

6.1.4 Intrinsic versus Extrinsic Geometry

Clearly it is the curvature of the surface that causes $\mathcal{E}(\Delta_1)$ and $\mathcal{E}(\Delta_2)$ to differ from their Euclidean value $\mathcal{E} = 0$. However, it cannot be the precise shape of the surface in space that is involved here. To see this, imagine that from the vegetable in [6.3] we were to cut out a patch of the skin containing Δ_1 . Suppose that this patch is made of fairly stiff material that does not stretch if we try to bend it a little. [As it happens, the skin of this vegetable is actually like this!] We can now gently bend the patch into infinitely many slightly different shapes: its so-called *extrinsic geometry* has been changed by our stretch-free bending. For example, the curves in space making up the edges of Δ_1 are no longer the same shape as before.

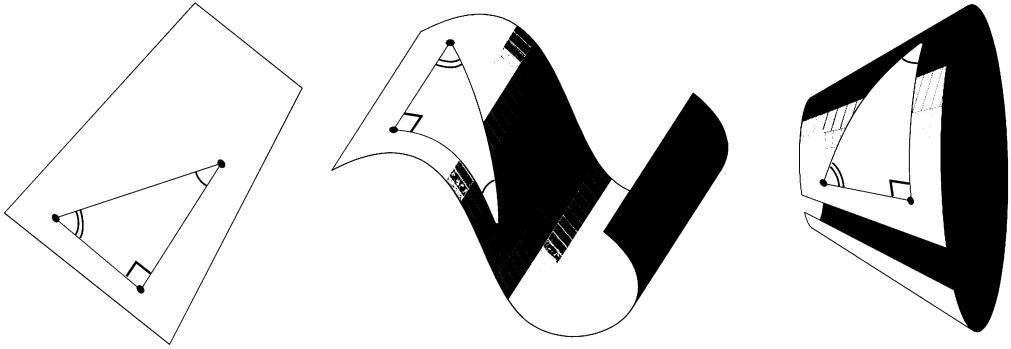


[6.3] **The intrinsic geometry of a surface** is the geometry experienced by small, 2-dimensional creatures living within the surface. For them, a “straight line” is the shortest route between two points, such as ab or cd . A “circle” of radius r is the set of points that lie at a fixed distance r from its centre. “Triangles” are no longer subject to the Parallel Axiom, so now their angles need not add up to π : $\mathcal{E}(\Delta_1) > 0$, and $\mathcal{E}(\Delta_2) < 0$.

On the other hand, if you were an intelligent ant living on this patch, no geometric experiment you could perform within the surface would reveal that any change had taken place whatsoever. We say that the *intrinsic geometry* has not changed. For example, the curves into which the edges of Δ_1 have been deformed are *still* the shortest routes on the surface. Correspondingly, the value of $\mathcal{E}$ is unaffected by stretch-free bending: $\mathcal{E}$ is governed by intrinsic (not extrinsic) curvature.

To highlight this fact, consider [6.4]. On the left is a flat piece of paper on which we have drawn a triangle T with angles $(\pi/2)$, $(\pi/6)$, $(\pi/3)$. Of course $\mathcal{E}(T) = 0$. Clearly we can bend such a flat piece of paper into either of the two (extrinsically) curved surfaces on the right⁴. However, *intrinsically* these surfaces have undergone no change at all—they are both as flat as a pancake! The illustrated triangles on these surfaces (into which T is carried by our stretch-free bending of the paper) are

⁴ Of course the conical example on the far right cannot be obtained by bending a rectangle.



[6.4] A bent piece of paper remains intrinsically flat. A flat piece of paper can be bent into all kinds of extrinsically curved shapes in space, but its internal, intrinsic geometry remains totally flat, and triangles constructed within the surface continue to obey all the normal laws of Euclidean geometry, such as Pythagoras's Theorem, and $\mathcal{E} = 0$.

identical to the ones that intelligent ants would construct using geodesics, and in both cases $\mathcal{E} = 0$: geometry on these surfaces is Euclidean.

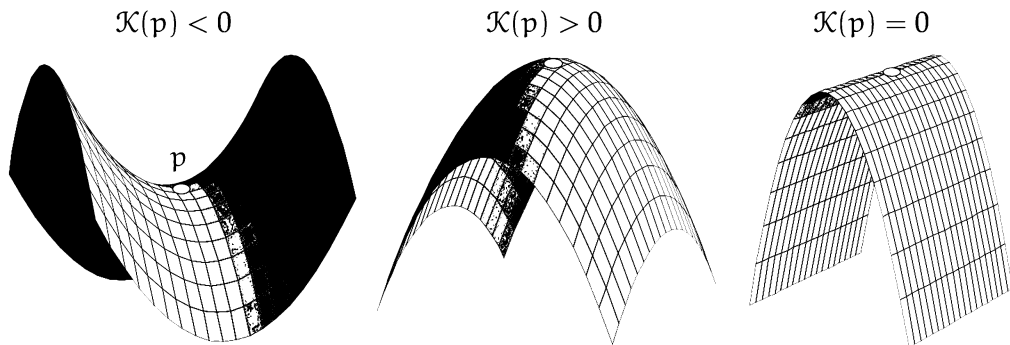
6.1.5 Gaussian Curvature

In 1827 Gauss published a beautiful and revolutionary analysis⁵ of the intrinsic and extrinsic geometry of surfaces, in which he revealed that remarkable connections exist between the two. Here we will simply state some of his most important conclusions, in their most general form. For explanations of these general results we refer you to works on differential geometry; see the recommendations at the end of this chapter. However, only special cases of the general results are needed to understand non-Euclidean geometry, and these will be separately verified in the course of this chapter.

For a surface such as [6.3], it is clear that some parts are more curved than others. Furthermore, the *kind* of bending also varies from place to place. To quantify the amount (and type) of bending of the surface at a point p , Gauss introduced a quantity $\mathcal{K}(p)$. This function $\mathcal{K}(p)$, whose precise definition will be given in a moment, is called the *Gaussian curvature*⁶. The greater the magnitude of $\mathcal{K}(p)$, the more curved the surface is at p . The sign of $\mathcal{K}(p)$ tells us qualitatively what the surface is like in the immediate neighbourhood of p . See [6.5]. If $\mathcal{K}(p) < 0$ then the neighbourhood of p resembles a saddle: it bends upwards in some directions, and downwards in others. If $\mathcal{K}(p) > 0$ then it bends the same way in all directions, like a piece of a sphere.

⁵ Gauss (1827).

⁶ Other names are *intrinsic curvature*, *total curvature*, or just plain *curvature*.



[6.5] The sign of the Gaussian curvature $\mathcal{K}(p)$ tells us the shape of the surface at p .

As we will now start to explain, it is no accident that we have used the same symbol to represent Gaussian curvature as we earlier used for the constant occurring in (6.4)—they are the same thing!

Gauss originally defined $\mathcal{K}(p)$ as follows. Let Π be a plane containing the normal vector $\mathbf{n}$ to the surface at p , and let κ be the (signed) curvature at p of the curve in which Π intersects the surface. The sign of κ depends on whether the centre of curvature is in the direction $\mathbf{n}$ or $-\mathbf{n}$. The so-called *principal curvatures* are the minimum $\kappa_{\min}$ and the maximum $\kappa_{\max}$ values of κ as Π rotates about $\mathbf{n}$. [Incidentally, Euler had previously made the important discovery that these principal curvatures occur in two *perpendicular* directions.] Gauss defined $\mathcal{K}$ as the product of the principal curvatures:

$$\mathcal{K} \equiv \kappa_{\min} \kappa_{\max}.$$

Note that this definition is in terms of the precise shape of the surface in space (extrinsic geometry). However, Gauss (1827) went on to make the astonishing discovery that $\mathcal{K}(p)$ actually measures the *intrinsic* curvature of the surface, that is, $\mathcal{K}$ is *invariant under bending*! Gauss was justifiably proud of this result, calling it the *Theorema Egregium* (Latin for “remarkable theorem”). As an example of the result, you may visually convince yourself that $\mathcal{K} = 0$ everywhere on each of the intrinsically flat surfaces in [6.4].

The intrinsic significance of $\mathcal{K}$ is exhibited in the following fundamental result: If Δ is an infinitesimal triangle of area dA located at the point p , then

$$\mathcal{E}(\Delta) = \mathcal{K}(p) dA. \quad (6.5)$$

Since $\mathcal{E}$ and dA are defined by the intrinsic geometry, so is $\mathcal{K} = (\mathcal{E}/dA)$. Once again, we refer you to works on differential geometry for a proof of (6.5).

It follows from (6.5) [see Ex. 1] that the angular excess of a non-infinitesimal triangle T is obtained by adding up (i.e., integrating) the Gaussian curvature over the interior of T :

$$\mathcal{E}(T) = \iint_T \mathcal{K}(p) \, dA. \quad (6.6)$$

As Beltrami recognized, and as we now explain, this lovely result of differential geometry brings us very close to a concrete interpretation of the non-Euclidean geometries.

6.1.6 Surfaces of Constant Curvature

Consider a surface such that $\mathcal{K}(p)$ has the same value $\mathcal{K}$ at every point p ; we call this a *surface of constant curvature*. For example, a plane is a surface of constant curvature $\mathcal{K} = 0$, as are the other surfaces in [6.4]; a sphere is an example (not the only one) of a surface of constant positive curvature; and the pseudosphere in [6.2] is an example (not the only one) of a surface of constant negative curvature.

In the case of a surface of constant curvature (and only in this case) we find that (6.6) takes the form,

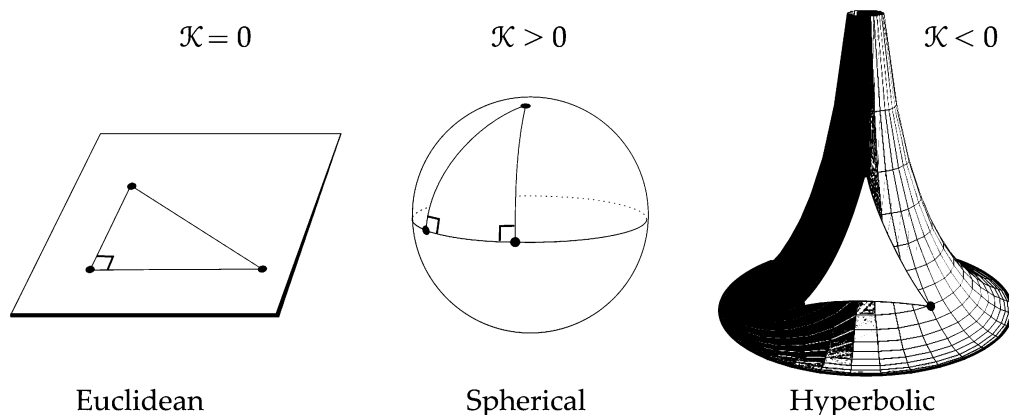
$$\mathcal{E}(T) = \mathcal{K} \iint_T dA = \mathcal{K} \mathcal{A}(T).$$

But this is identical to the fundamental formula (6.4) of non-Euclidean geometry! Thus, as Beltrami realized,

Euclidean, spherical, and hyperbolic geometry can all be interpreted concretely as the intrinsic geometry of surfaces of constant vanishing, positive, or negative curvature.

Figure [6.6] illustrates this using the simplest surfaces of each type. To obtain an added bonus, recall that we previously associated an absolute unit of length R with a non-Euclidean geometry by writing $\mathcal{K} = \pm(1/R^2)$. The bonus is that this length R now takes on vivid meaning: in spherical geometry R is simply the radius of the sphere, while in hyperbolic geometry it is the radius of the circular base of the pseudosphere (called the *radius* of the pseudosphere). These two interpretations will be justified later.

The requirement of constant curvature can be understood more intuitively by reconsidering the discussion at the end of Chapter 1. There we saw that a central idea in Euclidean geometry is that of a *group of motions* of the plane: one-to-one mappings that preserve the distance between all pairs of points. For example, two figures are congruent if and only if there exists a motion that carries the first into coincidence with the second. In order that this basic concept of equality be available in non-Euclidean geometry, we require that our surface admits an analogous group of motions. If we take one of the triangles on the surface in [6.3], it's clear that we cannot slide it to a new location and still have it fit the surface snugly, because the way in which the surface is curved at the new location is different: *variation in the curvature is the obstruction to motion*.



[6.6] Euclidean, Spherical, and Hyperbolic geometry are the intrinsic geometries of surfaces of constant Gaussian curvature.

This intuitive explanation can be clarified by appealing to (6.5). First, though, we wish to eliminate a possible confusion. The triangle on the flat plane in [6.6] can clearly be slid about and rotated freely, but what about the triangles on the (extrinsically) curved surfaces in [6.4]? After all, these surfaces are intrinsically flat, and so Beltrami would have us believe they are therefore just as good as the plane for doing Euclidean geometry. If we imagine these triangles as *completely* rigid then it's clear that if we try to move them to another location on the surface, they will no longer fit snugly against the surface. But if the triangle is instead cut out of a piece of ordinary (bendable but unstretchable) paper, then it *can* be slid about and rotated freely, always fitting perfectly against the surface. *This* is the kind of motion we are concerned with.

In order to clarify the connection between constant curvature and the existence of motions, consider an infinitesimal (bendable but unstretchable) triangle located at p . If its angular excess is $\mathcal{E}$ and its area is dA , then (6.5) tells us that the Gaussian curvature of the surface at p is given by $\mathcal{K}(p) = (\mathcal{E}/dA)$. Now suppose that there exists a motion that carries this triangle to an arbitrary point q on the surface. We may have to bend the triangle to make it fit against the surface at q , but since we are not allowed to stretch it, the values of $\mathcal{E}$ and dA do not alter. Thus $\mathcal{K}(q) = (\mathcal{E}/dA) = \mathcal{K}(p)$, and the surface has constant curvature.

Finally, let us return to the specific models of spherical and hyperbolic geometry shown in [6.6]. Clearly the triangle on the sphere can be slid about and rotated freely. In fact here, as on the plane, no bending is needed at all, because the sphere not only has constant intrinsic curvature, it also has constant extrinsic curvature.

What about hyperbolic geometry on the pseudosphere? It is certainly much less obvious, but the fact [to be proved later] that the pseudosphere has constant curvature guarantees that a bendable but unstretchable triangle *can* be slid about

and rotated freely, always fitting perfectly snugly against the surface. Exercise 15 shows how you can build your own pseudosphere; once built, you can verify this surprising claim experimentally.

6.1.7 The Connection with Möbius Transformations

As we established in Chapter 1, if the Euclidean plane is identified with $\mathbb{C}$ then its motions (and similarities) are represented by the particularly simple Möbius transformations of the form $M(z) = az + b$. One of the principal miracles we wish to explain in this chapter is that the motions of spherical and hyperbolic geometry are *also* Möbius transformations!

The most general (direct) motion of the sphere is a rotation about its centre. Stereographic projection onto $\mathbb{C}$ yields a conformal map of the sphere, and the rotations of the sphere thus become complex functions acting on this map. As we showed algebraically in Ex. 21 on page 211 of Chapter 3, they are the Möbius transformations of the form

$$M(z) = \frac{az + b}{-\bar{b}z + \bar{a}}.$$

This was first discovered by Gauss, around 1819. In the next section we will re-derive this result in a more illuminating way, and we will also explore the connection with Hamilton's "quaternions", which we first met on page 49.

Following the same pattern, it is also possible to construct conformal maps (in $\mathbb{C}$) of the pseudosphere, thereby transforming its motions into complex functions. One of the most convenient of these conformal maps is constructed in the unit disc. The motions of hyperbolic geometry then turn out to be the Möbius automorphisms of this circular map, which we first derived in (3.52) and then further investigated in Ex. 20 on page 210:

$$M(z) = \frac{az + b}{\bar{b}z + \bar{a}}.$$

This beautiful discovery was made by Henri Poincaré in 1882; see [Poincaré \(1985\)](#).

It seems magical enough that the motions of all three of the two-dimensional geometries are represented by special kinds of Möbius transformations, but there's more! In Chapter 3 [see (3.2) on page 138] we saw that the *general* Möbius transformation

$$M(z) = \frac{az + b}{cz + d}$$

has deep significance for physics: it corresponds to the most general Lorentz transformation of spacetime. Might it also have significance in non-Euclidean geometry? As we will explain at the end of this chapter, in 1883 Poincaré made the startling discovery that it represents the most general (direct) motion of *three-dimensional* hyperbolic space!

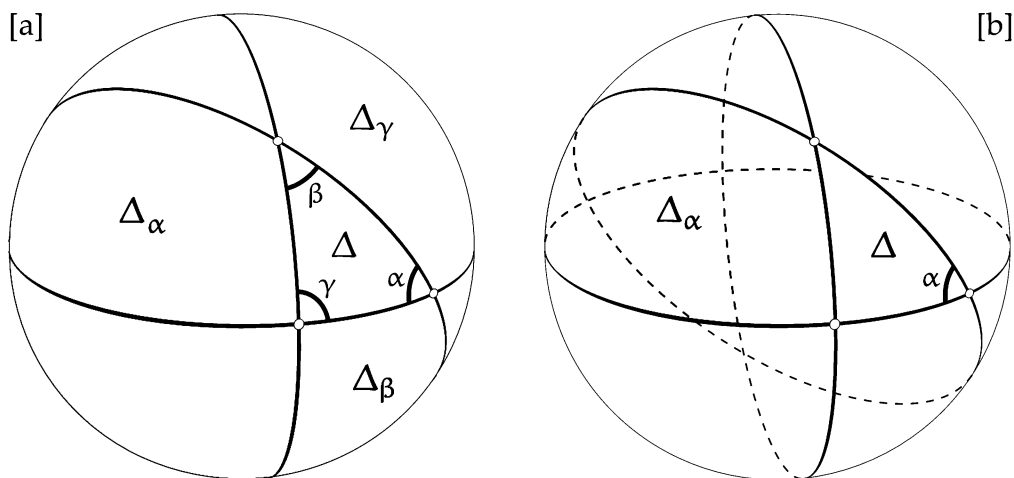
6.2 Spherical Geometry

6.2.1 The Angular Excess of a Spherical Triangle

The geodesics on the sphere are the *great circles*, that is the intersections of the sphere with planes through its centre. Thus if you were an ant living on the sphere, these great circles are what you would call “lines”.

Figure [6.7a] illustrates a general triangle Δ on a sphere of radius R obtained by joining three points using such “lines”. Without appealing to (6.6), which is a deep result in differential geometry, let us show directly that the angular excess $\mathcal{E}(\Delta)$ obeys the law (6.4), and that the constant $\mathcal{K}$ is indeed the Gaussian curvature, $\mathcal{K} = (1/R^2)$. The elegant argument that follows is usually attributed to Euler, who rediscovered it, but it was in fact first discovered by Thomas Harriot in 1603.⁷ Yes, this is the *same* Thomas Harriot who discovered the fundamentally important conformality of stereographic projection, (3.17)!

Prolonging the sides of Δ divides the surface of the sphere into eight triangles, the four triangles labelled Δ , Δ_α , Δ_β , Δ_γ each being paired with a congruent antipodal triangle. This is clearer in [6.7b]. Since the area of the sphere is $4\pi R^2$, we deduce that



[6.7] Harriot's 1603 proof that $\mathcal{E}(\Delta) = (1/R^2) \mathcal{A}(\Delta)$. Extending the sides of Δ that contain the angle α , we obtain the shaded wedge on the right, which occupies $(\alpha/2\pi)$ of the complete sphere, so $\mathcal{A}(\Delta) + \mathcal{A}(\Delta_\alpha) = 2\alpha R^2$; likewise for β and γ . Since each of the four triangles on the left has an antipodal partner of equal area—as can be seen more clearly on the right—it follows that together they occupy half the sphere: $\mathcal{A}(\Delta) + \mathcal{A}(\Delta_\alpha) + \mathcal{A}(\Delta_\beta) + \mathcal{A}(\Delta_\gamma) = 2\pi R^2$, and the result follows immediately.

⁷ See Stillwell (2010, §17.6).

$$\mathcal{A}(\Delta) + \mathcal{A}(\Delta_\alpha) + \mathcal{A}(\Delta_\beta) + \mathcal{A}(\Delta_\gamma) = 2\pi R^2. \quad (6.7)$$

On the other hand, it is clear in [6.7b] that Δ and Δ_α together form a wedge whose area is $(\alpha/2\pi)$ times the area of the sphere:

$$\mathcal{A}(\Delta) + \mathcal{A}(\Delta_\alpha) = 2\alpha R^2.$$

Similarly,

$$\mathcal{A}(\Delta) + \mathcal{A}(\Delta_\beta) = 2\beta R^2,$$

$$\mathcal{A}(\Delta) + \mathcal{A}(\Delta_\gamma) = 2\gamma R^2.$$

Adding these last three equations, we find that

$$3\mathcal{A}(\Delta) + \mathcal{A}(\Delta_\alpha) + \mathcal{A}(\Delta_\beta) + \mathcal{A}(\Delta_\gamma) = 2(\alpha + \beta + \gamma)R^2. \quad (6.8)$$

Finally, subtracting (6.7) from (6.8), we get

$$\mathcal{A}(\Delta) = (\alpha + \beta + \gamma - \pi)R^2.$$

In other words,

$$\mathcal{E}(\Delta) = \mathcal{K} \mathcal{A}(\Delta), \quad \text{where} \quad \mathcal{K} = (1/R^2), \quad (6.9)$$

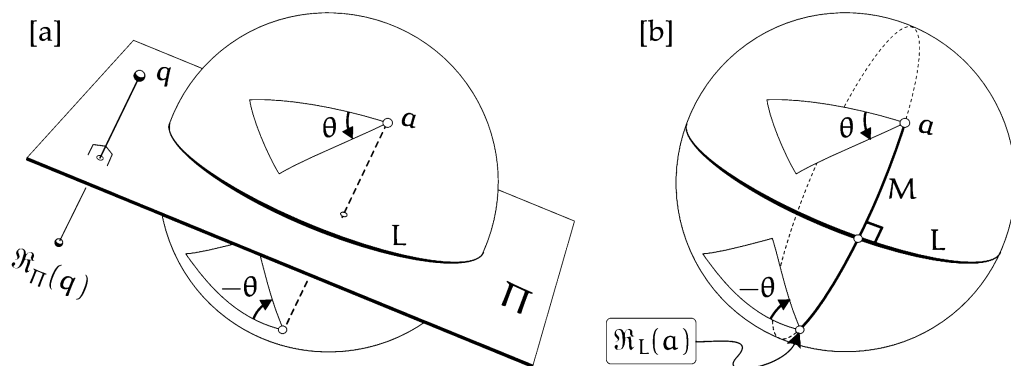
as was to be shown.

6.2.2 Motions of the Sphere: Spatial Rotations and Reflections

In order to understand the motions (i.e., one-to-one, distance-preserving mappings) of the sphere, we must first clarify the idea of “distance”. If two points a and b are not antipodal then there exists a unique line (great circle) L passing through them, and a and b divide L into two arcs of unequal length. The “distance” between the points can now be defined as the length of the shorter arc. But if the points *are* antipodal then every line through a automatically passes through b , and the distance between the points is defined to be the length πR of any of the semicircular arcs connecting them.

We can now generalize the Euclidean arguments given in the final section of Chapter 1. There we saw that a motion of the plane is uniquely determined by the images a', b', c' of any three points a, b, c not on a line: the image of P is the unique point P' whose distances from a', b', c' equal the distances of P from a, b, c . We leave it to you to check that this result (and the reason for it) is still true on the sphere.

On the sphere, as on the plane, we may consistently attribute a sense to angles—by convention an angle is positive if it is counterclockwise when viewed from *outside* the sphere. As happened in the plane, this leads to a division of spherical motions into two types: *direct* (i.e., conformal) motions, and *opposite* (i.e., anticonformal) motions.



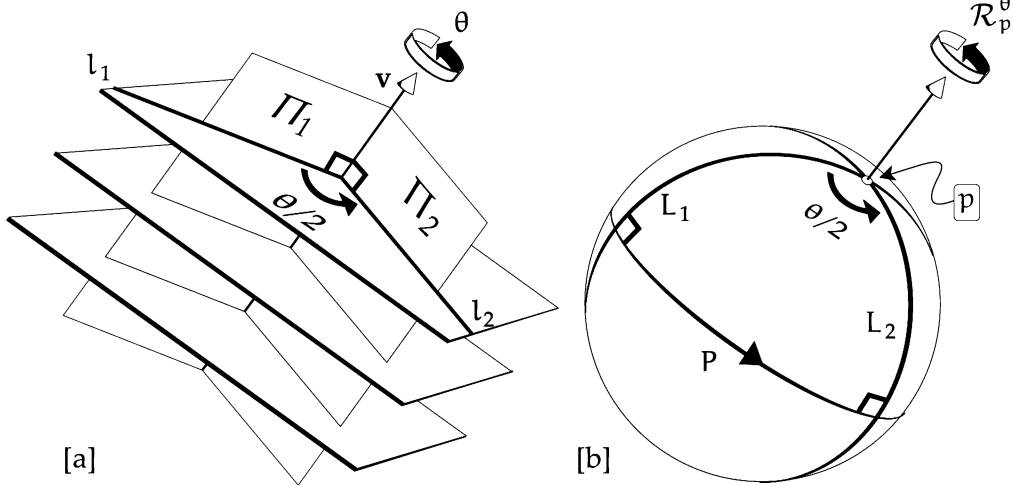
[6.8] Reflection of the sphere in a line. [a] If Π is the plane through the line (great circle) L , then reflection $\mathfrak{R}_\Pi$ of space across Π induces the anticonformal reflection $\mathfrak{R}_L$ of the sphere across L . [b] This reflection $\mathfrak{R}_L(a)$ can instead be expressed in intrinsic terms that make sense to the inhabitants of the sphere: travel some distance d along the orthogonal “line” M from a until you arrive at L , then travel an equal distance d again.

As in the plane, the simplest opposite motion of the sphere is *reflection* $\mathfrak{R}_L$ in a line L . This may be thought of as the transformation induced on the sphere by reflection $\mathfrak{R}_\Pi$ of space in the plane Π containing L . See [6.8a], which illustrates how the positive angle θ in the illustrated spherical triangle is reversed by $\mathfrak{R}_L$.

If you were an intelligent ant living on the sphere, the above construction of $\mathfrak{R}_L$ as the restriction of $\mathfrak{R}_\Pi$ to the sphere would be meaningless to you. However, it is not hard to re-express $\mathfrak{R}_L$ in intrinsically spherical terms. See [6.8b]. To reflect a in L , first draw the unique line M through a that cuts L at right angles⁸. If d is the distance we must crawl along M from a to reach L , then $\mathfrak{R}_L(a)$ is the point we reach after crawling a further distance of d . Of course M actually intersects L in two antipodal points, but we will arrive at the same $\mathfrak{R}_L(a)$ irrespective of which of these two points is used in the construction.

We now turn to direct motions. The obvious example of a direct motion is a rotation of the sphere about an axis V passing through its centre. Less obvious is the fact (to be proved shortly) that these rotations are the *only* direct motions. To avoid ambiguity in the description of such rotations, we introduce the following standard convention. First note that specifying the axis V is equivalent to specifying either of its antipodal intersection points (say p and q) with the sphere. Now pick one of these, say p . Suppose that the effect of the rotation on a small line-segment issuing from p is a positive rotation of θ —recall that this means *counterclockwise as seen from outside the sphere*. In this case the motion can be unambiguously described as a “positive rotation of θ about p ”; see [6.9b]. We will write this rotation as $\mathcal{R}_p^\theta$. Check for yourself that $\mathcal{R}_p^\theta = \mathcal{R}_q^{-\theta}$.

⁸ If L is thought of as the equator, then when a is one of the poles there are infinitely many M ’s—pick any one you like.



[6.9] Rotation of the sphere via two reflections. [a] If the planes Π_1 and Π_2 meet at angle $(\theta/2)$ then $(\mathfrak{R}_{\Pi_2} \circ \mathfrak{R}_{\Pi_1})$ is a rotation of space through angle θ about their line of intersection. [b] The induced transformation of the sphere is therefore a rotation $\mathcal{R}_p^\theta$ by θ about the intersection point p of the lines on the sphere: $\mathfrak{R}_{L_2} \circ \mathfrak{R}_{L_1} = \mathcal{R}_p^\theta$.

In Chapter 1 we saw that every direct motion of the plane was the composition of two reflections: a rotation if the lines intersected; a translation if the lines were parallel. We will now see that a similar phenomenon occurs on the sphere, but because *every* pair of lines intersect, the composition of two reflections is *always* a rotation—the sphere has no motions analogous to translations.

Figure [6.9a] illustrates the composition $(\mathfrak{R}_{\Pi_2} \circ \mathfrak{R}_{\Pi_1})$ of two reflections of space. Here the planes Π_1 and Π_2 intersect in a line with direction vector $\mathbf{v}$, and the angle from Π_1 to Π_2 is $(\theta/2)$. Restricting attention to any one of the shaded planes orthogonal to $\mathbf{v}$, we see that the transformation induced by $(\mathfrak{R}_{\Pi_2} \circ \mathfrak{R}_{\Pi_1})$ is $(\mathfrak{R}_{l_2} \circ \mathfrak{R}_{l_1})$, where l_1 and l_2 are the lines in which Π_1 and Π_2 intersect the plane. Since $(\mathfrak{R}_{l_2} \circ \mathfrak{R}_{l_1})$ is a rotation of the plane through θ about the intersection point of l_1 and l_2 , it is now clear that $(\mathfrak{R}_{\Pi_2} \circ \mathfrak{R}_{\Pi_1})$ is a rotation of space through angle θ about the axis $\mathbf{v}$.

Figure [6.9b] translates this idea into spherical terms. If Π_1 and Π_2 pass through the centre of the sphere, and the lines (great circles) in which they intersect the sphere are L_1 and L_2 , then

$$\mathfrak{R}_{L_2} \circ \mathfrak{R}_{L_1} = \mathcal{R}_p^\theta.$$

In other words,

A rotation $\mathcal{R}_p^\theta$ of the sphere about a point p through angle θ may be expressed as the composition of reflections in any two spherical lines that pass through p and contain the angle $(\theta/2)$. (6.10)

Note that there is precisely one line P that is mapped into itself by $\mathcal{R}_p^\theta$. If we orient P in agreement with the rotation (as illustrated) then we obtain a one-to-one correspondence between oriented lines and points: P is called the *polar line* of p , and p is called the *pole* of P .

In the case of the plane we used the analogue of (6.10) to show that the composition of two rotations about different points was equivalent (in general) to a single rotation about a third point; exceptionally, however, two rotations could result in a translation. As you might guess, in the case of the sphere there are no exceptions:

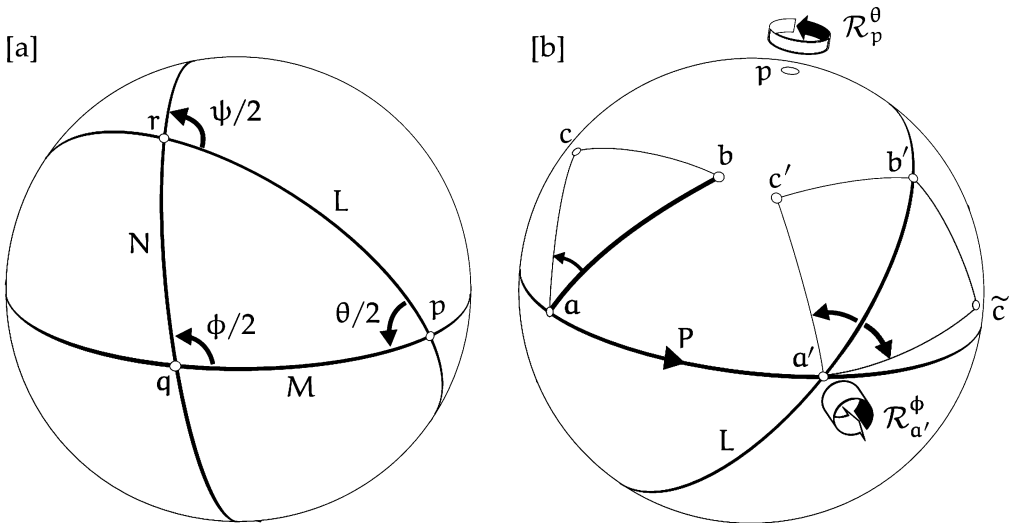
The composition of any two rotations of the sphere is equivalent to a single rotation. Thus the set of all rotations of the sphere forms a group. (6.11)

Figure [6.10a] shows how this may be established using exactly the same argument that was used in the plane. In order to find the net effect of $(\mathcal{R}_q^\phi \circ \mathcal{R}_p^\theta)$, draw the lines L, M, N in the illustrated manner. Then

$$\mathcal{R}_q^\phi \circ \mathcal{R}_p^\theta = (\mathfrak{R}_N \circ \mathfrak{R}_M) \circ (\mathfrak{R}_M \circ \mathfrak{R}_L) = \mathfrak{R}_N \circ \mathfrak{R}_L = \mathcal{R}_r^\psi.$$

This beautiful geometric method of composing spatial rotations was discovered by Olinde Rodrigues in 1840.

Note that in the plane the total amount of rotation produced by rotations of θ and ϕ is simply the sum $(\theta + \phi)$, but on the sphere we have a more complicated rule. If A is the area of the white spherical triangle, and $\mathcal{K} = (1/R^2)$ is the Gaussian curvature of the sphere, then the formula for the angular excess implies that



[6.10] [a] Composition of rotations: $\mathcal{R}_q^\phi \circ \mathcal{R}_p^\theta = (\mathfrak{R}_N \circ \mathfrak{R}_M) \circ (\mathfrak{R}_M \circ \mathfrak{R}_L) = \mathfrak{R}_N \circ \mathfrak{R}_L = \mathcal{R}_r^\psi$. **[b]** There is exactly one direct motion $\mathcal{M}$ (and exactly one opposite motion $\tilde{\mathcal{M}}$) that maps a given line-segment ab to another line-segment $a'b'$ of equal length. Furthermore, $\tilde{\mathcal{M}} = (\mathfrak{R}_L \circ \mathcal{M})$, where L is the line through a' and b' .

$$\psi = \theta + \phi - 2\mathcal{K}\mathcal{A}.$$

We may now complete the classification of the motions of the sphere. As we have remarked, there is precisely one motion of the sphere that carries a given spherical triangle abc to a given congruent image triangle. Figure [6.10b] helps to refine this result. Using the same logic as was used in the plane, we see [exercise] that

There is exactly one direct motion $\mathcal{M}$ (and exactly one opposite motion $\tilde{\mathcal{M}}$) that maps a given line-segment ab to another line-segment $a'b'$ of equal length. Furthermore, $\tilde{\mathcal{M}} = (\mathfrak{R}_L \circ \mathcal{M})$, where L is the line through a' and b' . (6.12)

Figure [6.10b] also shows how we may construct $\mathcal{M}$. Draw the line P through a and a' , and let p be its pole. With the appropriate value of θ , it's clear that $\mathcal{R}_p^\theta$ will carry the segment ab along P to a segment of equal length emanating from a' ; finally, an appropriate rotation $\mathcal{R}_{a'}^\phi$ about a' will carry this segment into $a'b'$. Thus $\mathcal{M} = (\mathcal{R}_{a'}^\phi \circ \mathcal{R}_p^\theta)$, which is equivalent to a single rotation by virtue of (6.11). Combining this fact with (6.12), we deduce that

Every direct motion of the sphere is a rotation, and every opposite motion is the composition of a rotation and a reflection. (6.13)

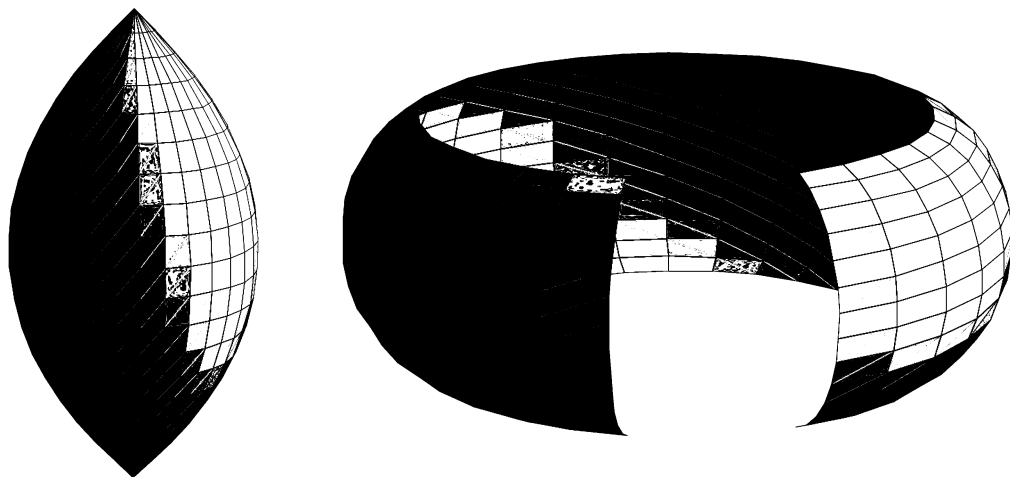
As a simple test of this result (and your grasp of it) consider the *antipodal mapping* that sends every point on the sphere to its antipodal point. Clearly this is a motion, but how does it accord with the above result?

6.2.3 A Conformal Map of the Sphere

The sphere merely provides one particularly simple model of what we have called spherical geometry. As Minding (1839) discovered, *any* surface of constant⁹ Gaussian curvature $\mathcal{K} = (1/R^2)$ has exactly the same intrinsic geometry as a sphere of radius R . To see that such surfaces exist, take a Ping-Pong ball and cut it in half: as you gently flex one of the hemispheres you obtain infinitely many surfaces whose intrinsic geometry is identical to the original sphere.

Figure [6.11] illustrates that even if we restrict attention just to surfaces of revolution, the sphere is not the only one of constant positive curvature. Though they hardly look like spheres, an intelligent ant living on either of these surfaces would never know that he wasn't living on a sphere. Well, that's almost true: eventually he might discover points at which the surface is not smooth, or else he might run into an edge. In 1899 H. Liebmann proved that if a surface of constant positive curvature does not suffer from these defects then it can *only* be a sphere.

⁹ If the curvature is *not* constant, two surfaces can have equal curvature at corresponding points and yet have different intrinsic geometry.



[6.11] Non-spherical surfaces of constant positive curvature. *There exist surfaces of constant positive curvature that are not a portion of the sphere, but Minding proved that their intrinsic geometry is identical to that of the sphere, at least so long as we stay away from the pointed tips on the left, or the edges on the right.*

The sphere also has the advantage of making it obvious that its intrinsic geometry admits a group of motions: in [6.11] it's certainly not clear that figures can be freely moved about and rotated on the surface without stretching them. Nevertheless, the above discussion shows that the actual shape of a surface in space is a distraction, and it would be better to have a more abstract model that captured the essence of all possible surfaces having the same intrinsic geometry.

By the “essence” we mean knowledge of the distance between any two points, for this and this alone determines the intrinsic geometry. In fact—and this is a fundamental insight of differential geometry—it is sufficient to have a rule for the *infinitesimal* distance between neighbouring points. Given this, we may determine the length of any curve as an infinite sum (i.e., integral) of the infinitesimal segments into which it may be divided. Consequently, we may also identify the “lines” of the geometry as shortest routes from one point to another, and we can likewise [exercise] determine angles.

This leads to the following strategy for capturing the essence of any curved surface S (not necessarily one of constant curvature). To avoid the distraction of the shape of the surface in space, we draw a map (in the sense of a geographical atlas) of S on a flat piece of paper. That is we set up a one-to-one correspondence between points $\hat{z}$ on S and points z on the plane, which we will think of as the *complex plane*.

Now consider the distance $d\hat{s}$ separating two neighbouring points $\hat{z}$ and $\hat{q}$ on S . In the map, these points will be represented by z and $q = z + dz$, separated by (Euclidean) distance $ds = |dz|$. Once we have a rule for calculating the actual separation $d\hat{s}$ on S from the apparent separation ds in the map, then (in principle) we know everything there is to know about the intrinsic geometry of S .

The rule giving $d\hat{s}$ in terms of ds is called the *metric*. In general $d\hat{s}$ depends on the direction of dz as well as its length ds : writing $dz = e^{i\phi} ds$,

$$d\hat{s} = \Lambda(z, \phi) ds. \quad (6.14)$$

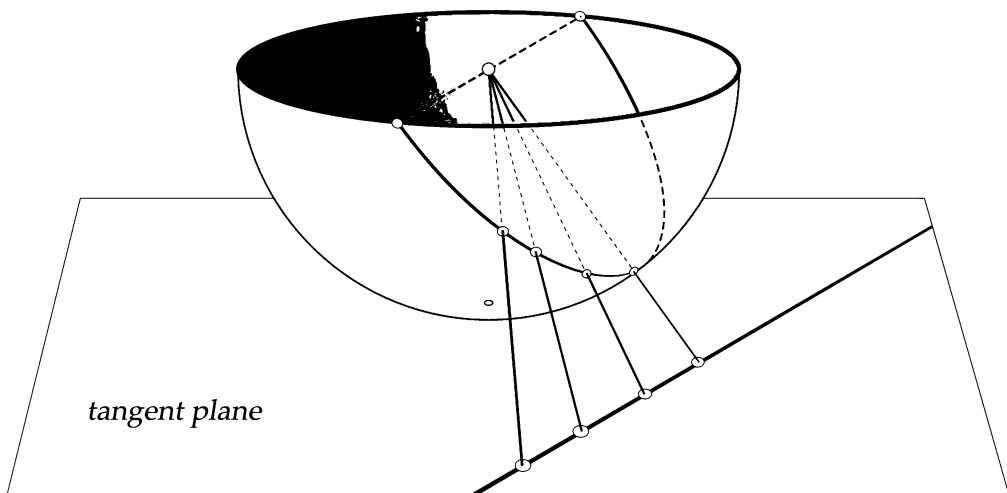
According to this formula, $\Lambda(z, \phi)$ is the amount by which we must expand the apparent separation ds in the map—located at z , and in the direction ϕ —to obtain the true separation $d\hat{s}$ on the surface S .

We will now carry out the above strategy for the sphere. It follows from (6.9) that it is impossible [exercise] to draw a map of the sphere that faithfully represents *every* aspect of its intrinsic geometry. How we choose to draw our map therefore depends on which features we wish to faithfully represent. For example, if we want lines (great circles) on the sphere to be represented by straight lines in the map, then we may employ the so-called *central projection*, in which points are projected from the centre of the sphere onto one of its tangent planes. This yields the so-called *projective map* or *projective model* of the sphere. See [6.12]. Here, the price that we pay for preserving the concept of lines is that angles are not faithfully represented: the angle at which two curves meet on the sphere is not (in general) the angle at which they meet on the map.

For most purposes it is much better to sacrifice straight lines in favour of preserving angles, thereby obtaining a *conformal map* of the surface. In terms of (6.14), a map is conformal if and only if the expansion factor Λ does not depend on the direction ϕ of the infinitesimal vector dz emanating from z :

$$d\hat{s} = \Lambda(z) ds. \quad (6.15)$$

[Recall that we established this fact in Chapter 4.] The great advantage of such a map is that an infinitesimal shape on the surface is represented in the map by a *similar* shape that differs from the original only in size: the one on S is just Λ times bigger.



[6.12] Central projection maps the lines of the sphere to the lines of the map.

In the case of the sphere we already know of a simple method of constructing a conformal map, namely, via stereographic projection. For simplicity's sake, *henceforth we shall take the sphere to have unit radius* so that it may be identified with the Riemann sphere Σ of Chapter 3. Unlike [6.12], the “lines” of this conformal map do not appear as straight lines. In fact it's not too hard to see [exercise] that great circles on Σ are mapped to circles in $\mathbb{C}$ that intersect the unit circle at opposite points.

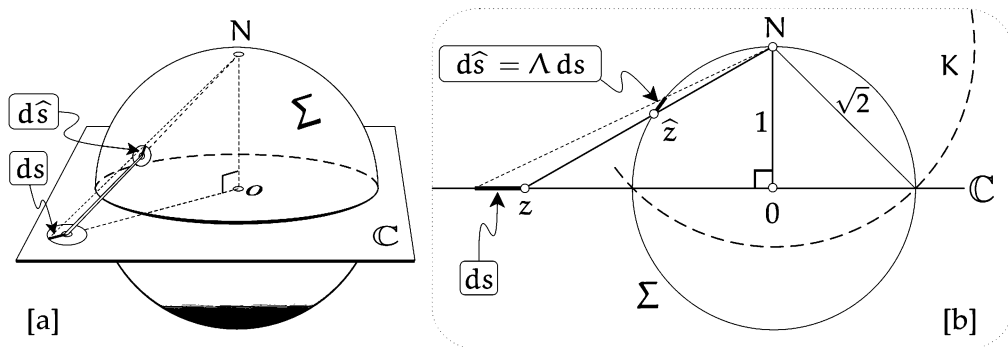
Formula (6.15) may be paraphrased as saying that a map is conformal if infinitesimal circles on S are represented in the map by infinitesimal *circles* (rather than ellipses). Of course stereographic projection satisfies this requirement since it preserves circles of all sizes. Figure [6.13a] illustrates this with an infinitesimal circle of radius $d\hat{s}$ on Σ being mapped to an infinitesimal circle of radius ds in $\mathbb{C}$. To complete the stereographic map we must find its associated metric function Λ —that is the ratio of the two radii in [6.13a].

Consider the vertical cross section of [6.13a] shown in [6.13b], and recall that we showed in Chapter 3 [see p. 161] that stereographic projection is a special case of inversion:

If K is the sphere of radius $\sqrt{2}$ centred at N , then stereographic projection is the restriction to $\mathbb{C}$ or Σ of inversion in K .

Next, consider (3.6) on p. 142, which describes the effect of inversion on the separation of two points. By taking the limit in which the two points coalesce, we may apply this result to [6.13b] to obtain [exercise]

$$d\hat{s} = \frac{2}{[Nz]^2} ds.$$



[6.13] Geometric derivation of the conformal metric under stereographic projection.

[a] A small, ultimately vanishing circle of radius $d\hat{s}$ on Σ is mapped to a circle of radius ds in $\mathbb{C}$. [b] Recalling that stereographic projection is the restriction to Σ of inversion in the sphere K , (3.6) yields $d\hat{s} \asymp \frac{2}{[Nz]^2} ds \asymp \frac{2}{1+|z|^2} ds$. This can also be obtained more directly, without using (3.6), by [exercise] choosing $d\hat{s}$ parallel to $\mathbb{C}$; see Needham (2021, p. 47) for the details.

This can also be obtained more directly, without using (3.6), by [exercise] choosing $d\hat{s}$ parallel to $\mathbb{C}$. Finally, applying Pythagoras' Theorem to the triangle $Nz0$, we obtain

$$d\hat{s} = \frac{2}{1 + |z|^2} ds. \quad (6.16)$$

This flat conformal map with metric (6.16) is the desired abstract depiction of all possible surfaces of constant Gaussian curvature $\mathcal{K} = +1$.

6.2.4 Spatial Rotations as Möbius Transformations

Quite generally, suppose that S is a surface of constant Gaussian curvature (so that it possesses a group of motions) and suppose we have drawn a conformal map of S with metric (6.15). Any motion of S will induce a corresponding transformation of this map in $\mathbb{C}$. Since direct motions of the curved surface are conformal, the conformality of the map implies that the induced complex functions must also be conformal and hence analytic. Purely in terms of $\mathbb{C}$, we may therefore identify a function $f(z)$ as a motion if it is analytic and it “preserves the metric” (6.15). That is, suppose that the analytic function $z \mapsto \tilde{z} = f(z)$ sends two infinitesimally separated points z and $(z + dz)$ to $\tilde{z}$ and $(\tilde{z} + d\tilde{z})$. Then $f(z)$ is a motion if and only if the image separation $d\tilde{s} = |d\tilde{z}|$ is related to the original separation $ds = |dz|$ by

$$\Lambda(\tilde{z}) d\tilde{s} = \Lambda(z) ds.$$

[Likewise, opposite motions of S correspond to the anticonformal mappings of $\mathbb{C}$ that satisfy this equation.] Since $d\tilde{z} = f'(z) dz$, this is equivalent to demanding that f satisfy the following differential equation:

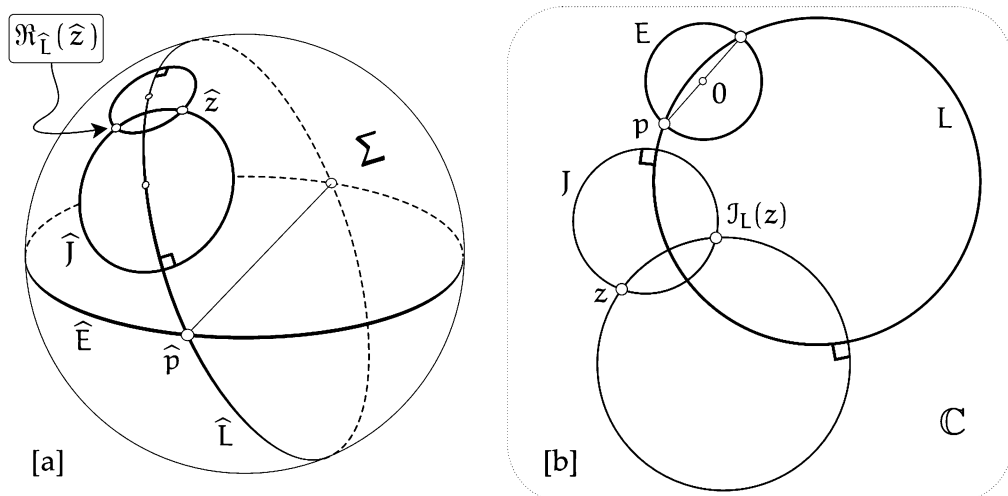
$$f'(z) = \frac{\Lambda(z)}{\Lambda[f(z)]}.$$

Returning to the particular case $S = \Sigma$, and to the particular conformal map obtained by stereographic projection, the direct motions of all possible surfaces of constant Gaussian curvature $\mathcal{K} = +1$ become the set of analytic complex functions that satisfy

$$f'(z) = \frac{1 + |f(z)|^2}{1 + |z|^2}. \quad (6.17)$$

In principle, we could find these complex functions without ever leaving $\mathbb{C}$. However, it is simpler and more illuminating to return to the motions of Σ , described by (6.10) and (6.13). When we apply stereographic projection to these motions, what complex functions are induced in $\mathbb{C}$?

The first step is clearly to find the complex function induced by a reflection $\mathfrak{R}_{\hat{\Gamma}}$ of Σ in a line $\hat{\Gamma}$. Consider [6.14a], which shows a new intrinsic method of constructing the reflection $\mathfrak{R}_{\hat{\Gamma}}(\hat{z})$ of a point $\hat{z}$ on Σ , namely [exercise], as the second intersection



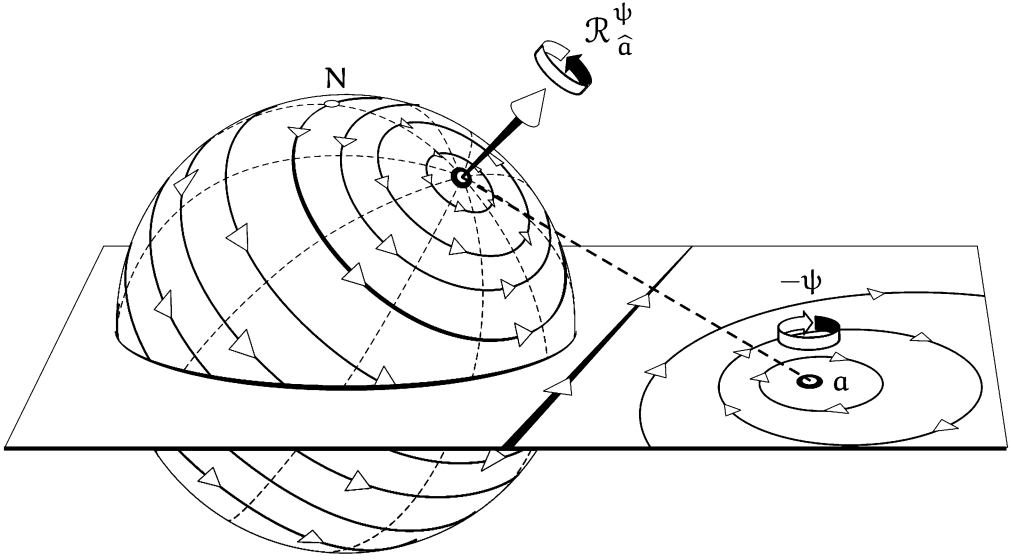
[6.14] Reflection of Σ in a line induces reflection (inversion) of $\mathbb{C}$ in the stereographic image of that line. [a] The reflection $\mathfrak{R}_{\hat{L}}(\hat{z})$ of a point $\hat{z}$ on Σ across the line $\hat{L}$ is the second intersection point of any two circles centred on $\hat{L}$ that pass through $\hat{z}$. [b] The same construction after stereographic projection to $\mathbb{C}$, revealing the standard orthogonal-circle construction of geometric inversion in L .

point of any two circles centred on $\hat{L}$ and passing through $\hat{z}$. Note that these two circles are orthogonal to $\hat{L}$. Figure [6.14b] shows what this construction looks like in the stereographic map. Since stereographic projection preserves circles and angles, the two circles orthogonal to $\hat{L}$ and passing through $\hat{z}$ are mapped to two circles orthogonal to L and passing through z . The second intersection point of these circles is thus the reflection $J_L(z)$ of z in L ! To sum up,

$$\text{Reflection of } \Sigma \text{ in a line induces reflection (inversion) of } \mathbb{C} \text{ in the stereographic image of that line.} \quad (6.18)$$

[For a different proof of (6.18), one that is perhaps even more natural than the one above, see Ex. 2.] As an important special case, note that if $\hat{L}$ is the intersection of Σ with the vertical plane through the real axis, then reflection of Σ in $\hat{L}$ induces complex conjugation, $z \mapsto \bar{z}$.

Now let's find the complex functions corresponding to rotations of Σ . Figure [6.15] illustrates a rotation $\mathcal{R}_a^\psi$ of Σ through angle ψ about the point $\hat{a}$. Let $\hat{b}$ be the antipodal point to $\hat{a}$, so that $b = -(1/\bar{a})$ [see (3.23), p. 167]. These points $\hat{a}$ and $\hat{b}$ lie on the axis of the rotation and remain fixed; correspondingly, a and b will be the fixed points of the induced transformation of $\mathbb{C}$. Furthermore, it is clear geometrically that the effect of the induced transformation on an infinitesimal neighbourhood of a is a rotation about a [exercise], and, by virtue of our conventions, the rotation angle is *negative* ψ .



[6.15] A rotation of the sphere stereographically projects to an Elliptic Möbius transformation of $\mathbb{C}$. According to (6.10), $\mathcal{R}_a^\psi = \mathcal{R}_{\widehat{L}_2} \circ \mathcal{R}_{\widehat{L}_1}$, where $\widehat{L}_1$ and $\widehat{L}_2$ are any two lines passing through $\widehat{a}$ such that the angle between them is $(\psi/2)$. Since stereographic projection preserves circles and angles, the images in $\mathbb{C}$ of these lines will be two circles passing through the fixed points a and b , and containing angle $(\psi/2)$ there. So (6.18) shows that the induced transformation of $\mathbb{C}$ is effected by reflecting across these two intersecting circles, yielding an Elliptic Möbius transformation with multiplier $m = e^{-i\psi}$.

According to (6.10),

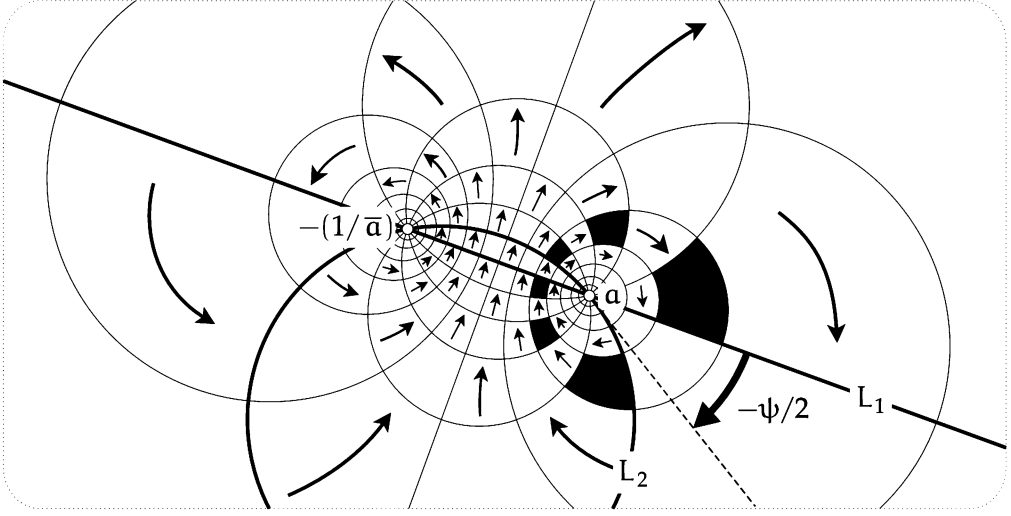
$$\mathcal{R}_a^\psi = \mathcal{R}_{\widehat{L}_2} \circ \mathcal{R}_{\widehat{L}_1},$$

where $\widehat{L}_1$ and $\widehat{L}_2$ are any two lines passing through $\widehat{a}$ (and hence also through $\widehat{b}$) such that the angle between them is $(\psi/2)$. Since stereographic projection preserves circles and angles, the images in $\mathbb{C}$ of these lines will be two circles L_1 and L_2 passing through the fixed points a and b , and containing angle $(\psi/2)$ there. It follows from (6.18) that the transformation $\mathcal{R}_a^\psi$ induced by the rotation $\mathcal{R}_a^\psi$ is

$$\mathcal{R}_a^\psi = \mathcal{J}_{L_2} \circ \mathcal{J}_{L_1}.$$

Thus $\mathcal{R}_a^\psi$ is a Möbius transformation! See [6.16], which illustrates a rotation of $\psi = (\pi/3)$. Referring back to (3.47) on p. 195, and recalling that the “multiplier” describes the local effect of a Möbius transformation in the immediate neighbourhood of a fixed point, we have found that

A rotation $\mathcal{R}_a^\psi$ of Σ stereographically induces an elliptic Möbius transformation $\mathcal{R}_a^\psi$ of $\mathbb{C}$. The fixed points of $\mathcal{R}_a^\psi$ are a and $-(1/\bar{a})$, and the multiplier m associated with a is $m = e^{-i\psi}$.



[6.16] The Elliptic Möbius transformation of $\mathbb{C}$ induced by rotating the sphere by $\psi = (\pi/3)$. The fixed points of the rotation of the sphere are the antipodal points $\hat{a}$ and $\hat{b}$ on the axis of rotation, and these map to the fixed points of the Möbius transformation, a and $b = -(1/\bar{a})$. The Möbius transformation is effected by reflecting in any two circles through these fixed points making angle $-(\psi/2)$, such as L_1 and L_2 .

A straightforward matrix calculation [see Ex. 4] based on (3.42), p. 189, yields the following explicit formula for the matrix of R_a^ψ :

$$[R_a^\psi] = \begin{bmatrix} e^{i(\psi/2)}|a|^2 + e^{-i(\psi/2)} & 2i a \sin(\psi/2) \\ 2i \bar{a} \sin(\psi/2) & e^{-i(\psi/2)}|a|^2 + e^{i(\psi/2)} \end{bmatrix}. \quad (6.19)$$

Note that this is in agreement with (3.39), p. 184: rotations of Σ induce Möbius transformations of the form

$$R_a^\psi(z) = \frac{Az + B}{-\bar{B}z + \bar{A}}. \quad (6.20)$$

By virtue of (6.13), this formula represents the most general direct motion of Σ . We have already noted that $z \mapsto \bar{z}$ corresponds to a reflection of Σ , and it follows [exercise] that the most general opposite motion is represented by a function of the form

$$z \mapsto \left(\frac{A\bar{z} + B}{-\bar{B}\bar{z} + \bar{A}} \right).$$

Figure [6.10b] provided a very elegant geometric method of composing rotations of space. The above analysis now opens the way to an equally elegant method of *computing* the net rotation produced by $(\mathcal{R}_p^\omega \circ \mathcal{R}_a^\psi)$. All we need do is compose the corresponding Möbius transformations:

$$[R_p^\omega \circ R_a^\psi] = [R_p^\omega] [R_a^\psi].$$

An otherwise tricky problem has been reduced to multiplying 2×2 matrices!

In practice, rotations are frequently expressed in terms of a unit vector $\mathbf{v}$ pointing along the axis of rotation, with $\hat{\mathbf{a}}$ at its tip. However, (6.19) is currently expressed in terms of the stereographic image $\mathbf{a}$ of the point $\hat{\mathbf{a}}$. Let us therefore re-express $[R_{\mathbf{a}}^{\psi}]$ in terms of the components l, m, n of the unit vector

$$\mathbf{v} = l\mathbf{i} + m\mathbf{j} + n\mathbf{k}, \quad l^2 + m^2 + n^2 = 1.$$

Referring back to (3.20) on p. 166, we see that $\mathbf{a}$ and $\mathbf{v}$ are related as follows:

$$\mathbf{a} = \frac{l + im}{1 - n} \quad \text{and} \quad |\mathbf{a}|^2 = \frac{1 + n}{1 - n}.$$

Substituting these expressions into (6.19), and removing the common factor of $2/(1 - n)$, we obtain [exercise]

$$[R_{\mathbf{v}}^{\psi}] = \begin{bmatrix} \cos(\psi/2) + in \sin(\psi/2) & (-m + il) \sin(\psi/2) \\ (m + il) \sin(\psi/2) & \cos(\psi/2) - in \sin(\psi/2) \end{bmatrix}. \quad (6.21)$$

You may check for yourself that this matrix is “normalized”: $\det [R_{\mathbf{v}}^{\psi}] = 1$. This makes life that much easier, for when we multiply two such matrices the resulting matrix will be of precisely the same form. Thus, by comparing the result with (6.21), we may read off the net rotation.

For example, suppose we perform a rotation of $(\pi/2)$ about $\mathbf{i}$, followed by a rotation of $(\pi/2)$ about $\mathbf{j}$. The Möbius matrix of the net rotation will therefore be

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 - i & -1 + i \\ 1 + i & 1 + i \end{bmatrix}. \quad (6.22)$$

Comparing this with (6.21), we see [exercise] that this is rotation of $\psi = (2\pi/3)$ about the axis $\mathbf{v} = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} - \mathbf{k})$.

6.2.5 Spatial Rotations and Quaternions

This is all rather elegant, but in fact the above method of composing rotations can be streamlined still further. To see how, let us resume the story of Hamilton’s *quaternions*, which were introduced at the close of Chapter 1.

On the morning of Monday, 16 October 1843, Hamilton went for a walk with his wife. In the back of his mind was a problem with which he had wrestled fruitlessly for more than ten years—the search for a three-dimensional analogue of the complex numbers, one that would permit vectors in space to be multiplied and divided. As we indicated in Chapter 1, Hamilton was unable to solve this problem for the simple reason that no such analogue exists. However, as he passed Brougham Bridge, he suddenly realized that the prize which had eluded him in three-dimensional space was indeed attainable in *four*-dimensional space!

In the two-dimensional complex plane, we may think of 1 and i as unit basis “vectors” in terms of which a general complex number may be expressed as

$z = a1 + b i$. The algebra of $\mathbb{C}$ amounts to stipulating that multiplication distributes over addition, that 1 is the identity (i.e., $1z = z1 = z$), and that $i^2 = -1$.

In four-dimensional space, Hamilton introduced four basis vectors **1**, **I**, **J**, **K** in terms of which a general vector $\mathbb{V}$ (which Hamilton called a quaternion) could be expressed as

$$\mathbb{V} = v\mathbf{1} + v_1 \mathbf{I} + v_2 \mathbf{J} + v_3 \mathbf{K}, \quad (6.23)$$

where the coefficients are all real numbers. To define the product of two such quaternions, Hamilton took **1** to be the identity, and he took **I**, **J**, **K** to be three different square roots of $-\mathbf{1}$, each analogous to i :

$$\mathbf{I}^2 = \mathbf{J}^2 = \mathbf{K}^2 = -\mathbf{1}. \quad (6.24)$$

As in ordinary algebra, Hamilton insisted that multiplication distribute over addition, but in order to render division possible he was forced to make a leap that was revolutionary in its time: non-commutative multiplication. More precisely, Hamilton postulated that

$$\mathbf{IJ} = \mathbf{K} = -\mathbf{JI}, \quad \mathbf{JK} = \mathbf{I} = -\mathbf{KJ}, \quad \mathbf{KI} = \mathbf{J} = -\mathbf{IK}. \quad (6.25)$$

These relations probably look familiar: they are formally identical to the vector products of the basis vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ in three-dimensional space. For example, $\mathbf{i} \times \mathbf{j} = \mathbf{k} = -\mathbf{j} \times \mathbf{i}$. We can use this analogy between $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and **I**, **J**, **K** to express the product of two quaternions in a particularly simple way.

First, let's use the analogy to simplify the notation (6.23). As in ordinary algebra, we suppress the identity **1** in the first term and write $v\mathbf{1} = v$, which Hamilton called the *scalar part* of $\mathbb{V}$. Next we collect the remaining three terms into $\mathbf{V} \equiv v_1 \mathbf{I} + v_2 \mathbf{J} + v_3 \mathbf{K}$, which Hamilton called the *vector part* of $\mathbb{V}$. Thus (6.23) becomes

$$\mathbb{V} = v + \mathbf{V}.$$

In the special case where the scalar part v vanishes, Hamilton called $\mathbb{V} = \mathbf{V}$ a *pure quaternion*. Historically, the concept of a pure quaternion was the forerunner of the idea of an ordinary vector in space. In fact the very word "vector" was coined by Hamilton in 1846 as a synonym for a "pure quaternion".

If we multiply $\mathbb{V}$ by another quaternion $\mathbb{W} = w + \mathbf{W}$, then (6.24) and (6.25) imply [exercise] that

$$\mathbb{V}\mathbb{W} = (vw - \mathbf{V} \cdot \mathbf{W}) + (v\mathbf{W} + w\mathbf{V} + \mathbf{V} \times \mathbf{W}). \quad (6.26)$$

In particular, if $\mathbb{V}$ and $\mathbb{W}$ are pure (i.e., $v = 0 = w$) then this reduces to

$$\mathbb{V}\mathbb{W} = -\mathbf{V} \cdot \mathbf{W} + \mathbf{V} \times \mathbf{W}. \quad (6.27)$$

Historically, this formula constituted the very first appearance in mathematics of the concepts of the dot and cross products. Thus, initially, these vectorial operations

were viewed as merely two facets (the scalar and vector parts) of quaternion multiplication. However, it did not take physicists long to realize that the scalar product and the vector product were each important in their own right, independently of the quaternions from which they had both sprung.

Further results on quaternions will be derived in the exercises; here we wish only to explain the connection between quaternions and rotations of space. This connection hinges on the idea of a *binary rotation*, which means a rotation of space through an angle of π . The appropriateness of the word “binary” stems from the fact that if the same binary rotation is applied twice then the result is the identity.

According to (6.21), the Möbius transformation corresponding to the binary rotation about the axis $\mathbf{v} = l\mathbf{i} + m\mathbf{j} + n\mathbf{k}$ is

$$[\mathbb{R}_{\mathbf{v}}^{\pi}] = \begin{bmatrix} in & -m + il \\ m + il & -in \end{bmatrix}.$$

Now, forgetting about quaternions for a moment, let us redefine $\mathbf{1}$ to be the identity matrix, and $\mathbf{I}, \mathbf{J}, \mathbf{K}$ to be the binary rotation matrices about $\mathbf{i}, \mathbf{j}, \mathbf{k}$, respectively. Thus

$$\mathbf{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{I} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}.$$

As a simple check, note that the Möbius transformation corresponding to the Möbius matrix $\mathbf{K}$ is $K(z) = -z$. Make sure you can see why this is as it should be.

Now we can state the surprising connection with quaternions: *under matrix multiplication, these binary rotation matrices obey [exercise] exactly the same laws (6.24) and (6.25) as Hamilton's $\mathbf{I}, \mathbf{J}, \mathbf{K}$.* It follows that quaternion multiplication is equivalent to multiplying the corresponding 2×2 matrices obtained by replacing Hamilton's $\mathbf{1}, \mathbf{I}, \mathbf{J}, \mathbf{K}$ with the matrices above. Conversely, the general rotation matrix $[\mathbb{R}_{\mathbf{v}}^{\psi}]$ in (6.21) can be expressed [exercise] as the quaternion

$$\mathbb{R}_{\mathbf{v}}^{\psi} = \cos(\psi/2) + \mathbf{V} \sin(\psi/2), \quad (6.28)$$

where $\mathbf{V} = l\mathbf{I} + m\mathbf{J} + n\mathbf{K}$. This elegant formula is much easier to remember than (6.21)!

To compose two rotations of space, we need only multiply the corresponding quaternions. For example, the calculation (6.22)—in which a rotation of $(\pi/2)$ about $\mathbf{i}$ was followed by a rotation of $(\pi/2)$ about $\mathbf{j}$ —now becomes

$$\frac{1}{\sqrt{2}}(1 + \mathbf{J}) \frac{1}{\sqrt{2}}(1 + \mathbf{I}) = \frac{1}{2}(1 + \mathbf{I} + \mathbf{J} - \mathbf{K}).$$

Once again, but more easily than before, we deduce that this is rotation of $\psi = (2\pi/3)$ about the axis $\mathbf{v} = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} - \mathbf{k})$.

Quaternions also yield a very compact formula for the effect of $\mathbb{R}_{\mathbf{v}}^{\psi}$ on the position vector $\mathbf{P} = X\mathbf{i} + Y\mathbf{j} + Z\mathbf{k}$ of a point in space. Suppose that $\mathbb{R}_{\mathbf{v}}^{\psi}$ rotates $\mathbf{P}$ to $\tilde{\mathbf{P}}$. If

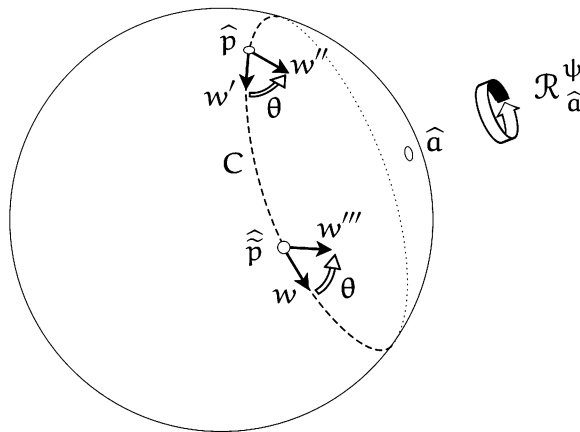
we represent $\mathbf{P}$ by the pure quaternion $\mathbb{P} = X\mathbf{I} + Y\mathbf{J} + Z\mathbf{K}$, and likewise represent $\tilde{\mathbf{P}}$ as $\tilde{\mathbb{P}}$, then

$$\tilde{\mathbb{P}} = \mathbb{R}_{\mathbf{v}}^{\psi} \mathbb{P} \mathbb{R}_{\mathbf{v}}^{-\psi}. \quad (6.29)$$

This result was first published by Arthur Cayley in 1845, though he later conceded priority to Hamilton. Not only is the result elegant, it is also practical. For example, [Hoggar \(1992\)](#) discusses how (6.29) can be used to smooth the motion of a rotating object in a computer animation, while [Horn \(1991\)](#) has used it in research connected with robotic vision!¹⁰

Here we will give the most intuitive explanation of (6.29) that we have been able to think of; Exs. 7, 8 give two more. Begin by noting that any multiple of $\mathbf{P}$ is rotated to the same multiple of $\tilde{\mathbf{P}}$. To establish (6.29) in general, it is therefore sufficient to establish it for the case where $\mathbf{P}$ and $\tilde{\mathbf{P}}$ are unit vectors whose tips $\hat{\mathbf{p}}$ and $\hat{\tilde{\mathbf{p}}}$ lie on the unit sphere. As before, let $\hat{\mathbf{a}}$ be the point at the tip of $\mathbf{v}$.

Consider the following composition of three rotations: $(\mathcal{R}_{\hat{\mathbf{a}}}^{\psi} \circ \mathcal{R}_{\hat{\mathbf{p}}}^{\theta} \circ \mathcal{R}_{\hat{\mathbf{a}}}^{-\psi})$. Certainly this is equivalent to a single rotation, and [6.17] helps us to see what it is. Let C be the invariant circle of $\mathcal{R}_{\hat{\mathbf{a}}}^{\psi}$ passing through $\hat{\mathbf{p}}$ and $\hat{\tilde{\mathbf{p}}}$, and let w be an infinitesimal vector emanating from $\hat{\tilde{\mathbf{p}}}$ and tangent to C . Note that any vector emanating from a point on C will be carried by $\mathcal{R}_{\hat{\mathbf{a}}}^{\pm\psi}$ into a vector making the same



[6.17] Geometric derivation of the quaternion rotation formula. A rotation of θ about $\hat{\tilde{\mathbf{p}}}$, turning w into w''' , can be expressed in terms of an equal rotation about $\hat{\mathbf{p}}$, because the effect of $\mathcal{R}_{\hat{\mathbf{p}}}^{\theta} = \mathcal{R}_{\hat{\mathbf{a}}}^{\psi} \circ \mathcal{R}_{\hat{\tilde{\mathbf{p}}}}^{\theta} \circ \mathcal{R}_{\hat{\mathbf{a}}}^{-\psi}$ is $w \mapsto w' \mapsto w'' \mapsto w'''$. Expressed in terms of quaternions, $\mathbb{R}_{\hat{\mathbf{p}}}^{\theta} = \mathbb{R}_{\mathbf{v}}^{\psi} \mathbb{R}_{\tilde{\mathbf{p}}}^{\theta} \mathbb{R}_{\mathbf{v}}^{-\psi}$. Finally, putting $\theta = \pi$, the binary rotations $\mathbb{R}_{\hat{\mathbf{p}}}^{\pi}$ and $\mathbb{R}_{\hat{\tilde{\mathbf{p}}}}^{\pi}$ become the pure quaternions $\mathbb{P}$ and $\tilde{\mathbb{P}}$, thereby proving the quaternion rotation formula discovered by Hamilton and rediscovered by Cayley: $\tilde{\mathbb{P}} = \mathbb{R}_{\mathbf{v}}^{\psi} \mathbb{P} \mathbb{R}_{\mathbf{v}}^{-\psi}$.

¹⁰ NOTE for the 25th Anniversary Edition: There have no doubt been countless other important applications of this idea over the past 25 years, but I have not researched them for this new edition.

angle with C . This justifies the illustrated effect $w \mapsto w' \mapsto w'' \mapsto w'''$ of the three rotations. Thus the net effect $w \mapsto w'''$ is a rotation of θ about $\hat{p}$:

$$\mathcal{R}_{\hat{p}}^{\theta} = \mathcal{R}_{\hat{a}}^{\psi} \circ \mathcal{R}_{\hat{p}}^{\theta} \circ \mathcal{R}_{\hat{a}}^{-\psi}.$$

This geometric fact may be expressed in terms of Möbius matrices, or equivalently in terms of quaternions:

$$\mathbb{R}_{\hat{p}}^{\theta} = \mathbb{R}_{\mathbf{v}}^{\psi} \mathbb{R}_{\hat{p}}^{\theta} \mathbb{R}_{\mathbf{v}}^{-\psi}.$$

Finally, if we put $\theta = \pi$ then the binary rotations $\mathbb{R}_{\hat{p}}^{\pi}$ and $\mathbb{R}_{\tilde{p}}^{\pi}$ are simply the pure quaternions $\mathbb{P}$ and $\tilde{\mathbb{P}}$, so we are done.

Further Reading. For more on the historical significance of (6.29), see [Altmann \(1989\)](#); for the details of how Hamilton was led to quaternions, see [Waerden \(1985\)](#); for discussion of the connections with modern mathematics and physics, see [Penrose and Rindler \(1984\)](#), [Yaglom \(1988\)](#), [Stillwell \(1992\)](#), and [Penrose \(2005\)](#).

6.3 Hyperbolic Geometry

6.3.1 The Tractrix and the Pseudosphere

Having studied the intrinsic geometry of surfaces of constant positive Gaussian curvature, we now turn to the intrinsic geometry of surfaces of constant *negative* curvature. Just as there are infinitely many surfaces with $\mathcal{K} > 0$, so there are infinitely many with $\mathcal{K} < 0$. Beltrami called such surfaces *pseudospherical*. According to the previously stated result of Minding, all pseudospherical surfaces having the same negative value of $\mathcal{K}$ possess the same intrinsic geometry. To begin to understand hyperbolic geometry, it is therefore sufficient to examine *any* pseudospherical surface. For our purposes, the simplest one is the pseudosphere, so let us explain how this surface may be constructed.

Try the following experiment. Take a small heavy object, such as a paperweight, and attach a length of string to it. Now place the object on a table and drag it by moving the free end of the string along the edge of the table. You will see that the object moves along a curve like that in [6.18a], where the Y-axis represents the edge of the table. This curve is called the *tractrix*, and the Y-axis (which the curve approaches asymptotically) is called the *axis*. The tractrix was first investigated by Newton, in 1676.

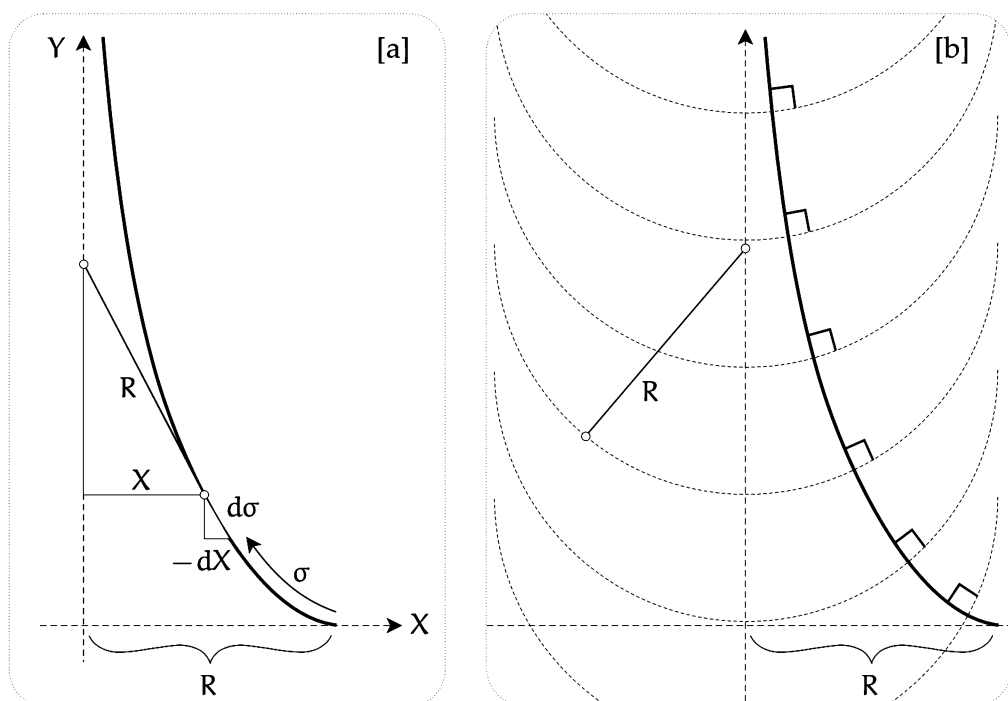
If the length of the string is R , then it follows that the tractrix has the following geometric property: *the segment of the tangent from the point of contact to the Y-axis has constant length R* . This was Newton's definition of the tractrix. As an interesting aside, it follows [exercise] that the tractrix can be constructed as shown in [6.18b], namely, as an orthogonal trajectory through the family of circles of radius R centred

on the axis. This provides a good method of quickly sketching a fairly accurate tractrix.

Returning to [6.18a], let σ represent arc length along the tractrix, with $\sigma = 0$ corresponding to the starting position $X = R$ of the object we are dragging. Just as the object is about to pass through (X, Y) , let dX denote the infinitesimal change in X that occurs while the object moves a distance $d\sigma$ along the tractrix. From the similarity of the illustrated triangles, we deduce that

$$\frac{dX}{d\sigma} = -\frac{X}{R} \implies X = R e^{-\sigma/R}. \quad (6.30)$$

The *pseudosphere* of radius R may now be simultaneously defined and constructed as the surface obtained by rotating the tractrix about its axis. Remarkably, this surface was investigated as early as 1693 (by Christiaan Huygens), two centuries prior to its catalytic role in the acceptance of hyperbolic geometry.



[6.18] Newton's tractrix. [a] Tie a small paperweight to a piece of string of length R . On a table top, start with the string running along one edge, at right angles to the other edge, the Y -axis. If you move the free end of the string along the Y -axis edge of the table, the paperweight will be dragged along the illustrated curve, called the tractrix, which Newton first investigated in 1676. We see that $\frac{dX}{d\sigma} = -\frac{X}{R}$ and so $X = R e^{-\sigma/R}$. [b] The tractrix can also be constructed as an orthogonal trajectory through the family of circles of radius R centred on the Y -axis.

6.3.2 The Constant Negative Curvature of the Pseudosphere*

In this optional section we offer a purely geometric proof that the pseudosphere does indeed have constant negative Gaussian curvature. More precisely, we will use the *extrinsic* definition of $\mathcal{K}$ as the product of the principal curvatures to show that *the pseudosphere of radius R has constant negative curvature $\mathcal{K} = -(1/R^2)$* . Later we will give a purely intrinsic demonstration of this fact, so you won't miss too much if you skip the following argument.

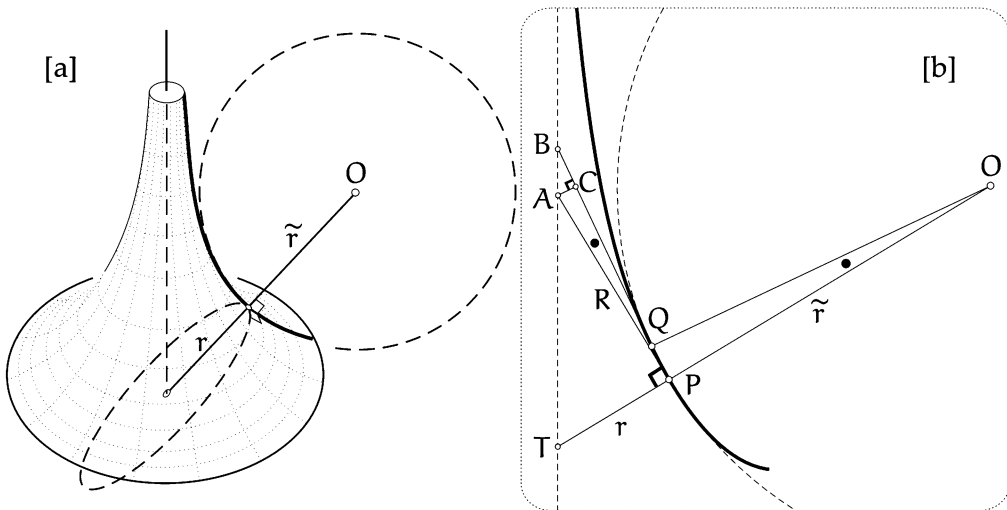
Let r and $\tilde{r}$ be the two principal radii of curvature of the pseudosphere of radius R . As with any surface of revolution, it follows by symmetry¹¹ that

$$\begin{aligned}\tilde{r} &= \text{radius of curvature of the generating tractrix,} \\ r &= \text{the segment of the normal from the surface to the axis,}\end{aligned}$$

as illustrated in [6.19a]. The problem of determining the Gaussian curvature

$$\mathcal{K} = -\frac{1}{r\tilde{r}}$$

is thereby reduced to a problem in plane geometry, which is solved in [6.19b].



[6.19] Geometric proof that the pseudosphere has constant negative Gaussian curvature. [a] The principal directions of maximum and minimum curvature will always occur in the orthogonal directions in which a surface has local mirror symmetry. [See [Needham \(2021\)](#).] Clearly, one direction in which the pseudosphere has mirror symmetry is straight up the pseudosphere, along a meridian tractrix generator. The other principal direction is the orthogonal sideways direction along a circle of latitude. Thus, $\mathcal{K} = -1/(r\tilde{r})$. [b] From this figure we can deduce [see text] that the pseudosphere does indeed have constant negative Gaussian curvature, $\mathcal{K} = -(1/R^2)$.

¹¹ For a full discussion of all these ideas, see [Needham \(2021\)](#).

By definition, the tractrix in this figure has tangents of constant length R . At the neighbouring points P and Q , figure [6.19b] illustrates two such tangents, PA and QB , containing angle $\bullet$. The corresponding normals PO and QO therefore contain the same angle $\bullet$. Note that AC has been drawn perpendicular to QB .

Now let's watch what happens as Q coalesces with P , which itself remains fixed. In this limit, O is the centre of the circle of curvature, PQ is an arc of this circle, and AC is an arc of a circle of radius R centred at P . Thus,¹²

$$\tilde{r} \asymp OP \quad \text{and} \quad \frac{PQ}{OP} \asymp \bullet \asymp \frac{AC}{R} \implies \frac{AC}{PQ} \asymp \frac{R}{\tilde{r}}.$$

Next we appeal to the defining property $PA = R = QB$ of the tractrix to deduce [exercise] that as Q coalesces with P ,

$$BC \asymp PQ.$$

Finally, using the fact that as Q coalesces with P the triangle ABC is ultimately similar to the triangle TAP , we deduce that

$$\frac{r}{R} \asymp \frac{AC}{BC} \asymp \frac{AC}{PQ} \asymp \frac{R}{\tilde{r}}.$$

Behold!¹³

$$\mathcal{K} = -\frac{1}{r\tilde{r}} = -\frac{1}{R^2}.$$

6.3.3 A Conformal Map of the Pseudosphere

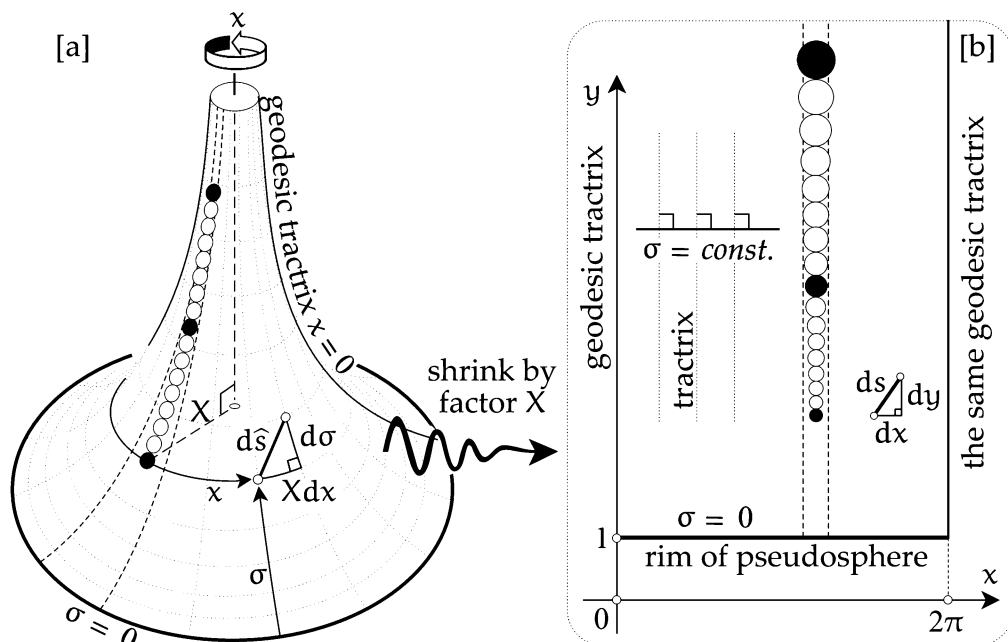
Our next step is to construct a conformal map of the pseudosphere. Recall the benefits of such a map in the case of a sphere: (1) it simultaneously describes all surfaces of curvature $\mathcal{K} = +1$; (2) it provides an elegant and practical description of the motions as Möbius transformations. Both of these benefits persist in the present case of negatively curved surfaces; in particular, the (direct) motions of hyperbolic geometry *again* turn out to be Möbius transformations!

For simplicity's sake, *henceforth we shall take the radius of the pseudosphere to be* $R = 1$, so our map will represent pseudospherical surfaces of curvature $\mathcal{K} = -1$. [NOTE: This is the conventional choice in almost all texts on hyperbolic geometry.] As a first step towards a conformal map, [6.20a] introduces a fairly natural coordinate system (χ, σ) on the pseudosphere.

The first coordinate χ measures angle around the axis of the pseudosphere, say restricted to $0 \leq \chi < 2\pi$. The second coordinate σ measures arc length along each tractrix generator (as in [6.18a]). Thus the curves $\chi = \text{const.}$ are the tractrix

¹² Once again, " $\asymp$ " denotes Newton's concept of *ultimate equality*; see the new Preface.

¹³ One of my most treasured possessions is an email from Bill Thurston saying that he *liked* my Newtonian proof of this fundamentally important fact!



[6.20] Constructing a conformal map of the pseudosphere. [a] Having specialized to the standard case, $R = 1$, we have $X = e^{-\sigma}$. [b] Let us decide to use the angle x around the axis of symmetry as the x -coordinate in this map. If we insist that the map be conformal then all small (ultimately vanishing) distances must be scaled equally, regardless of direction. Since the sideways movement $X dx$ on the left is divided by X in passing to the map on the right, all distances must be divided by X , so $\frac{dy}{d\sigma} = \frac{1}{X} = e^{\sigma}$ and therefore $y = e^{\sigma} + k$. Choosing $k = 0$, we find that the distance $d\hat{s}$ on the pseudosphere is related to the distance ds in the map by the conformal metric formula, $d\hat{s} = ds/y = \sqrt{dx^2 + dy^2}/y$.

generators of the pseudosphere [note that these are clearly geodesics], and the curves $\sigma = \text{const.}$ are circular cross sections of the pseudosphere [note that these are clearly *not* geodesics]. Since the radius of such a circle is the same thing as the X -coordinate in [6.18a], it follows from (6.30) that

The radius X of the circle $\sigma = \text{const.}$ passing through the point (x, σ) is given by $X = e^{-\sigma}$.

In our map, let us choose the angle x as our horizontal axis, so that the tractrix generators of the pseudosphere are represented by vertical lines. See [6.20b]. Thus a point on the pseudosphere with coordinates (x, σ) will be represented in the map by a point with Cartesian coordinates (x, y) , which we will soon think of as the complex number $z = x + iy$.

If our map were not required to be special in any way, then we could simply choose $y = y(x, \sigma)$ to be an arbitrary function of x and σ . In stark contrast to

this, our requirement that the map be *conformal* leaves (virtually) no freedom in the choice of the y -coordinate. Let's try to understand this.

Firstly, the tractrix generators $x = \text{const.}$ are orthogonal to the circular cross sections $\sigma = \text{const.}$, so the same must be true of their images in our conformal map. Thus the image of $\sigma = \text{const.}$ must be represented by a horizontal line $y = \text{const.}$, and from this we deduce that $y = y(\sigma)$ must be a function solely of σ .

Secondly, on the pseudosphere consider the arc of the circle $\sigma = \text{const.}$ (of radius X) connecting the points (x, σ) and $(x + dx, \sigma)$. By the definition of x , these points subtend angle dx at the centre of the circle, so their separation on the pseudosphere is $X dx$, as illustrated. In the map, these two points have the same height and are separated by distance dx . Thus in passing from the pseudosphere to the map, this particular line-segment is shrunk by factor X . [We say "shrunk" because we're dividing by X , but since $X \leq 1$ this is actually an expansion.] However, since the map is conformal, an infinitesimal line-segment emanating from (x, σ) in *any* direction must be multiplied by the *same* factor $(1/X) = e^\sigma$. In other words, the metric is

$$d\hat{s} = X ds.$$

Thirdly, consider the uppermost black disc on the pseudosphere shown in [6.20a]. Think of this disc as infinitesimal, say of diameter ϵ . In the map, it will be represented by *another disc*, whose diameter (ϵ/X) may be interpreted more vividly as the angular width of the original disc as seen by an observer on the pseudosphere's axis. Now suppose we repeatedly translate the original disc towards the pseudosphere's rim, moving it a distance ϵ each time. Figure [6.20a] illustrates the resulting chain of touching, congruent discs. As the disc moves down the pseudosphere, it recedes from the axis, and its angular width as seen from the axis therefore diminishes. Thus the image disc in the map appears to gradually shrink as it moves downward, and the equal distances 8ϵ between the successive black discs certainly do not appear equal in the map.

Having developed a feel for how the map works, let's actually calculate the y -coordinate corresponding to the point (x, σ) on the pseudosphere. From the above observations (or directly from the requirement that the illustrated triangles be similar) we deduce that

$$\frac{dy}{d\sigma} = \frac{1}{X} = e^\sigma \implies y = e^\sigma + \text{const.}$$

The standard choice of this constant is 0, so that

$$y = e^\sigma = (1/X).$$

Thus the entire pseudosphere is represented in the map by the shaded region lying above the line $y = 1$ (which itself represents the pseudosphere's rim), and the metric associated with the map is

$$d\hat{s} = \frac{ds}{y} = \frac{\sqrt{dx^2 + dy^2}}{y}. \quad (6.31)$$

For future use, also note that an infinitesimal rectangle in the map with sides dx and dy represents a similar infinitesimal rectangle on the pseudosphere with sides (dx/y) and (dy/y) . Thus the apparent area $dx \, dy$ in the map is related to the true area dA on the pseudosphere by

$$dA = \frac{dx \, dy}{y^2}. \quad (6.32)$$

6.3.4 Beltrami's Hyperbolic Plane

In the Introduction we gave the impression that Beltrami had succeeded in interpreting abstract hyperbolic geometry as the intrinsic geometry of the pseudosphere. This is really not possible, and it is *not* what Beltrami claimed.

The abstract hyperbolic geometry discovered by Gauss, Bolyai, and Lobachevsky is understood to take place in a *hyperbolic plane* that is exactly like the Euclidean plane, *except* that lines within it obey the hyperbolic axiom (6.3):

Given a line L and a point p not on L , there are at least two lines through p that do not meet L .

The constant negative curvature of the pseudosphere ensures that it faithfully represents all consequences of this axiom that deal only with a finite region of the hyperbolic plane. An example of such a consequence is the theorem that the angular excess of a triangle is a negative multiple of its area, and this does indeed hold on the pseudosphere.

However, the pseudosphere will not do as a model of the *entire* hyperbolic plane, because it departs from the Euclidean plane in two unacceptable ways:

- The pseudosphere is akin to a cylinder instead of a plane. For example, a closed loop in the plane can always be shrunk to a point, but a loop on the pseudosphere that wraps around the axis cannot be.
- In the hyperbolic plane, as in the Euclidean plane, a line-segment can be extended indefinitely in either direction. We have already remarked that the tractrix generators of the pseudosphere are clearly geodesic, and we would therefore like to interpret them as hyperbolic lines. But although such a tractrix extends indefinitely *up* the pseudosphere, in the other direction it terminates when it hits the rim.

Beltrami pointed out that the first of these problems can be resolved as follows. Imagine the pseudosphere covered by a thin stretchable sheet. To obtain the map in [6.20b], we cut this sheet along a tractrix generator and unwrap it onto the shaded

region. Of course to make it lie flat and fit into this rectangular region, the sheet must be stretched—the metric (6.31) tells us *how much* stretching must be applied to each part. But now imagine the sheet as wrapping round and round the pseudosphere infinitely many times¹⁴, like an endless roll of cling film¹⁵. By unwrapping this infinitely long sheet (stretching as we go) we can now cover the entire region above $y = 1$. According to this interpretation, a particle travelling along a horizontal line in the map would correspond to a particle travelling round and round a circle $\sigma = \text{const.}$ on the pseudosphere, executing one complete revolution for each movement of 2π along the line.

Now let us explain how the conformal map solves our second problem—the pseudosphere's edge. In terms of *extrinsic* geometry, this edge is an insurmountable obstacle: we cannot extend the pseudosphere smoothly beyond this edge while preserving its constant curvature. However, we only care about the pseudosphere's *intrinsic* geometry, and we have seen that if we measure distance using $d\hat{s} = \frac{ds}{y}$, this is identical to the region $y > 1$ in [6.21].

Imagining yourself as a tiny two-dimensional being living in [6.21], walking down a line $x = \text{const.}$ is exactly the same thing as walking down a tractrix on the pseudosphere. Of course on the pseudosphere your walk is rudely interrupted at some point $\hat{p}$ on the rim ($\sigma = 0$), corresponding to a point p on the line $y = 1$. But in the map this point p is just like any other, and there is absolutely nothing preventing you from continuing your walk all the way down to the point q on $y = 0$.

Why stop at q ? The answer is that you will never even get that far, because q is infinitely far from p ! Suppose that you are the illustrated small disc on the line $y = 2$, and that I am standing outside your hyperbolic world, watching as you walk at a steady pace towards $y = 0$. Of course you remain the same hyperbolic size as you walk, but to *me* you appear to shrink. This is made particularly vivid by the illustrated Euclidean interpretation [exercise] of your hyperbolic size $d\hat{s} = \frac{ds}{y}$:

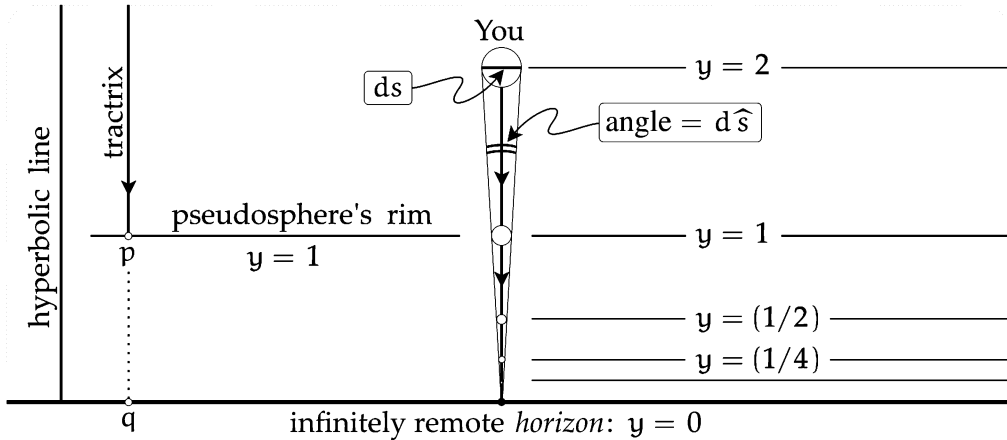
*The hyperbolic diameter of an infinitesimal disc centred at $(x + iy)$ is
the angle it subtends at the point x on the real axis.* (6.33)

Thus your apparent size must shrink so that you subtend a constant angle, and although all your hyperbolic strides are the same length, to me they look shorter and shorter, and you appear to be travelling more and more slowly.

For example, suppose you are walking at a steady speed of $\ln 2$. As illustrated, integration of (dy/y) shows [exercise] that you reach $y = 1$ after one unit of time, $y = (1/2)$ after two units of time, $y = (1/4)$ after three units of time, etc. Thus, viewed from outside your world, each successive unit of time only halves your

¹⁴ Stillwell (1996) points out that this is probably the very first appearance in mathematics of what topologists now call a *universal cover*.

¹⁵ For Americans, read "plastic wrap".



[6.21] The pseudosphere has an edge, but the hyperbolic plane is uniform and infinite. Suppose “You” start at the point on the pseudosphere corresponding to $y = 2$ and walk down a tractrix generator at a steady rate of $\ln 2$. Then in the conformal map you reach $y = 1$ (the pseudosphere’s rim—the edge of your world!) after just one unit of time. But if we imagine you living in the map, perceiving distances around you via the hyperbolic metric, then you can continue your downward journey, arriving at $y = (1/2)$ after two units of time, $y = (1/4)$ after three units of time, etc. Although you actually stay the same size, in the map you appear to shrink, for your true size is the angle $d\hat{s}$ that you subtend at the horizon. Thus, viewed from outside your world, you shrink with each successive unit of time, only halving your distance from the horizon, $y = 0$; therefore, you will never reach it. Your world is infinite and uniform: no place or direction seems any different from any other.

distance from $y = 0$, and therefore you will never reach it. [An appropriate name for this phenomenon might be “Zeno’s Revenge”!]

We now possess a concrete model of the *hyperbolic plane*, namely, the entire shaded half-plane $y > 0$ with metric $d\hat{s} = \frac{ds}{y}$. The points on the real axis are infinitely far from ordinary points and are not (strictly speaking) considered part of the hyperbolic plane. They are called *ideal points*, or *points at infinity*. The complete line $y = 0$ of points at infinity will be called the *horizon*¹⁶.

Studying hyperbolic geometry by means of this map is like studying spherical geometry via a stereographic map, without ever having seen an actual sphere. This is not as bad as it sounds. After all, by constructing geographical maps through terrestrial measurements, man developed a good understanding of the surface of the Earth centuries before venturing into space and gazing down on its roundness!

Still, it would be nice to have the analogue of a globe instead of a mere atlas. The pseudosphere only models a portion of the hyperbolic plane, but might there

¹⁶ For reasons that will be clear shortly, another name is the *circle at infinity*.

exist a different surface that is isometric to the *entire* hyperbolic plane? Sadly, in 1901 Hilbert proved¹⁷ that every pseudospherical surface necessarily has an edge beyond which it cannot be smoothly extended while preserving its constant negative curvature. Thus the upper half-plane with metric (6.31) is as good a depiction of the hyperbolic plane as we are going to get.

However, just as an atlas uses different kinds of maps to represent the surface of the Earth, so we can and will use different types of maps to represent the hyperbolic plane. The particular map we have obtained is conventionally called the *Poincaré upper half-plane*, but there is also one called the *Poincaré disc*, and another called the *Klein disc*. Poincaré obtained the first two models in 1882, while Klein obtained the third in 1871.

We cannot let the names of these models pass without comment. Anyone with even a passing interest in the history of mathematics will know that ideas are frequently (usually?) named after the wrong person. In fact¹⁸, *the three models above were all discovered by Beltrami!* As we shall see, Beltrami obtained these three models (in 1868, 14 years before Poincaré) in a beautifully unified way, from a fourth model consisting of a map drawn on a hemisphere. And in case you're wondering, yes, the hemisphere model is Beltrami's, too!

In this *25th Anniversary Edition* we shall dogedly attempt to set history straight, as we have previously attempted to do in VDGF, by giving both men *equal* credit, and renaming the conformal maps as the *Beltrami–Poincaré* half-plane and disc models.

6.3.5 Hyperbolic Lines and Reflections

Before we get going, let's indicate *where* we are going, focusing just on direct motions. In Euclidean geometry, every direct motion is the composition of reflections in two lines. We have seen that the same is true in spherical geometry, and we will soon show that it is again true in hyperbolic geometry. Since two Euclidean lines must intersect or be parallel, there are just two kinds of direct Euclidean motions: rotations and translations. The absence of parallel lines on the sphere implies that its direct motions can only be rotations. Conversely, the multitude of parallel lines in the hyperbolic plane yields a geometry that is *richer* than Euclid's, containing rotations, translations, and a *third* kind of motion that has no Euclidean counterpart.

To avoid confusion, let us use the prefix “h-” to distinguish hyperbolic concepts from their Euclidean descriptions in the map. For example, an “h-line” will mean a “hyperbolic line” (i.e., a geodesic), while a “line” will refer to an ordinary straight

¹⁷ Hilbert (1965, Vol. 2, pp. 437–448), with English translation available in Hilbert (1902).

¹⁸ See Milnor (1982), Beltrami (1868a), Beltrami (1868b), and Stillwell (2010).

line in the map. Let us also define $\mathcal{H}\{z_1, z_2\}$ to be the h-distance (measured using $d\hat{s} = \frac{ds}{y}$) between z_1 and z_2 . For example, if dz is infinitesimal, then

$$\mathcal{H}\{z + dz, z\} = \frac{|dz|}{\text{Im } z}.$$

Finally, let us define an h-circle of h-radius ρ and h-centre c to be the locus of points z such that $\mathcal{H}\{z, c\} = \rho$.

Since tractrix generators of the pseudosphere are clearly geodesic, vertical lines in the map should also be geodesic, i.e., they should be examples of h-lines. Figure [6.22a] confirms this directly by showing that

The (unique) shortest route between two vertically separated points is the vertical line-segment L connecting them. (6.34)

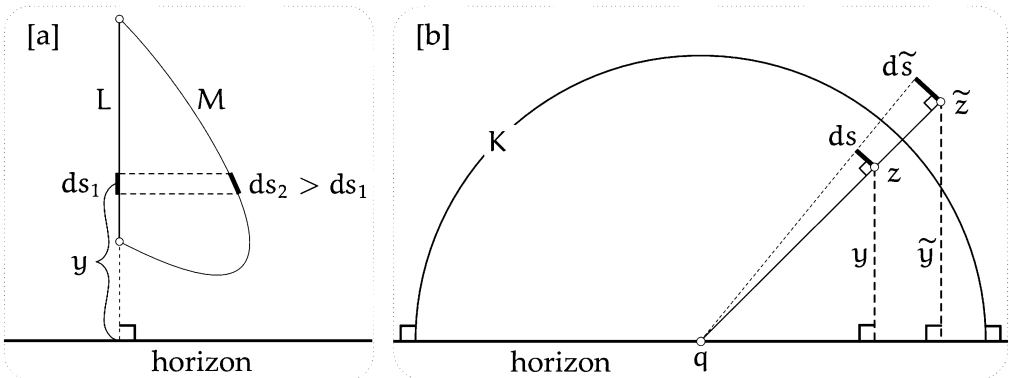
To see this, compare L with any other route, such as M. Let ds_1 be an infinitesimal segment of L at height y , and let ds_2 be the corresponding element of M cut off by horizontal lines through the ends of ds_1 . Since

$$d\hat{s}_1 = \frac{ds_1}{y} < \frac{ds_2}{y} = d\hat{s}_2,$$

the total hyperbolic length of L is less than M's. Done. From this we can deduce that

$$\mathcal{H}\{(x + iy_1), (x + iy_2)\} = |\ln(y_1/y_2)|. \quad (6.35)$$

Through a given point of the pseudosphere we obviously have geodesics in all directions, not just tractrix generators; what do these more general h-lines look like in the map? The answer is very beautiful and unexpected:



[6.22] [a] Vertical lines are h-lines (geodesics), because $d\hat{s}_1 = ds_1/y < ds_2/y = d\hat{s}_2$. [b] Inversion in a semicircle orthogonal to the horizon preserves h-distance. Under $z \mapsto \tilde{z} = \mathcal{I}_K(z)$, we see that $d\hat{s} = d\tilde{s}/\tilde{y} = ds/y = d\hat{s}$. Since distance is preserved in this particular direction, it must be preserved in all directions.

Every h-line is either a half-line orthogonal to the horizon, or else a semicircle orthogonal to the horizon. (6.36)

Before we prove this, it's important to realize that if you were an inhabitant of the hyperbolic plane, there would be no way for you to distinguish between the semi-circular h-lines and the vertical h-lines: every line is exactly like every other, it's just our map that makes them look different. What about the fact that the semicircles have two ends on the horizon, whereas the vertical h-lines appear to only have one? The answer is that, in addition to the points on the real axis, there is one more point at infinity, and all the vertical h-lines meet there. According to (6.31), as we move upward along two neighbouring, vertical h-lines, the h-distance between them dies away like $(1/y)$, and they converge to a single point at infinity; this is particularly vivid on the pseudosphere. Finally, note that even in terms of the map, a vertical h-line may be viewed as just a special case of a semicircular h-line by allowing the radius to tend to infinity.

We will prove (6.36) by first establishing another equally beautiful fact, one that is fundamental to all that follows:

Inversion in a semicircle orthogonal to the horizon is an opposite motion of the hyperbolic plane. (6.37)

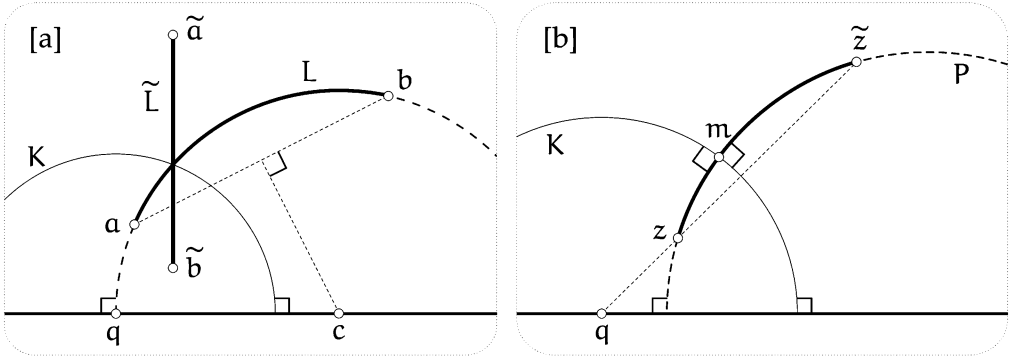
To see why this is true, consider the inversion $z \mapsto \tilde{z} = \mathcal{I}_K(z)$ illustrated in [6.22b]. We need to show that $\mathcal{I}_K(z)$ does not alter the h-length $d\hat{s}$ of any infinitesimal line-segment ds emanating from z . However, because our model of the hyperbolic plane is *conformal*, we need only show that $\mathcal{I}_K(z)$ preserves the h-length of any *single* ds , in a direction of our choosing. Choosing ds orthogonal to the radius qz of K (as illustrated), the anticonformality of inversion implies that the image $d\tilde{s}$ is also orthogonal to this radius. Thus, by virtue of the illustrated similar triangles, it follows [exercise] that

$$d\hat{\tilde{s}} = \frac{d\tilde{s}}{\tilde{y}} = \frac{ds}{y} = d\hat{s},$$

as was to be shown.

To establish (6.36), consider [6.23a]. First, the figure shows that two points a and b [$\text{Re}(a) \neq \text{Re}(b)$] can always be joined by a unique arc L of a semicircle orthogonal to the real axis: to construct the centre c , simply draw the perpendicular bisector of ab . As illustrated, let q be one of the ends of this semicircle. Now we need to show that L is the shortest (smallest h-length) route from a to b .

We show this by applying an inversion $z \mapsto \tilde{z} = \mathcal{I}_K(z)$, where K is any circle centred at q . This carries the arc L into a vertical line-segment $\tilde{L}$, and (6.37) tells us that $\tilde{L}$ and L have equal h-length. More generally, any route M from a to b has the same h-length as the route $\tilde{M} = \mathcal{I}_K(M)$ from $\tilde{a}$ to $\tilde{b}$. Thus if L were not the shortest



[6.23] [a] Geometric proof that semicircles orthogonal to the horizon are h-lines (geodesics). Construct L as shown, then invert it in any circle K centred at q , obtaining $\mathcal{I}_K(L) = \tilde{L}$, as shown. Since $\tilde{L}$ has been proven to be the shortest route from $\tilde{a}$ to $\tilde{b}$, and since $\mathcal{I}_K$ preserves h -distance, L must be the shortest route from a to b . **[b] If K is an h-line, then $\mathcal{I}_K = \mathfrak{R}_K$ is (literal) h -reflection in K .** For if $\tilde{z} = \mathcal{I}_K(z)$ then all circles through z and $\tilde{z}$ cut K orthogonally, and so the illustrated h -line P does, too. The h -lengths of zm and $m\tilde{z}$ are equal, for they are interchanged by the h -distance-preserving $\mathcal{I}_K$, so $\tilde{z} = \mathfrak{R}_K(z)$, as claimed.

route from a to b , then $\tilde{L}$ would not be the shortest route from $\tilde{a}$ to $\tilde{b}$, in violation of (6.34). Done.

Incidentally, note that this construction also enables us (in principle) to calculate the h -distance between any two points in the hyperbolic plane:

$$\mathcal{H}\{a, b\} = \mathcal{H}\{\tilde{a}, \tilde{b}\} = \left| \ln \left(\frac{\text{Im } \tilde{a}}{\text{Im } \tilde{b}} \right) \right|,$$

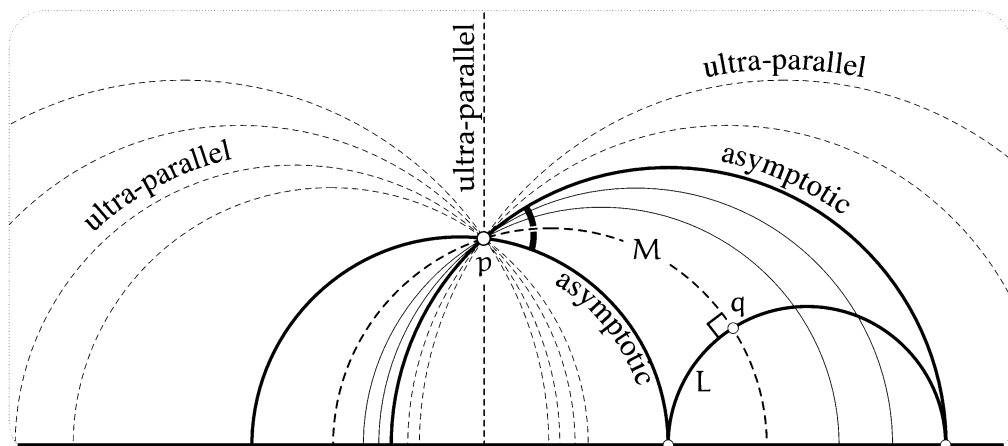
by virtue of (6.35). Later we shall be able to derive a more explicit formula.

The fact that a semicircle orthogonal to the real axis is an h -line strongly suggests the following re-interpretation of (6.37): hyperbolic plane in the h -line

Inversion in a semicircle K orthogonal to the horizon is a reflection $\mathfrak{R}_K$ of the hyperbolic plane in the h -line K . (6.38)

In symbols, $\mathfrak{R}_K(z) = \mathcal{I}_K(z)$. Before proving this, let's be clear what we *mean* by reflection. Just as we would in Euclidean and spherical geometry, we begin the construction of $\mathfrak{R}_K(z)$ by drawing the h -line P that passes through z and cuts K perpendicularly, say at m . Then $\mathfrak{R}_K(z)$ is defined to be the point on P that is the same h -distance from m as z .

To prove (6.38), consider [6.23b], in which $\tilde{z} = \mathcal{I}_K(z)$. First recall that every circle through z and $\tilde{z}$ is automatically orthogonal to K . In particular, the unique h -line through z and $\tilde{z}$ must be orthogonal to K , and hence it is the desired " P " of the previous paragraph. Finally, recall that $\mathcal{I}_K$ maps P into itself, swapping the segments zm and $\tilde{z}m$. Thus, since $\mathcal{I}_K$ is a motion, these two h -line segments have equal h -length, as was to be shown.



[6.24] The hyperbolic plane satisfies the Hyperbolic Axiom, for we see that there are infinitely many h -lines [shown dashed] through p that fail to meet the h -line L : they are said to be *ultra-parallel* to L . The two h -lines that meet L at the horizon are called *asymptotic*. The h -distance of p from L is defined as in Euclidean geometry, as the length of the unique line segment pq that meets L at right angles.

Conversely, if we are *given* any two points z and $\tilde{z}$, then we may draw the perpendicular h -bisector K , and $\mathfrak{R}_K$ swaps z and $\tilde{z}$. Also note that z and its reflection $\tilde{z} = \mathfrak{R}_K(z)$ are the same h -distance from *every* point k on K , just as in Euclidean and spherical geometry. This is easily proved: since $\mathcal{I}_K$ is a motion, and $\tilde{k} = \mathcal{I}_K(k) = k$, it follows that $\mathcal{H}\{z, k\} = \mathcal{H}\{\tilde{z}, \tilde{k}\} = \mathcal{H}\{\tilde{z}, k\}$.

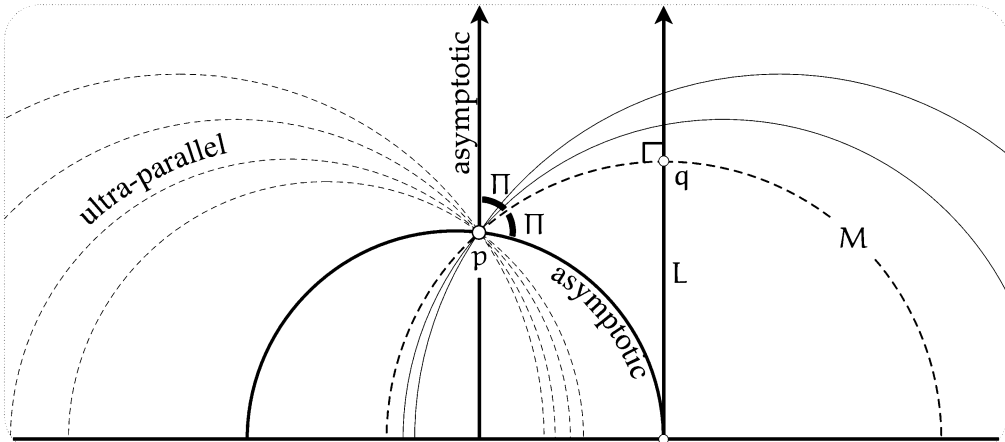
It is becoming clear that hyperbolic geometry has much in common with Euclidean geometry. However, now that we know what h -lines look like, [6.24] shows that hyperbolic geometry really is non-Euclidean: there are infinitely many h -lines through p [shown dashed] that do not meet the h -line L . Such h -lines are said to be *ultra-parallel* to L .

Separating the ultra-parallels from the h -lines that do intersect L , we see that there are precisely two h -lines that fail to meet L anywhere within the hyperbolic plane proper, but that do meet it on the horizon. These two h -lines are called *asymptotic*¹⁹.

As in Euclidean geometry, the figure makes it clear that there is precisely one h -line M passing through p that cuts L at right angles (say at q). In fact [exercise] M may be constructed as the unique h -line through p and $\mathfrak{R}_L(p)$. The existence of M makes it possible to define the distance of a point p from a line L in the usual way, namely, as the h -length of the segment pq of M .

Since M and L are orthogonal, $\mathfrak{R}_M = \mathcal{I}_M$ maps L into itself, swapping the two ends on the horizon. It follows [exercise] that $\mathfrak{R}_M$ swaps the two asymptotic lines,

¹⁹ Another commonly used name is *parallel*.



[6.25] The same geometry as the previous figure, but in the case where L is a vertical half-line.

and that M bisects the angle at p contained by the asymptotic lines. The angle between M and either asymptotic line is called the *angle of parallelism*, and is usually denoted Π . As one rotates the line M about p , its intersection point on L moves off towards infinity, and Π tells you how far you can rotate M before it starts missing L entirely.

Finally, [6.25] merely serves to illustrate the same concepts and terminology as [6.24], but in the case where the h -line L happens to be represented as a vertical half-line instead of a semicircle.

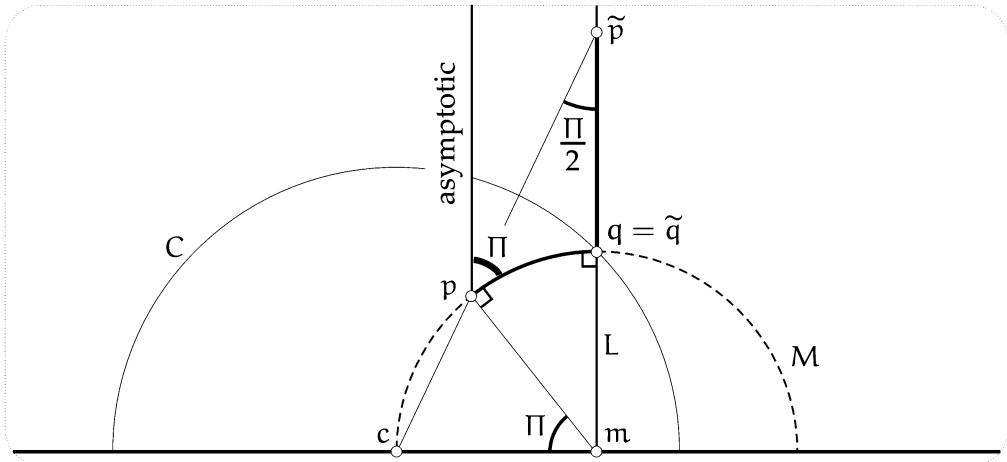
6.3.6 The Bolyai–Lobachevsky Formula*

This brief, optional subsection nicely illustrates how the preceding ideas may be used to solve a significant, concrete problem: finding the angle of parallelism, Π .

In Euclidean geometry the analogue of the two asymptotic lines is the unique parallel line through p , and since this is perpendicular to M , the analogue of Π is a right angle. On the other hand, in hyperbolic geometry it is clear that Π is always acute, and that its value decreases as the distance $D \equiv \mathcal{H}\{p, q\}$ of p from L increases. More precisely, both Bolyai and Lobachevsky obtained this result:

$$\text{Bolyai–Lobachevsky Formula : } \quad \tan(\Pi/2) = e^{-D},$$

and from this they were able to derive many of their other results. We now give a simple geometric proof of this so-called *Bolyai–Lobachevsky Formula*. Greenberg (2008) has called this “one of the most remarkable formulas in the whole of mathematics”, but for us it will be of only incidental interest.



[6.26] Geometric proof of the Bolyai–Lobachevsky Formula. To find the h -length D of the arc pq , apply $z \mapsto \tilde{z} = \mathfrak{R}_C(z)$. This carries the arc pq into the illustrated vertical line-segment $\tilde{p}q$. Applying (6.35), $D = \left| \ln \left(\frac{[qm]}{[pm]} \right) \right| = \left| \ln \left(\frac{[cm]}{[pm]} \right) \right| = |\ln \tan(\Pi/2)| = -\ln \tan(\Pi/2)$, the last equality following from the fact that Π is acute. Thus, $\tan(\Pi/2) = e^{-D}$, as was to be shown.

First note that it is sufficient to establish the formula using [6.25], rather than [6.24]. This is because we may transform [6.24] into [6.25] by performing an inversion (i.e., a hyperbolic reflection) in any semicircle centred at one of the ends of L .

Figure [6.26] reproduces the essential elements of [6.25]. In order to find the h-length D of the arc pq , let us apply the h-reflection $z \mapsto \tilde{z} = \mathfrak{R}_C(z)$, where C is the illustrated semicircle that is centred at the end c of M , and that passes through q . This carries the arc pq into the illustrated vertical line-segment $\tilde{p}q$. By virtue of (6.35), it only remains to find the ratio of the y -coordinates of q and $\tilde{p}$, i.e., the ratio of the Euclidean distances $[qm]$ and $[\tilde{p}m]$.

From the fact that the radius pm is orthogonal to the circle M it follows [exercise] that the angle pmc equals Π . It then follows [exercise] that the angle $c\tilde{p}m$ equals $(\Pi/2)$, as illustrated. Thus

$$D = \left| \ln \left(\frac{[q\mathbf{m}]}{[\tilde{p}\mathbf{m}]} \right) \right| = \left| \ln \left(\frac{[c\mathbf{m}]}{[\tilde{p}\mathbf{m}]} \right) \right| = |\ln \tan(\Pi/2)| = -\ln \tan(\Pi/2),$$

where the last equality follows from the fact that $\tan(\Pi/2) < 1$, because Π is acute. Thus $\tan(\Pi/2) = e^{-D}$, as was to be shown.

6.3.7 The Three Types of Direct Motion

As we have pointed out, the “Poincaré upper half-plane” was first discovered by Beltrami, and in this *25th Anniversary Edition* we are biting the bullet (as we already

did in VDGF) calling it instead the *Beltrami–Poincaré upper half-plane*. What Poincaré *does* deserve sole credit for—enormous credit!—is the realization that hyperbolic geometry is intimately connected with complex analysis. The cornerstone of this connection is the fact that the (direct) motions of the hyperbolic plane are Möbius transformations. Let us outline how this comes about.

If L_1 and L_2 are two h-lines, then the composition

$$\mathcal{M} \equiv \mathfrak{R}_{L_2} \circ \mathfrak{R}_{L_1}$$

of h-reflection in these lines will be a direct motion of the hyperbolic plane. Since every h-reflection is represented in the map by inversion in a circle, we immediately deduce that any direct motion of the form $\mathcal{M}$ is represented by a (non-loxodromic) Möbius transformation $M(z)$. Furthermore, later we will show that *every* direct motion is of the form $\mathcal{M}$; indeed, we will even give an explicit geometric construction for decomposing an arbitrary direct motion into two h-reflections. Supposing this already done, we see that *every direct motion is represented as a (non-loxodromic) Möbius transformation*.

Conversely, suppose that $M(z)$ is an *arbitrary* Möbius transformation that maps the upper half-plane to itself. Then it follows that $M(z)$ must map the real axis (the horizon) into itself. But a loxodromic Möbius transformation cannot possess such an invariant line: its strangely shaped invariant curves were illustrated in [3.32] on p. 188. Thus $M(z)$ is non-loxodromic, and from (3.49), p. 198, we deduce that $M(z)$ is the composition of inversion in two circles orthogonal to the real axis. Thus *the most general Möbius transformation of the upper half-plane to itself represents a direct hyperbolic motion of the type $\mathcal{M}$ above*.

One way to discover the algebraic form of these Möbius transformations is to use the formula (3.4), p. 141: inversion in a circle K centred at the point q on the real axis, and of radius R , is given by

$$J_K(z) = \frac{q\bar{z} + (R^2 - q^2)}{\bar{z} - q}.$$

Composing two such functions, we find [exercise] that a motion of type $\mathcal{M}$ corresponds to a Möbius transformation

$$M(z) = \frac{az + b}{cz + d}, \quad \text{where } a, b, c, d \text{ are real, and } (ad - bc) > 0. \quad (6.39)$$

Recall that in Ex. 25, p. 212, you showed that this is the form of the most general Möbius transformation of the upper half-plane to itself. Thus we have agreement with the conclusion of the previous paragraph.

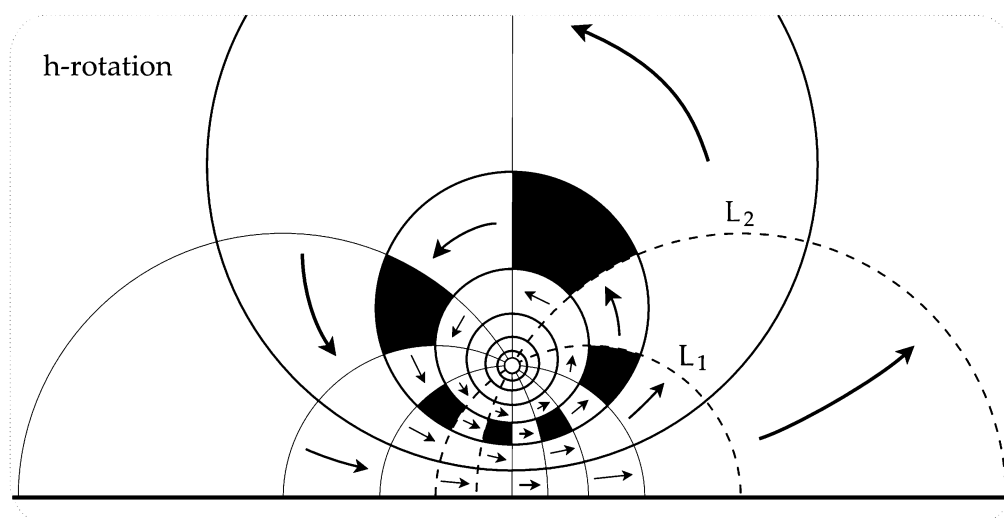
So much for the overview—now let's look in detail at the direct motions $\mathcal{M}$. We know from [6.24] or [6.25] that there are just three possible configurations for the h-lines L_1 and L_2 , and correspondingly $\mathcal{M} \equiv \mathfrak{R}_{L_2} \circ \mathfrak{R}_{L_1}$ is one of three fundamentally different types:

- (i) If the h-lines *intersect*, then $\mathcal{M}$ is called a *hyperbolic rotation*.
- (ii) If the h-lines are *asymptotic*, then $\mathcal{M}$ is a new kind of motion (peculiar to hyperbolic geometry) called a *limit rotation*.
- (iii) If the h-lines are *ultra-parallel*, then $\mathcal{M}$ is called a *hyperbolic translation*.

We can now reap the rewards of all our hard work in Chapter 3, for these three types of motion are just the three types of non-loxodromic Möbius transformation: (i) h-rotations are the “elliptic” ones; (ii) limit rotations are the “parabolic” ones; and (iii) h-translations are the “hyperbolic”²⁰ ones. At this point, you might find it helpful to reread the discussion of these Möbius transformations at the end of Chapter 3.

We already understand these Möbius transformations, so it only remains to look at them afresh, through hyperbolic spectacles. That is, imagine that you belong to the race of *Beltrami–Poincarites*—tiny, intelligent, two-dimensional beings who inhabit the hyperbolic plane. To you and your fellow Beltrami–Poincarites, h-lines *really are* straight lines, the real axis *really is* infinitely far away, etc. What will you see if the above motions are applied to your world?

Let us begin with h-rotations. Figure [6.27] illustrates the elliptic Möbius transformation—let’s call it $\mathcal{R}_a^\Phi$ —that arises in the case where the h-lines intersect at a , and the angle from L_1 to L_2 is $(\Phi/2)$. [We have chosen to illustrate $\Phi = (\pi/3)$.] Thus $\mathcal{R}_a^\Phi$ has fixed points a and $\bar{a}$, and the multiplier associated with a is $m = e^{i\Phi}$. As in



[6.27] A Hyperbolic Rotation, results from reflecting across intersecting h-lines, L_1 and L_2 .

²⁰ Try not to be confused by this unrelated use of the word “hyperbolic”.

Chapter 3, each shaded “rectangle” is mapped by $\mathcal{R}_a^\phi$ to the next one in the direction of the arrows—some of these regions have been filled with black to emphasize this.

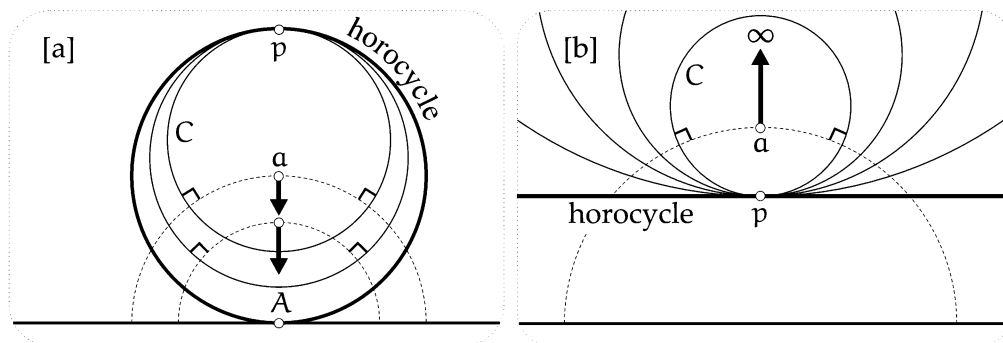
Consider how all this looks to you and your fellow Beltrami–Poincarites. For example, *you* see each black “rectangle” as being exactly the same shape and size as every other. To understand $\mathcal{R}_a^\phi$ better, we begin by noting that (in terms of the map) its effect on an infinitesimal neighbourhood of a is just a Euclidean rotation of ϕ about a . But since the map is conformal, this implies that a Poincarite standing at a will *also* see his immediate neighbourhood undergoing a rotation of ϕ .

More remarkably, however, the Poincarite at a will see the *entire* hyperbolic plane undergoing a perfect rotation of ϕ . Every h -line segment ap he constructs emanating from a is transformed by $z \mapsto \tilde{z} = \mathcal{R}_a^\phi(z)$ into another h -line segment $a\tilde{p}$ of equal length, making angle ϕ with the original. If the Poincarite gradually increases ϕ from 0 to 2π , then he sees $\tilde{p}$ tracing out an *h -circle* centred at a , while in the map we see $\tilde{p}$ miraculously tracing out a *Euclidean circle*! Thus the illustrated Euclidean circles orthogonal to the h -lines through a are all genuine hyperbolic circles, and a is their common h -centre. Let us record this remarkable result, adding a detail that is not too hard to prove [exercise]:

Every h -circle is represented in the map by a Euclidean circle, and its h -centre is the intersection of any two h -lines orthogonal to it. Algebraically, the h -circle with h -centre $a = (x + iy)$ and h -radius ρ is represented by the Euclidean circle with centre $(x + iy \cosh \rho)$ and radius $y \sinh \rho$.

As a stepping stone to the limit rotations, [6.28] introduces a new type of curve in the hyperbolic plane. On a line L in Euclidean geometry, let p be a fixed point, let a be a moveable point, and let C be the circle centred at a that passes through p . If we let a recede to infinity along L , then the limiting form of C is a *line* (through p and perpendicular to L). Figure [6.28a] shows that it’s a different story in the hyperbolic plane. As a recedes towards the infinitely remote point A on the real axis, the limiting form of C is a (Euclidean) circle that touches the real axis at A . This is neither an ordinary h -circle, nor an h -line: it is a new type of curve called a *horocycle*. Figure [6.28b] shows that horizontal (Euclidean) lines are also horocycles. Note that if K is any circle centred at A then the h -reflection $\mathcal{R}_K = \mathcal{J}_K$ transforms [6.28a] into [6.28b]. Thus the Beltrami–Poincarites cannot distinguish between these two types of horocycle.

Now consider [6.29], which illustrates the parabolic Möbius transformation that results from h -reflection in h -lines L_1 and L_2 that are asymptotic at A . Referring to [6.27] and [6.28], you can now understand why this is called a limit rotation: it may be viewed as the limit of the h -rotation $\mathcal{R}_a^\phi$ as a tends to the point A on the horizon. Note some of the interesting features of this picture: the invariant curves are horocycles touching at A ; each such horocycle is orthogonal to every h -line that



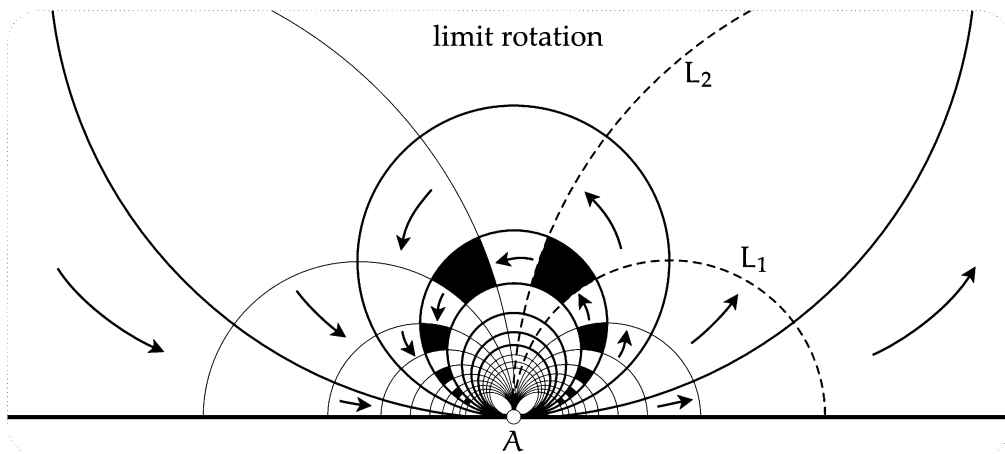
[6.28] Horocycles. [a] Remarkably, an h -circle C appears in the map as an ordinary Euclidean circle, but its centre a is the intersection of any two h -lines that meet C orthogonally. Keeping p fixed, let us keep increasing the h -radius of C indefinitely. The centre a gradually moves toward the point A on the horizon, and the circle ultimately becomes the illustrated horocycle. [b] If we instead let the centre a move upward indefinitely, the limiting horocycle takes the form of a horizontal line. If K is any circle centred at A then the h -reflection $\mathfrak{R}_K = \mathcal{I}_K$ transforms the first kind of horocycle in [a] into this new one, so inhabitants of the hyperbolic plane cannot tell them apart!

ends at A ; and any two such horocycles cut off the same h -length on every h -line that ends at A .

In terms of the map, the simplest limit rotation occurs when the asymptotic h -lines L_1 and L_2 are represented as vertical Euclidean half-lines, say separated by Euclidean distance $(\alpha/2)$. In this case, $\mathcal{M} = (\mathfrak{R}_{L_2} \circ \mathfrak{R}_{L_1})$ is represented in the map by the composition of two Euclidean reflections in parallel lines. Thus $\mathcal{M}$ is just a Euclidean translation $z \mapsto (z + \alpha)$ of the upper half-plane, and the invariant curves are horizontal lines, which are again horocycles, but now of the form shown in [6.28b]. Note that this Euclidean translation is *not* an h -translation. This is particularly clear if we visualize the effect of $\mathcal{M}$ on the pseudosphere, where it becomes a rotation through angle α about the pseudosphere's axis.

Figure [6.30] illustrates the third and final type of motion, the h -translation (hyperbolic Möbius transformation) resulting from h -reflection in two ultra-parallel h -lines. First note that there is precisely one h -line L that is orthogonal to both L_1 and L_2 . Unlike a Euclidean translation, this h -line L is the *only* h -line that is mapped into itself; it is called the *axis* of the h -translation. Despite this difference, the name “ h -translation” is appropriate, for every point on the h -line L is moved the same h -distance (say δ) along L . If we assume that the axis L has a direction assigned to it, then we may unambiguously denote this h -translation by $\mathcal{T}_L^\delta$.

In Euclidean geometry, the invariant curves of a translation are the parallel lines in the direction of the translation. However, [6.30] shows that the invariant curves of $\mathcal{T}_L^\delta$ are not h -lines, but rather arcs of Euclidean circles connecting the ends e_1 and



[6.29] A Hyperbolic Limit Rotation results from reflecting across asymptotic h -lines, L_1 and L_2 . This transformation has no analogue in Euclidean geometry, but it can be thought of as the limit of the h -rotation $\mathcal{R}_A^\Phi$ as α tends to the point A on the horizon. Note that the invariant curves are horocycles.

e_2 of L . These are called the *equidistant* curves of L , because every point on such a curve is the same h -distance from the h -line L . Make sure you can see this.

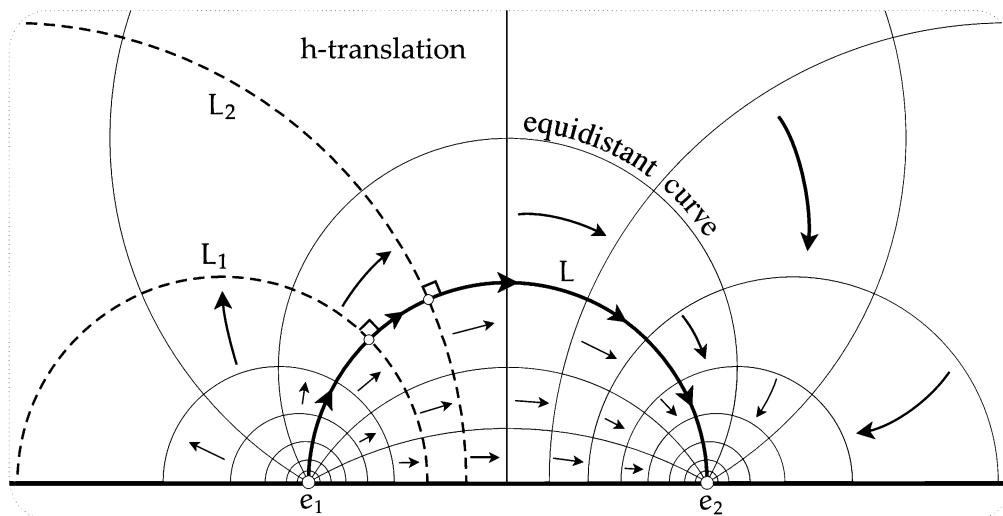
In terms of the map, the simplest h -translation occurs when the ultra-parallel h -lines L_1 and L_2 are represented by concentric Euclidean semicircles, say centred at the origin for convenience. In this case, the two h -reflections (i.e., inversions) yield a central dilation $z \mapsto kz$, where k is the real expansion factor. The axis of this h -translation is the vertical line through the origin (the y -axis), and the equidistant curves are all other (Euclidean) lines through the origin (cf. [6.20] and [6.21]). Note that this Euclidean expansion is a similarity transformation of the map, but it is *not* a similarity transformation of the hyperbolic plane—there are none!

Having completed our survey of these three types of direct motion, it's important to note that they not only look very different in terms of their effect on the map, but they also have unique fingerprints in terms of the intrinsic hyperbolic geometry. To put this another way, *Beltrami–Poincarites can tell these motions apart*. For example, of the three, only h -rotations have invariant h -circles, and only h -translations have an invariant h -line.

6.3.8 Decomposing an Arbitrary Direct Motion into Two Reflections

We will now show that the h -rotations, limit rotations, and h -translations are the *only* direct motions of the hyperbolic plane. That is, an *arbitrary* direct motion $\mathcal{M}$ can always be decomposed into two h -reflections: $\mathcal{M} \equiv (\mathcal{R}_{L_2} \circ \mathcal{R}_{L_1})$.

The first step is a familiar lemma: *an arbitrary hyperbolic motion $\mathcal{M}$ (not necessarily direct) is uniquely determined by its effect on any three non-collinear points*. As in



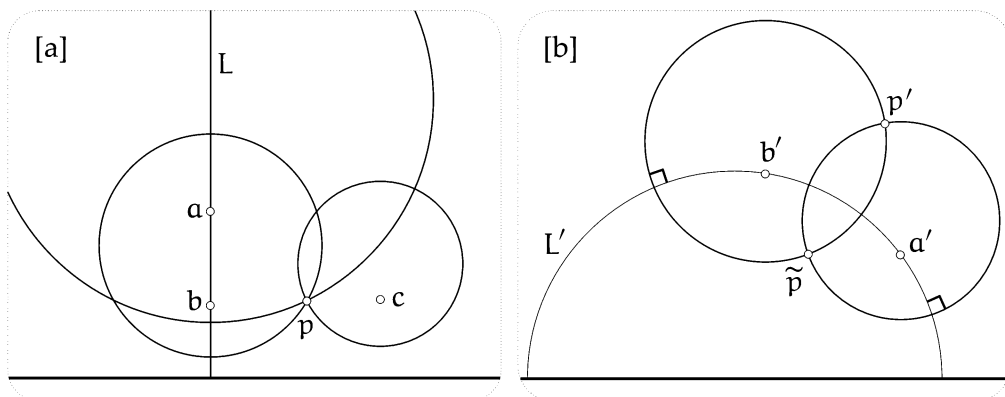
[6.30] A Hyperbolic Translation results from reflecting across non-intersecting (ultra-parallel) h -lines, L_1 and L_2 . Unlike Euclidean geometry, there is only one h -line L orthogonal to L_1 and L_2 , and it is the only invariant h -line. The other invariant curves are arcs of Euclidean circles connecting the ends e_1 and e_2 , and they are not h -lines. They are called equidistant curves, because every point on such a curve is the same h -distance from the h -line L .

In Euclidean geometry, this will be established if we can show that the location of a point p is uniquely determined by its h -distances from any three non-collinear points a, b, c . Consider [6.31a], in which we have supposed (for simplicity's sake only) that the h -line L through a and b is represented by a vertical line in the map. Through the point p , draw h -circles centred at a, b , and c . Since c does not lie on L (by assumption), we see that p is the *only* point at which the three circles intersect. Done.

Now suppose that an arbitrary motion carries two points a and b to the points a' and b' in [6.31b]. By the above result, the motion will be determined once we know the image of any third point p not on the line L through a and b . Drawing the illustrated h -circles with h -centres a' and b' and with h -radii $\mathcal{H}\{a, p\}$ and $\mathcal{H}\{b, p\}$, we see that the two intersection points p' and $\tilde{p}$ are the only possible images for p . Furthermore, since the h -line L' through a' and b' is necessarily orthogonal to the h -circles centred at those points, we also see that p' and $\tilde{p}$ are symmetric with respect to L' , i.e., $\tilde{p} = \mathcal{I}_{L'}(p') = \mathfrak{R}_{L'}(p')$. Thus we have shown that

(6.40)

There is exactly one direct motion $\mathcal{M}$ (and exactly one opposite motion $\tilde{\mathcal{M}}$) that maps a given h -line segment ab to another h -line segment $a'b'$ of equal h -length. Furthermore, $\tilde{\mathcal{M}} = (\mathfrak{R}_{L'} \circ \mathcal{M})$, where L' is the h -line through a' and b' .



[6.31] An arbitrary hyperbolic motion $\mathcal{M}$ is uniquely determined by its effect on any three non-collinear points. [a] The location of a point p is uniquely determined by its h -distances from any three non-collinear points a, b, c . [b] Let $\mathcal{M}$ be an arbitrary motion, and let $a' = \mathcal{M}(a)$ and $b' = \mathcal{M}(b)$. Then $\mathcal{M}$ will be determined once we know the image of any third point p not on the line L through a and b . Drawing the illustrated h -circles with h -centres a' and b' and with h -radii $\mathcal{H}\{a', p\}$ and $\mathcal{H}\{b', p\}$, we see that the two intersection points p' and $\tilde{p}$ are the only possible images for p , and that these two possibilities are simply reflections of each other in L' . This completes the proof of (6.40).

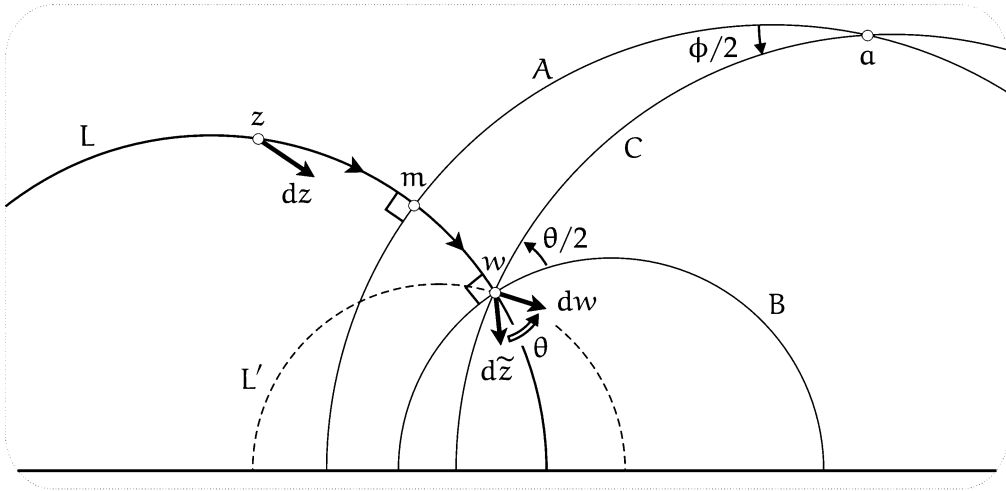
We shall now give an explicit geometric construction for decomposing an arbitrary direct motion $\mathcal{M}$ into two h -reflections. First note that (6.40) implies that $\mathcal{M}$ is determined by its effect on any two points, *no matter how close together they are*. Though it is not essential, the following construction is particularly clear if we choose the points to be infinitesimally separated.

Let us therefore take the two given points to be z and $(z + dz)$, and their given images under $\mathcal{M}$ to be $w = \mathcal{M}(z)$ and $(w + dw) = \mathcal{M}(z + dz)$. Figure [6.32] illustrates this idea. Our task is to find two h -reflections that will simultaneously carry z to w , and dz to dw . [Incidentally, since $\mathcal{M}$ must be conformal, it can be thought of as an analytic function, so we may write $dw = \mathcal{M}'(z) dz$.]

First, carry z to w using the h -translation $\mathcal{T}_L^\delta$, where $\delta = \mathcal{H}\{z, w\}$, and L is the unique h -line from z to w . Note that since $\mathcal{T}_L^\delta$ is conformal, it carries dz to an infinitesimal vector $d\tilde{z}$ (of equal h -length) making the same angle with L as dz . Next, apply the h -rotation $\mathcal{R}_w^\theta$, where θ is the angle from $d\tilde{z}$ to dw . This leaves w where it is, and it rotates $d\tilde{z}$ to dw . Since the net transformation carries z to w , and dz to dw , it must be $\mathcal{M}$:

$$\mathcal{M} = \mathcal{R}_w^\theta \circ \mathcal{T}_L^\delta.$$

Implicitly, this formula decomposes $\mathcal{M}$ into four h -reflections, because $\mathcal{T}_L^\delta$ and $\mathcal{R}_w^\theta$ can both be decomposed into two h -reflections. However, [6.32] illustrates that we can always arrange for two of the four h -reflections to cancel. Defining m to



[6.32] Decomposing an arbitrary direct motion $\mathcal{M}$ into two h -reflections. *Imagine two points close together, z and $z + dz$, being mapped to their images w and $w + dw$. First carry z to w with the unique h -translation $\mathcal{T}_L^\delta$ (where $\delta = \mathcal{H}\{z, w\}$) along the h -line from z to w . This carries dz conformally to $d\tilde{z}$, still making the same angle with L . Now rotate $d\tilde{z}$ about w by θ to obtain dw . Thus, $\mathcal{M} = \mathcal{R}_w^\theta \circ \mathcal{T}_L^\delta$. This implicitly decomposes $\mathcal{M}$ into four reflections, but we can always arrange for the middle two to cancel. Here, $\mathcal{M} = (\mathcal{R}_C \circ \mathcal{R}_B) \circ (\mathcal{R}_B \circ \mathcal{R}_A) = \mathcal{R}_C \circ \mathcal{R}_A = \mathcal{R}_a^\phi$, but clearly this construction may just as easily yield an A and a C that are asymptotic or ultra-parallel, in which case $\mathcal{M}$ is a limit rotation or an h -translation.*

be the h -midpoint of the h -line segment zw , draw h -lines A and B orthogonal to L and passing through m and w , respectively. Then $\mathcal{T}_L^\delta = (\mathcal{R}_B \circ \mathcal{R}_A)$. If we now draw an h -line C through w making angle $(\theta/2)$ with B , then $\mathcal{R}_w^\theta = (\mathcal{R}_C \circ \mathcal{R}_B)$. Thus, as we set out to show, every direct motion can be decomposed into two h -reflections:

$$\mathcal{M} = (\mathcal{R}_C \circ \mathcal{R}_B) \circ (\mathcal{R}_B \circ \mathcal{R}_A) = \mathcal{R}_C \circ \mathcal{R}_A.$$

In the illustrated example, it so happens that the h -lines A and C intersect, and so the motion is an h -rotation: $\mathcal{M} = \mathcal{R}_a^\phi$, where a is the intersection of A and C , and $(\phi/2)$ is the angle between them. However, it is clear that this construction may just as easily yield an A and a C that are asymptotic or ultra-parallel, in which case $\mathcal{M}$ is a limit rotation or an h -translation.

Summarizing what we have shown, and recalling (6.39),

Every direct motion of the hyperbolic plane is the composition of two h -reflections, and is thus an h -rotation, a limit rotation, or an h -translation. In the Beltrami–Poincaré upper half-plane, all such motions are presented by Möbius transformations of the form

$$M(z) = \frac{az + b}{cz + d}, \quad \text{where } a, b, c, d \text{ are real, and } (ad - bc) > 0.$$

Finally, returning to [6.32] and appealing to (6.40), the unique opposite motion $\tilde{M}$ carrying z to w and dz to dw is given by three h-reflections:

$$\tilde{M} = \mathfrak{R}_{L'} \circ \mathfrak{R}_C \circ \mathfrak{R}_A.$$

Here L' is the illustrated h-line passing through w and $(w + dw)$, i.e., passing through w in the direction dw . This decomposition does not, however, yield the simplest geometric interpretation of $\tilde{M}$; for that, and for the formula describing the general opposite motion, see Ex. 24.

6.3.9 The Angular Excess of a Hyperbolic Triangle

Joining three points in the hyperbolic plane with h-line segments yields (by definition) a *hyperbolic triangle*. Our objective will be to show that the angular excess $\mathcal{E}(T)$ of such a hyperbolic triangle T is given by

$$\mathcal{E}(T) = (-1)\mathcal{A}(T). \quad (6.41)$$

As we pointed out in the Introduction, this says (amongst other things) that the angles of Δ always add up to less than π , and that no matter how large we make Δ , its area can never exceed π . Referring to the differential geometry result (6.6), we also see that in establishing this formula we will have provided an intrinsic²¹ proof of the fact that the hyperbolic plane is a surface of constant negative curvature $\mathcal{K} = -1$.

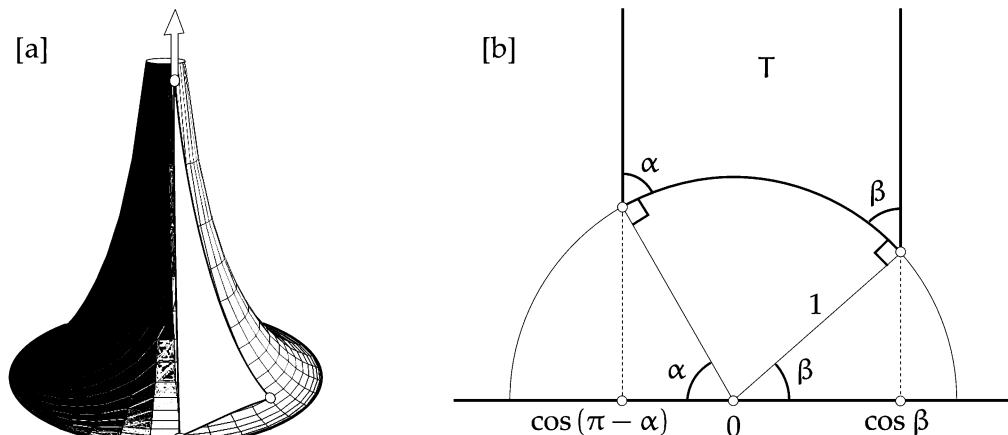
We have already remarked that Christiaan Huygens investigated the pseudosphere as early as 1693, and to get acquainted with hyperbolic area we will now confirm one of his surprising results: *the pseudosphere has finite area*. In the upper half-plane the pseudosphere is represented by the shaded region $\{0 \leq x < 2\pi, y \geq 1\}$ shown in [6.20], and (6.32) implies that this region of infinite Euclidean area does indeed have finite hyperbolic area:

$$\mathcal{A}(\text{pseudosphere}) = \iint dA = \int_{x=0}^{2\pi} \int_{y=1}^{\infty} \frac{dx \, dy}{y^2} = \int_{x=0}^{2\pi} dx \int_{y=1}^{\infty} \frac{dy}{y^2} = 2\pi,$$

as Huygens discovered.

Figure [6.33a] illustrates a triangle on the pseudosphere. If the uppermost vertex moves up the pseudosphere indefinitely, then the angle at that vertex tends to zero, and the edges meeting at that vertex tend to asymptotic lines, namely, tractrix generators meeting at infinity. Such a limiting triangle, two of whose edges are asymptotic, is called an *asymptotic triangle*. In order to establish (6.41) for ordinary triangles, we first establish it for asymptotic triangles. Figure [6.33b] illustrates

²¹ Recall that earlier we used the pseudosphere to give an *extrinsic* proof.



[6.33] [a] An asymptotic triangle results when the top vertex moves up the pseudosphere indefinitely. [b] The angular excess of an asymptotic triangle. Suppose the finite edge of the asymptotic triangle T is an arc of the unit circle. A simple calculation [see text] then shows that $\mathcal{A}(T) = \pi - \alpha - \beta$. Taking the third angle of T to be zero, this indeed accords with $\mathcal{E}(T) = (-1)\mathcal{A}(T)$.

such a triangle T in the upper half-plane, the asymptotic tractrix generators becoming vertical half-lines. By Huygens' result, T clearly has a finite area $\mathcal{A}(T)$, and because the asymptotic edges meet at angle zero, the result we wish to establish is $\mathcal{A}(T) = (\pi - \alpha - \beta)$.

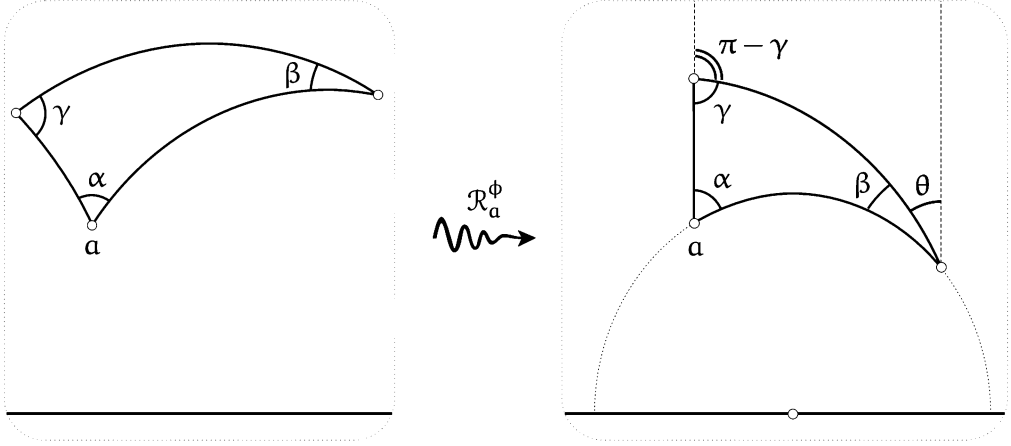
To simplify the derivation of this result, [6.33b] supposes that the finite edge of T is an arc of the unit circle. This does not involve any loss of generality, because an arc of a circle of radius r centred at $x = X$ may be transformed into an arc of the unit circle by applying the limit rotation $z \mapsto (z - X)$, followed by the h -translation $z \mapsto (z/r)$. From [6.33b] we now deduce that

$$\mathcal{A}(T) = \int_{x=\cos(\pi-\alpha)}^{\cos \beta} \left[\int_{y=\sqrt{1-x^2}}^{\infty} \frac{dy}{y^2} \right] dx = \int_{x=\cos(\pi-\alpha)}^{\cos \beta} \frac{dx}{\sqrt{1-x^2}},$$

and writing $x = \cos \theta$ then yields the desired result:

$$\mathcal{A}(T) = \int_{\pi-\alpha}^{\beta} \frac{-\sin \theta d\theta}{\sin \theta} = \pi - \alpha - \beta.$$

On the left of [6.34] is a general triangle, say of area $\mathcal{A}$. By applying a suitable h -rotation about one of the vertices, we can bring one of the edges into a vertical position, as illustrated on the right of [6.34]. This makes it clear that the area $\mathcal{A}$ of the triangle may be viewed as the difference of the areas of two asymptotic triangles: one with angles α and $(\beta + \theta)$; the other with angles $(\pi - \gamma)$ and θ . Finally, applying the above result for asymptotic triangles, we deduce (6.41):



[6.34] The angular excess of a general hyperbolic triangle. [a] A general hyperbolic triangle of area A can be rotated so that one edge becomes vertical. [b] But now A is clearly the difference of the areas of two asymptotic triangles: one with angles α and $(\beta + \theta)$; the other with angles $(\pi - \gamma)$ and θ . It follows immediately that $\mathcal{E}(T) = (-1)A(T)$, thereby confirming that the hyperbolic plane has constant negative Gaussian curvature $\mathcal{K} = -1$.

$$\begin{aligned} A &= [\pi - \alpha - (\beta + \theta)] - [\pi - (\pi - \gamma) - \theta] \\ &= \pi - \alpha - \beta - \gamma \\ &= -\mathcal{E}. \end{aligned}$$

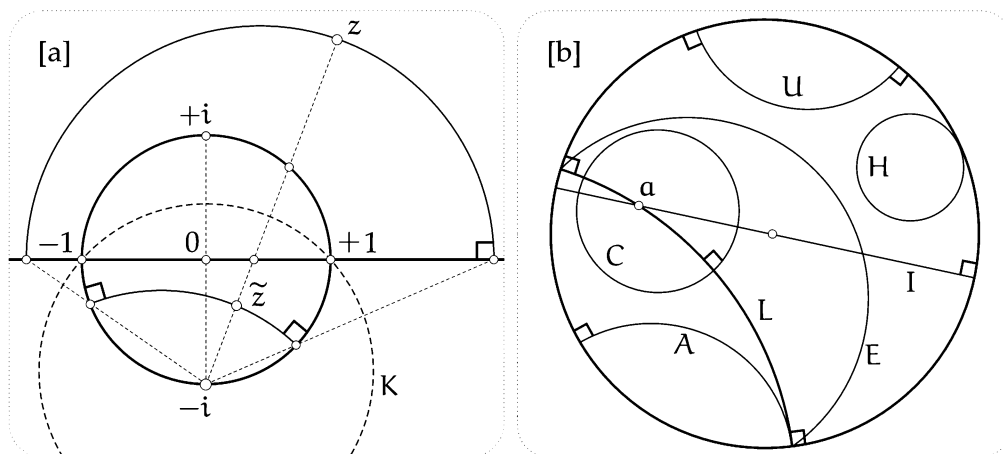
6.3.10 The Beltrami–Poincaré Disc

In addition to the upper half-plane model, [Beltrami \(1868b\)](#) constructed another extremely useful conformal map of the hyperbolic plane, this time inside the unit disc. Fourteen years later Poincaré rediscovered this map, which is now universally (and unfairly) known as the *Poincaré disc*. Following the example we set ourselves in VDGF, let us now attempt to set the record straight by given both men *equal* credit, calling it the *Beltrami–Poincaré disc*.

Figure [6.35a] illustrates the first step of the construction, which is to map the entire upper half-plane into the unit disc by means of the inversion

$$z \mapsto \tilde{z} = \mathcal{I}_K(z),$$

where K is the illustrated circle centred at $-i$ and passing through ± 1 . In order for this disc to represent the hyperbolic plane, its metric must be inherited from the upper half-plane. That is, we must define the h -separation $\mathcal{H}\{\tilde{a}, \tilde{b}\}$ of two points in the disc to be the h -separation $\mathcal{H}\{a, b\}$ of their preimages in the upper half-plane. Note that this implies [exercise] that the h -lines of the disc are precisely the images of h -lines in the upper half-plane.



[6.35] Mapping the Beltrami-Poincaré Half-Plane to the Beltrami-Poincaré Disc.

[a] The anticonformal inversion $z \mapsto \tilde{z} = \mathcal{I}_K(z)$ maps the upper half-plane to the interior of the unit disc. To retain the hyperbolic geometry, we define the h -separation $\mathcal{H}\{\tilde{a}, \tilde{b}\}$ of two points in the disc to be the h -separation $\mathcal{H}\{a, b\}$ of their preimages in the upper half-plane. H -lines now become arcs of circles meeting the unit circle at right angles, and the horizon is now the unit circle itself, also known as the circle at infinity. **[b]** Composing this inversion with conjugation, we recover a conformal mapping to the unit disc, which we call the Beltrami-Poincaré disc model of the hyperbolic plane, in which h -lines now take the form of arcs of circles orthogonal to the horizon, such as L , A , U .

Before moving on, try staring at [6.35a] until the following details become clear: (i) ± 1 remain fixed and i is mapped to 0 ; (ii) the entire shaded part of the upper half-plane is mapped to the shaded bottom half of the unit disc; (iii) the remaining part of the upper half-plane (i.e., the top half of the unit disc) is mapped into itself; (iv) h -lines in the disc are the images of h -lines in the upper half-plane, and these are arcs of circles orthogonal to the unit circle; (v) the entire horizon of the hyperbolic plane is represented by the unit circle, with the common point at infinity of vertical h -lines in the upper half-plane being represented by $-i$.

At this point we have obtained a map of the hyperbolic plane within the unit disc. However, since $\mathcal{I}_K(z)$ is anticonformal, so is our map: angles in the upper half-plane are currently represented by equal but *opposite* angles in the disc. If we now apply $z \mapsto \bar{z}$, which reflects the disc across the real axis into itself, then angles are reversed a second time, and we obtain the conformal Beltrami-Poincaré disc.

The net transformation from the Beltrami-Poincaré upper half-plane to the Beltrami-Poincaré disc is thus the composition of $z \mapsto \mathcal{I}_K(z)$ and $z \mapsto \bar{z}$, and this is a Möbius transformation, say $D(z)$. Since $D(z)$ maps i to 0 and $-i$ to ∞ , it is clear that $D(z)$ must be proportional to $(z - i)/(z + i)$. Finally, recalling that a Möbius transformation is uniquely determined by its effect on three points, and noting that ± 1 remain fixed, we deduce [exercise] that

$$D(z) = \frac{iz + 1}{z + i}. \quad (6.42)$$

Alternatively, this may be derived by brute force [exercise] using the formula for inversion, (3.4), p. 141.

Since $D(z)$ preserves angles and circles, it is easy to transfer the basic types of curve in the hyperbolic plane from the Beltrami–Poincaré upper half-plane to the Beltrami–Poincaré disc. Figure [6.35b] illustrates that h-lines are represented by arcs of circles orthogonal to the unit circle (such as L , A , U), including diameters such as I . Incidentally, since the horizon is now represented by the unit circle, you can understand why the horizon is also called the *circle at infinity*.

The terminology for h-lines is the same as before: I intersects L , A is asymptotic to L , U is ultra-parallel to L , and a Euclidean circular arc E connecting the ends of L is an equidistant curve of L . It is also easy to see that a Euclidean circle C lying strictly inside the unit disc represents an h-circle, though its h-centre a does not generally coincide with its Euclidean centre. Finally, the horocycles in [6.28a] and [6.28b] are represented in the Beltrami–Poincaré disc by circles such as H that touch the unit circle.

Now let us find the metric in the Beltrami–Poincaré disc. Ex. 19 shows how this may be done by brute calculation, but the following geometric approach²² is much more enlightening and powerful. First, [6.36a] recalls the earlier observation (6.33): if ds is the infinitesimal Euclidean length of a horizontal line-element emanating from z , then the angle between L and E is its hyperbolic length $d\hat{s} = [ds/\text{Im}(z)]$.

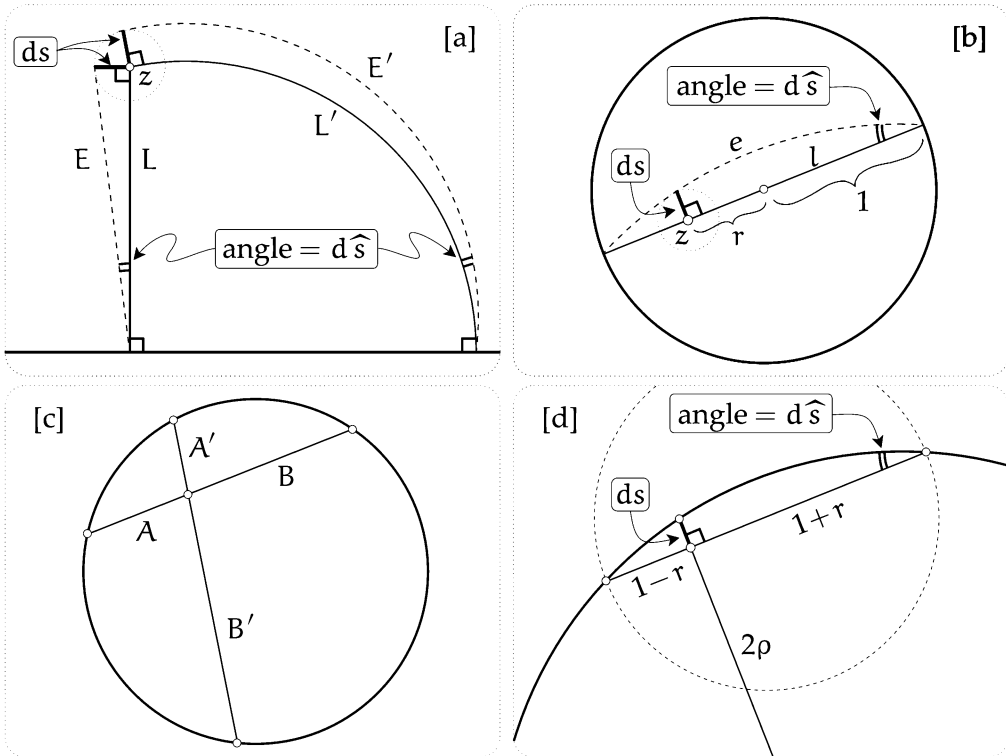
Note that in purely hyperbolic terms, L is an h-line orthogonal to ds , and E is an equidistant curve of L . If we apply an h-rotation $\mathcal{R}_z^\phi$ then L is carried into another h-line L' , and E is carried into an equidistant curve E' of L' , and the angle between L' and E' is the same as before. Thus we have the following general construction:

Through one end of ds , draw the h-line l orthogonal to ds , and through the other end of ds draw the equidistant curve e . Then the h-length $d\hat{s}$ of ds is the angle of intersection (on the horizon) of l and e . (6.43)

Now the beauty of interpreting $d\hat{s}$ as an angle in this way is that the Möbius transformation D to the Beltrami–Poincaré disc is *conformal*, and so the above construction of $d\hat{s}$ is valid there too!

Figure [6.36b] illustrates an infinitesimal disc of Euclidean radius ds centred at $z = r e^{i\theta}$ in the Beltrami–Poincaré disc. Because the map is conformal, the h-length $d\hat{s}$ of ds is independent of the direction of ds , so we may simplify the construction (6.43) by choosing ds orthogonal to the diameter l through z . The equidistant curve e is then the illustrated arc of a Euclidean circle through the ends of l .

²² We merely rediscovered this angular interpretation of hyperbolic distance, which we believe originates with Thurston (1997). However, our explanation (and our applications) of the idea differ somewhat from Thurston's.



[6.36] Geometric derivation of the disc metric. [a] If ds is the infinitesimal Euclidean length of a horizontal line-element emanating from z , then the angle between the orthogonal h -line L and the equidistant curve E is its hyperbolic length $d\hat{s} = [ds / \text{Im}(z)]$. But performing a rotation about z we see that $d\hat{s}$ may equally be measured as the angle between the h -line L' and its equidistant curve E' . [b] Since the mapping from the half-plane to the disc is conformal, this angular interpretation of hyperbolic distance applies here, too! Choose ds orthogonal to the diameter l through z . The equidistant curve e is then the illustrated arc of a Euclidean circle of radius ρ through the ends of l , and $\rho d\hat{s} = 1$. [c] $AB = A'B'$. [d] So, $2\rho ds = (1-r)(1+r) = 1 - |z|^2$, and the hyperbolic disc metric is therefore $d\hat{s} = \frac{2}{1-|z|^2} ds$.

To turn this picture of $d\hat{s}$ into a formula, begin by noting that if ρ is the radius of the circle containing the arc e , then [draw a picture!]

$$\rho d\hat{s} = 1.$$

Next recall (or prove) the familiar property of circles illustrated in [6.36c], namely, that all chords passing through a fixed interior point are divided into two parts whose lengths have constant product: $AB = A'B'$. Applying this result to the copy of [6.36b] shown in [6.36d], we obtain [exercise]

$$2\rho ds = (1-r)(1+r) = 1 - |z|^2.$$

Thus the metric of the Beltrami–Poincaré disc is

$$d\hat{s} = \frac{2}{1-|z|^2} ds. \quad (6.44)$$

Note the remarkable similarity to the formula for the metric of the sphere under stereographic projection, (6.16)!

Since the Euclidean line-segment connecting 0 to z is also an h -line segment, we can now find the h -separation of these points by simple integration along the line-segment:

$$\mathcal{H}\{0, z\} = \int_0^{|z|} \frac{2 dr}{1-r^2} = \int_0^{|z|} \left[\frac{1}{1+r} + \frac{1}{1-r} \right] dr,$$

and so

$$\mathcal{H}\{0, z\} = \ln \left(\frac{1+|z|}{1-|z|} \right). \quad (6.45)$$

As a simple check of this formula, note that as z moves toward the unit circle (the horizon), $\mathcal{H}\{0, z\}$ tends to infinity, as it should.

6.3.11 Motions of the Beltrami–Poincaré Disc

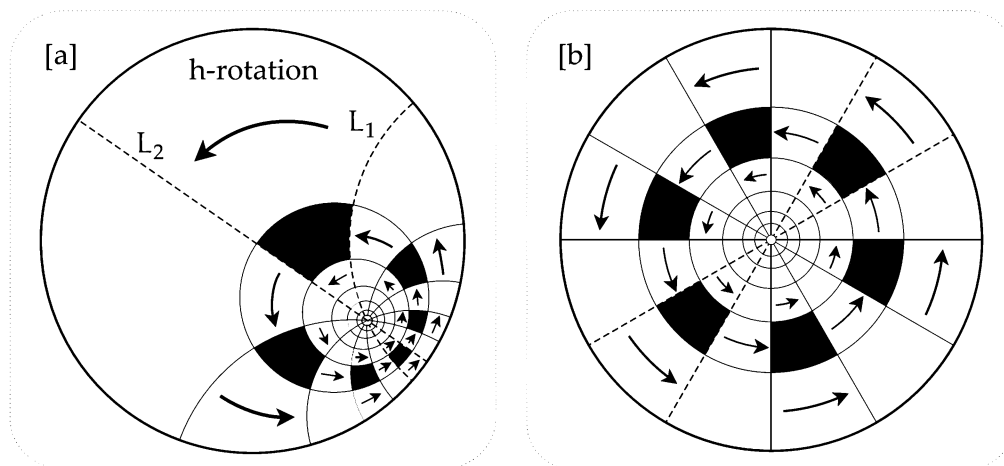
In the upper half-plane we found that every direct motion was the composition of two h -reflections, and every opposite motion was the composition of three h -reflections. Since the intrinsic geometry of the Beltrami–Poincaré disc is identical to the upper half-plane, this result must still be true, so it only remains to find out what h -reflection means in the disc. In the upper half-plane we saw that h -reflection in an h -line K meant geometric inversion in K , *and the same is true in the Beltrami–Poincaré disc!*

This is easy to understand. In the upper half-plane, q is the h -reflection of p in K means that p and q are symmetric (in the sense of inversion) in K . In order to make the Beltrami–Poincaré disc isometric to the upper half-plane, we insisted that the mapping $z \mapsto \tilde{z} = D(z)$ preserve hyperbolic distance. In particular, $\tilde{q}$ is the h -reflection of $\tilde{p}$ in $\tilde{K}$. But $D(z)$ is a *Möbius transformation*, and so the Symmetry Principle [see p. 168] implies that $\tilde{p}$ and $\tilde{q}$ are symmetric in $\tilde{K}$, as was to be shown.

Thus every direct motion $\mathcal{M}$ of the Beltrami–Poincaré disc has the form

$$\mathcal{M} = \mathfrak{R}_{L_2} \circ \mathfrak{R}_{L_1} = \mathcal{I}_{L_2} \circ \mathcal{I}_{L_1},$$

where L_1 and L_2 are h -lines, namely, arcs of circles orthogonal to the unit circle. As in the upper half-plane, every direct motion is therefore a non-loxodromic Möbius transformation. We already know that there are just three hyperbolically distinguishable types of direct motion, and the distinction between them in terms of L_1 and L_2 is the same as before: we get an h -rotation when they intersect, a limit rotation when they are asymptotic, and an h -translation when they are ultra-parallel.



[6.37] A Hyperbolic Rotation results from reflecting across intersecting h -lines, L_1 and L_2 . **[a]** The typical case. **[b]** If the rotation is about the centre, then it reduces to an ordinary Euclidean rotation.

We will discuss the formula for these Möbius transformations in a moment, but first let's draw pictures of them.

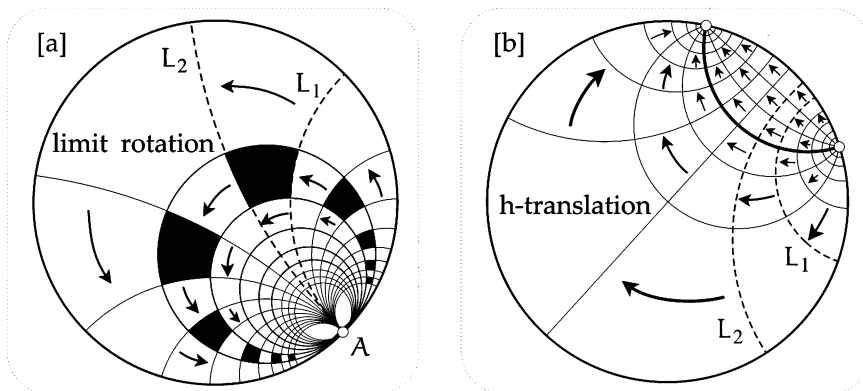
Figure [6.37a] shows a typical h -rotation; note the appearance of h -circles with a common h -centre. Figure [6.37b] illustrates the pleasant fact that if L_1 and L_2 intersect at the origin (in which case they are Euclidean diameters) then the resulting h -rotation manifests itself as a Euclidean rotation.

In this connection, we offer a word of warning. As Euclidean beings, we suffer from an almost overwhelming temptation to regard the centre of the Beltrami–Poincaré disc as being special in some way. One must therefore constantly remind oneself that to the Beltrami–Poincarites who inhabit the disc, every point is indistinguishable from every other point. In particular, the Beltrami–Poincarites do not see any difference between [6.37a] and [6.37b].

Figure [6.38a] illustrates a typical limit rotation generated by an L_1 and an L_2 that are asymptotic at a point A on the horizon. Once again note that the invariant curves are horocycles touching at A , and that these are orthogonal to the family of h -lines that are asymptotic at A .

Finally, [6.38b] illustrates a typical h -translation. Once again, note that there is precisely one invariant h -line [shown in bold], and that the invariant equidistant curves are arcs of circles through the ends of this axis.

From our work in the upper half-plane we know that the three types of motion pictured above are the *only* direct motions of the Beltrami–Poincaré disc, and we now turn to the formula that describes them. We know that every direct motion is a Möbius transformation that maps the unit disc into itself, and at the end of



[6.38] [a] A **Limit Rotation** results from reflecting across asymptotic h -lines, L_1 and L_2 . [b] A **Translation** results from reflecting across ultra-parallel h -lines, L_1 and L_2 .

Chapter 3, with malice aforethought, we investigated these “Möbius automorphisms” of the unit disc. We found [see (3.52), p. 201] that the formula representing the most general one $M_a^\phi(z)$ is

$$M_a^\phi(z) = e^{i\phi} M_a(z), \quad \text{where} \quad M_a(z) = \frac{z - a}{\bar{a}z - 1}.$$

Thus M_a^ϕ is the composition of M_a and a rotation of ϕ about the origin.

Recall that M_a swaps a and 0 : $M_a(a) = 0$ and $M_a(0) = a$. More generally, M_a swaps every pair of points $z, M_a(z)$: the transformation is *involutory*. This is explained by [6.39a], which recalls the result illustrated in [3.39], p. 203:

$$M_a = \mathcal{I}_B \circ \mathcal{I}_A,$$

where B is the diameter through a , and where A is the circle centred at $(1/\bar{a})$ that is orthogonal to the unit circle.

Hyperbolic geometry gives us a fresh perspective on this result: the intersection point m of A and B is the h -midpoint of 0 and a , and A itself is the perpendicular h -bisector of $0a$. Furthermore, the inversions in A and B are h -reflections. Thus M_a is the composition of two h -reflections in perpendicular h -lines through m , and so

The unique Möbius automorphism M_a that swaps a and 0 is the h -rotation $\mathcal{R}_m^\pi$ through angle π about the h -midpoint m of the h -line segment $0a$.

An immediate benefit of this insight is that we can now easily find the formula for the h -separation of any two points, a and z . The h -rotation M_a brings a to the origin, and we already know the formula (6.45) for the h -distance of a point from there:

$$\mathcal{H}\{a, z\} = \mathcal{H}\{M_a(a), M_a(z)\} = \mathcal{H}\{0, M_a(z)\} = \ln \left(\frac{1 + |M_a(z)|}{1 - |M_a(z)|} \right),$$

and so

$$\mathcal{H}\{a, z\} = \ln \left(\frac{|\bar{a}z - 1| + |z - a|}{|\bar{a}z - 1| - |z - a|} \right). \quad (6.46)$$

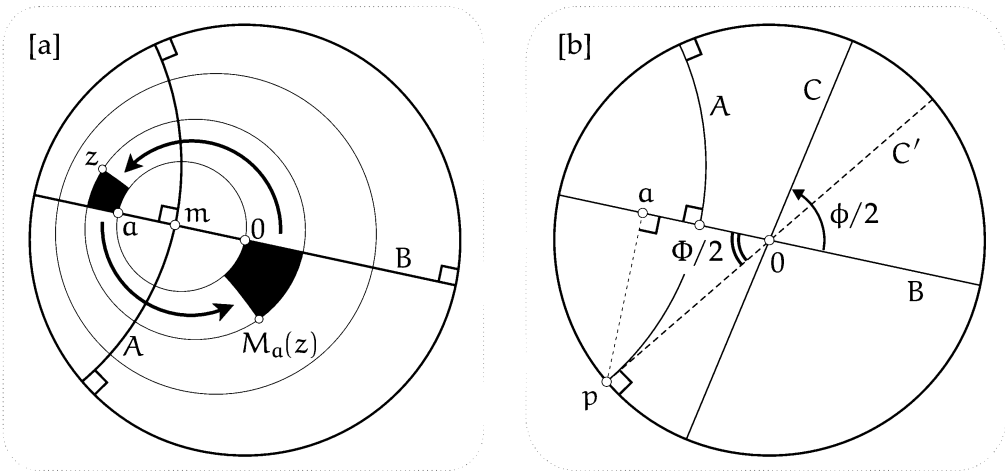
Now let us resume and complete our discussion of M_a^ϕ . As illustrated in [6.37b], the Euclidean rotation $z \mapsto e^{i\phi}z$ represents the h-rotation $\mathcal{R}_0^\phi$. Thus the most general Möbius automorphism of the disc may be interpreted as the composition of two h-rotations:

$$M_a^\phi = \mathcal{R}_0^\phi \circ \mathcal{R}_m^\pi.$$

Figure [6.39b] shows how to compose these h-rotations, using the same idea as was used in both Euclidean and spherical geometry. The h-rotation $\mathcal{R}_0^\phi$ is the composition of h-reflections in any two h-lines through 0 (diameters) containing angle $(\phi/2)$. Thus, choosing the first h-line to be B, and calling the second h-line C, we deduce that

$$M_a^\phi = (\mathcal{R}_C \circ \mathcal{R}_B) \circ (\mathcal{R}_B \circ \mathcal{R}_A) = \mathcal{R}_C \circ \mathcal{R}_A.$$

Thus M_a^ϕ is an h-rotation, limit rotation, or h-translation according as A and C are intersecting, asymptotic, or ultra-parallel.



[6.39] Geometric interpretation and classification of the general Möbius automorphism, M_a^ϕ . [a] The unique Möbius automorphism $M_a = \mathcal{I}_B \circ \mathcal{I}_A$ that swaps a and 0 is the h-rotation $\mathcal{R}_m^\pi$ through angle π about the h-midpoint m of the h-line segment $0a$. [b] The most general Möbius automorphism is $M_a^\phi = \mathcal{R}_0^\phi \circ \mathcal{R}_m^\pi = \mathcal{I}_C \circ \mathcal{I}_B \circ \mathcal{I}_B \circ \mathcal{I}_A = \mathcal{I}_C \circ \mathcal{I}_A$. Thus M_a^ϕ is an h-rotation, limit rotation, or h-translation according as A and C are intersecting, asymptotic, or ultra-parallel.

Thinking of a as fixed and ϕ as variable, the critical value $\phi = \Phi$ separating the h -rotations from the h -translations occurs when C is in the position C' [shown dashed] asymptotic to A at p . It is not hard to see [exercise] that the triangle pa_0 is right angled, and so it follows that $\cos(\Phi/2) = |a|$, or

$$\Phi = 2 \cos^{-1} |a|.$$

This explains the result (3.54), p. 204, which you proved algebraically in Ex. 27, p. 212.

To sum up,

The most general Möbius automorphism M_a^ϕ of the disc is a direct hyperbolic motion, and it is (i) an h -rotation if $\phi < \Phi$; (ii) a limit rotation if $\phi = \Phi$; and (iii) an h -translation if $\phi > \Phi$.

Finally, recall from Ex. 20, p. 210, that the set of Möbius transformations of the form M_a^ϕ is identical to the set of the form

$$M(z) = \frac{Az + B}{\overline{B}z + \overline{A}}, \quad \text{where } |A| > |B|.$$

Comparing this with (6.20), we see that there are striking similarities not only between the conformal metrics of the sphere and the hyperbolic plane, but also between the Möbius transformations that represent their direct motions.

6.3.12 The Hemisphere Model and Hyperbolic Space

Figure [6.40a] illustrates how we may obtain two new models of the hyperbolic plane. Following Beltrami (1868b), let us stereographically project the Beltrami–Poincaré disc from the south pole S of the Riemann sphere onto the northern hemisphere. Defining the h -separation of two points to be the h -separation of their preimages in the disc, we have a new *conformal* map of the hyperbolic plane, called the *hemisphere model*. The h -lines of this model are the images of h -lines in the disc, and since stereographic projection preserves circles as well as angles, we deduce [exercise] that *h -lines are (semi-circular) vertical sections of the hemisphere*. What do equidistant curves and horocycles look like?

The hemisphere was Beltrami's primary model of the hyperbolic plane, and it was by applying the above stereographic projection to this hemisphere that he discovered the Beltrami–Poincaré disc. In fact by projecting his hemisphere in different ways, Beltrami obtained (in a unified way) almost all the models in current use.

For example, by projecting the hemisphere vertically down onto the complex plane (see [6.40a]) he obtained a new model of the hyperbolic plane inside the unit disc. This is traditionally called the “Klein model”, but, again, we shall rename it the *Beltrami–Klein model*. Less controversially, it is also called the *projective model*. Since a small circle on the hemisphere is clearly projected to an ellipse in the disc, the

Beltrami–Klein model is *not* conformal. This is a serious disadvantage, but it is compensated for by the fact that the vertical sections of the hemisphere are projected to (Euclidean) straight lines: *h-lines in the Beltrami–Klein model are straight Euclidean chords of the unit circle*. Note the analogy with figure [6.12], in which geodesics on the sphere are represented by straight lines in the map; Ex. 14 reveals that this analogy is more than superficial.

Other properties of the Beltrami–Klein model will be explored in the exercises, but right now we have bigger fish to fry! Up to this point we have focused on developing the geometry of the hyperbolic *plane*, the negatively curved counterpart of the Euclidean plane. The geometry of this Euclidean plane may be thought of as being inherited from the geometry of three-dimensional Euclidean space. That is, if (X, Y, Z) are Cartesian coordinates in this space, then the Euclidean distance ds between two infinitesimally separated points is given by

$$ds = \sqrt{dX^2 + dY^2 + dZ^2},$$

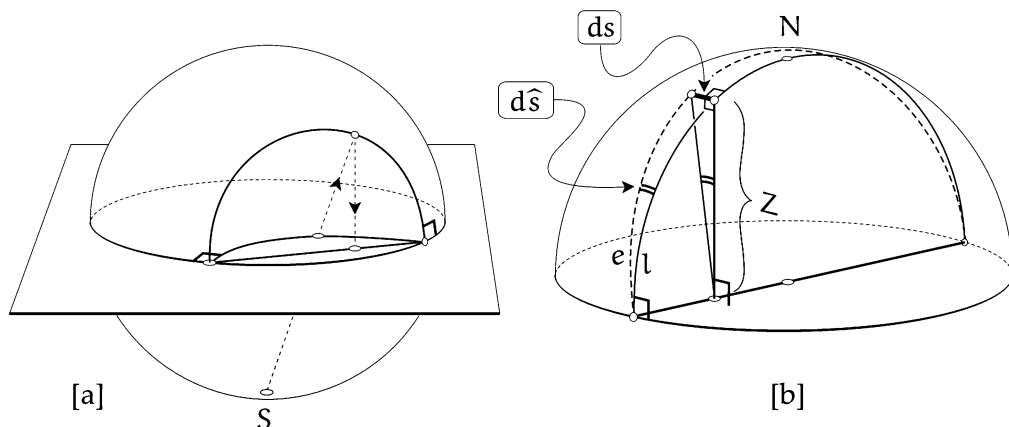
and two-dimensional Euclidean geometry is obtained by restricting this formula to the points of an ordinary plane.

The question therefore arises whether there might exist a negatively curved (whatever that might mean) counterpart of three-dimensional Euclidean space, such that the geometry induced on each “plane” within this space would automatically be the geometry of the hyperbolic plane. We shall now show that this three-dimensional *hyperbolic space* does indeed exist.

To do so, let us find the metric of the hemisphere model. Because the stereographic projection of the Beltrami–Poincaré disc onto the hemisphere is conformal, it follows that $d\hat{s}$ is once again given by the construction (6.43). Since $d\hat{s}$ is independent of the direction of ds on the hemisphere, we may once again simplify the construction by choosing ds in an auspicious direction. In the Beltrami–Poincaré disc the best choice of ds was orthogonal to the diameter through the point of interest, and the best choice on the hemisphere is simply the stereographic projection of this configuration.

Thus in [6.40b] we have chosen the h -line l to be the vertical section of the hemisphere passing through the north pole and the point from which ds emanates. Thus l and e are both halves of great circles: the plane of l is vertical, the plane of e is inclined at angle $d\hat{s}$ to the vertical, and the intersection of these planes is the illustrated diameter of the unit circle lying directly beneath l .

Now let the coordinates of the point at which we have drawn ds be (X, Y, Z) , where the X and Y axes coincide with the real and imaginary axes of $\mathbb{C}$, so that Z measures the height of the point above $\mathbb{C}$. Since ds is orthogonal to l , and since the vertical plane of l is orthogonal to the hemisphere, we see that ds is *horizontal*. Thus the angle that ds subtends at the point $(X, Y, 0)$ directly beneath it is (ds/Z) . But this angle is just the angle between the planes of l and e ! Thus the metric of the hemisphere model is



[6.40] [a] The Beltrami Hemisphere Model is obtained by stereographically projecting the Beltrami–Poincaré disc model from the south pole, S , onto the northern hemisphere. Since stereographic projection preserves circles and angles, h -lines are semi-circular vertical sections of the hemisphere. Projecting vertically downwards yields the non-conformal Beltrami–Klein model (aka projective model) of the hyperbolic plane, in which h -lines now appear as straight Euclidean chords of the unit circle. **[b]** Conformality implies that the angular interpretation of hyperbolic distance given in [6.36] applies here, too. Therefore, the hyperbolic metric on the hemisphere is $d\hat{s} = (ds/Z)$.

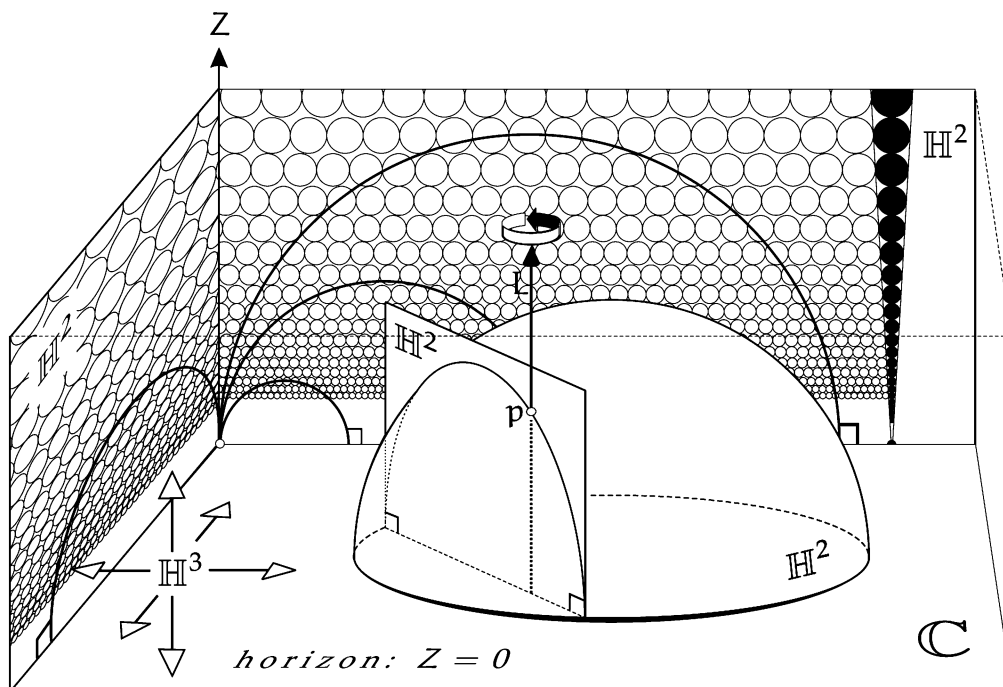
$$d\hat{s} = \frac{ds}{Z}. \quad (6.47)$$

This formula only describes the h -separation of points on the hemisphere, but there is nothing preventing us from using it to *define* the h -separation of *any* two infinitesimally separated points in the three-dimensional region $Z > 0$. This region lying above $\mathbb{C}$, with h -distance defined by (6.47), is called the *half-space model* of three-dimensional *hyperbolic space*, denoted $\mathbb{H}^3$. Without going into detail, it is clear from (6.47) that the points of $\mathbb{C}$ are infinitely h -distant from points that lie strictly above $\mathbb{C}$. Thus $\mathbb{C}$ represents the two-dimensional *horizon* or *sphere at infinity* of hyperbolic space.

At the moment it is a mere tautology that the geometry induced on the hemisphere by (6.47) is that of a hyperbolic plane. To begin to see that there is real meat on this idea, let us consider some simple motions of hyperbolic space. Clearly $d\hat{s}$ is unaltered by a *translation parallel to $\mathbb{C}$* , so this is a motion. It is also clear that $d\hat{s}$ is unaltered by a *dilation* $(X, Y, Z) \mapsto (kX, kY, kZ)$ centred at the origin. More generally, a *dilation centred at any point of $\mathbb{C}$* will preserve $d\hat{s}$, so this too is a motion.

By applying these two types of motion to the origin-centred unit hemisphere that we have been studying, we see that

$$\text{In the half-space model, } \mathbb{H}^3, \text{ every hemisphere orthogonal to } \mathbb{C} \text{ is a hyperbolic plane, } \mathbb{H}^2. \quad (6.48)$$

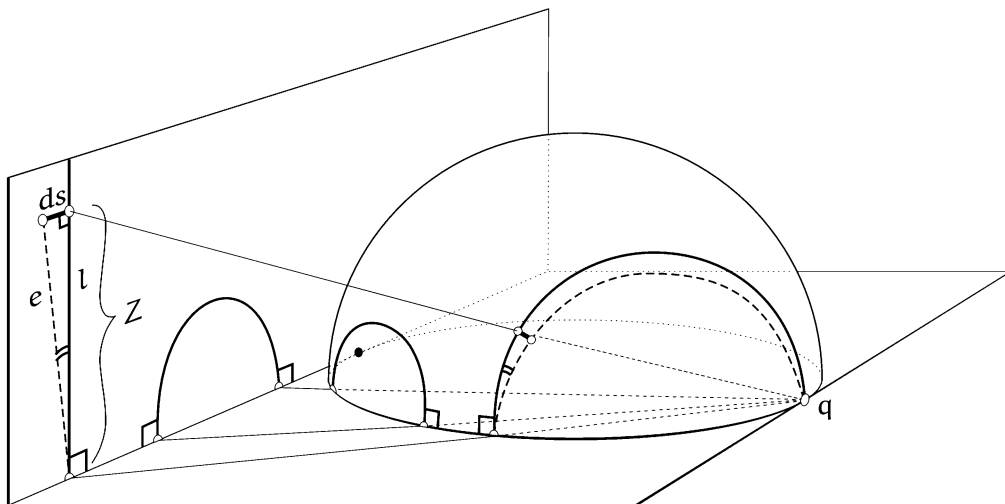


[6.41] Hyperbolic 3-space, $\mathbb{H}^3$. Hyperbolic planes appear as vertical half-planes and as hemispheres orthogonal to $\mathbb{C}$. The intersection of two such $\mathbb{H}^2$ planes is a hyperbolic line: a vertical half-line, or a semicircle orthogonal to $\mathbb{C}$.

In Euclidean geometry the intersection of two planes is a line, and this suggests that an h-line should be the intersection of two hyperbolic planes. Thus we anticipate that *every semicircle orthogonal to $\mathbb{C}$ is an h-line*, for every such semicircle is the intersection of two hemispheres orthogonal to $\mathbb{C}$. Note that this agrees with what we already know: the h-lines of the hemisphere model *are* semicircles orthogonal to $\mathbb{C}$. Figure [6.41] attempts to make this all much more vivid,²³ showing the vertical $\mathbb{H}^2$ walls filled with circles that are all the same hyperbolic size; compare this to [6.20], page 337.

Let us return to two-dimensional geometry for a moment. Figure [6.42] illustrates how Beltrami obtained the upper half-plane model from his hemisphere model. From a point q on the rim of the hemisphere, we stereographically project onto the tangent plane at the point antipodal to q —actually, any plane tangent to this one would do equally well. Since this preserves circles and angles, we see that a typical h-line of the hemisphere is mapped to a semicircle orthogonal to the bottom edge of the half-plane, while an h-line passing through q is mapped to a vertical line.

²³ This figure was not in the original edition of VCA. I drew it about 15 years later, for VDGF, where it appears as figure [6.6].



[6.42] How Beltrami discovered the upper half-plane model. From a point q on the rim of the hemisphere, stereographically project onto the tangent plane at the point antipodal to q . Since this preserves circles and angles, we see that a typical h -line of the hemisphere is mapped to a semicircle orthogonal to the bottom edge of the half-plane, while an h -line passing through q is mapped to a vertical line. The angular interpretation of hyperbolic distance given in [6.36] applies here, too. We therefore recover the familiar hyperbolic metric in the upper half-plane: $d\hat{s} = (ds/Z)$.

Well, since these are the h -lines, it certainly *looks* like we have obtained the Beltrami–Poincaré upper half-plane, but to make sure, let's check that its metric is really given by (6.31). Since stereographic projection is conformal, we may yet again use the construction (6.43). Choosing l to be the image of an h -line through q , the figure immediately reveals that the metric is $d\hat{s} = (ds/Z)$. Apart from a change of notation, this is indeed the same as (6.31).

We have thus returned to the half-plane that began our journey, but we have returned wiser than when we left. Looking at (6.47) we now recognize this half-plane orthogonal to $\mathbb{C}$ as a hyperbolic plane within hyperbolic space. This reveals the true role of the stereographic projection in [6.42].

We know that stereographic projection from q is just the restriction to the hemisphere of inversion $\mathcal{I}_K$ in a sphere K centred at q . Using the same argument (figure [6.22b]) as in the plane, we see that $\mathcal{I}_K$ preserves the metric (6.47), so it is a *motion* of hyperbolic space, carrying h -lines into h -lines and carrying h -planes into h -planes. Furthermore, (6.48) tells us that K is a hyperbolic plane in this hyperbolic space, and we therefore suspect that $\mathcal{I}_K$ is *reflection* in this h -plane. This can be confirmed [exercise] by generalizing the argument in [6.23b]. Thus we have the following generalization of (6.38):

Inversion in a hemisphere K orthogonal to the horizon is reflection $\mathfrak{R}_K$ of hyperbolic space in the h -plane K .

It is beyond the scope of this book to explore the motions of hyperbolic space²⁴. However, let us at least describe one particularly beautiful result.

Just as an arbitrary direct motion of an h-plane is the composition of two h-reflections in h-lines within it, so an arbitrary direct motion of hyperbolic space is the composition of *four* reflections in h-planes within it. Thus, in the half-space model with horizon $\mathbb{C}$, such a motion is the composition of four inversions in spheres centred on $\mathbb{C}$. If we restrict attention to the points of $\mathbb{C}$ then inversion in such a sphere K is equivalent to two-dimensional inversion of $\mathbb{C}$ in the equatorial circle in which K intersects $\mathbb{C}$. Conversely, inversion of $\mathbb{C}$ in a circle k extends uniquely to an inversion of space: simply construct the sphere with equator k .

Finally, then, every direct motion of hyperbolic space can be uniquely represented in terms of $\mathbb{C}$ (the horizon) as the composition of inversion in four circles, and *this is none other than the most general Möbius transformation*,

$$z \mapsto M(z) = \frac{az + b}{cz + d},$$

of the complex plane! Poincaré discovered this wonderful fact in 1883; see [Poincaré \(1985\)](#).

We have seen that the direct motions of the hyperbolic plane, the Euclidean plane and the sphere are subgroups of this group of general Möbius transformations. As we shall now see, this fact has a remarkable geometrical explanation.

Hilbert's result on surfaces of constant negative curvature shows that three-dimensional Euclidean geometry cannot accommodate a model of the hyperbolic plane. Amazingly, however, three-dimensional hyperbolic space *does* contain surfaces whose intrinsic geometry is *Euclidean*! In fact these surfaces are the *horospheres* that generalize the horocycles. Analogously to [6.28], horospheres are Euclidean spheres that touch $\mathbb{C}$, as well as planes $Z = \text{const.}$ that are parallel to $\mathbb{C}$.

Vertical planes orthogonal to $\mathbb{C}$ *look* flat in our model of hyperbolic space, but in reality they are intrinsically curved hyperbolic planes. However, a horosphere $Z = \text{const.}$ not only looks flat, it really *is* flat. For its metric, inherited from the metric (6.47) of the surrounding space, is just

$$d\hat{s} = (\text{constant}) ds,$$

and this is the metric of a Euclidean plane!

The motions of Euclidean plane geometry may now be viewed as the motions of hyperbolic space that map this intrinsically flat horosphere into itself. Clearly, these are the composition of reflections in vertical planes, i.e., h-reflections in h-planes orthogonal to the horosphere. In this manner, the direct motions of the Euclidean plane manifest themselves on the horizon $\mathbb{C}$ as a subgroup of the Möbius transformations.

²⁴ [Thurston \(1997\)](#) is an excellent source of information on these motions.

As for spherical geometry, we begin by defining an *h-sphere* as the set of points at constant *h*-distance from a given point (the *h*-centre). It is not hard to see that these *h*-spheres are represented in the half-space model by Euclidean spheres, though their *h*-centres do not coincide with their Euclidean centres.

Though it is not immediately obvious in this model, it can be shown [see Ex. 27] that the intrinsic geometry of such an *h*-sphere is the same as that of an ordinary sphere (of different radius) in Euclidean space. As with the horosphere, the motions of this *h*-sphere may be viewed as the motions of hyperbolic space that map the *h*-sphere into itself. Again, these are the composition of *h*-reflections in *h*-planes orthogonal to the *h*-sphere, and again we arrive at a subgroup of the Möbius transformations.

Clearly the motions of the hyperbolic plane may also be viewed in this way, and so we have arrived at a fitting high point on which to end this chapter:

Two-dimensional hyperbolic, Euclidean, and spherical geometry are all subsumed by three-dimensional hyperbolic geometry.

Further Reading. For a short, masterful overview of Differential Geometry, see Penrose (1978). For a complete, elementary, and *geometrical* treatment of Differential Geometry (and Differential Forms), in the same spirit as this book, see Needham (2021). For more on its fundamental applications to physics, particularly General Relativity, see Misner, Thorne, and Wheeler (1973), Wald (1984), Ludvigsen (1999), Penrose (2005), Thorne and Blandford (2017), and Schutz (2022). For more on the use of Differential Forms in Differential Geometry, see do Carmo (1994) and O'Neill (2006), and for their applications to physics see Schutz (1980), Arnol'd (1989), and Dray (2015). For more on hyperbolic geometry itself, see Stillwell (1992), Stillwell (1996), Stillwell (2010), and Thurston (1997). Finally, for the connections between topology and differential geometry, see Hopf (1956).

6.4 Exercises

- 1 Draw a geodesic triangle Δ on the surface of a suitable fruit or vegetable. Now draw a geodesic segment from one of the vertices to an arbitrary point of the opposite side. This divides Δ into two geodesic triangles, say Δ_1 and Δ_2 . Show that the angular excess function $\mathcal{E}$ is *additive*, i.e., $\mathcal{E}(\Delta) = \mathcal{E}(\Delta_1) + \mathcal{E}(\Delta_2)$. By continuing this process of subdivision, deduce that (6.5) implies (6.6).
- 2 Explain (6.18) by generalizing the argument that was used to obtain the special case (3.18), on p. 162. That is, think of reflection of the sphere in terms reflection of space in a plane Π , as in [6.8], p. 318. Also, think of stereographic projection as the restriction to the sphere of the three-dimensional inversion $\mathcal{I}_K$, where K is the sphere of radius $\sqrt{2}$ centred at the north pole of Σ (see [6.13b]). Now let α be a point on Σ , and consider the effect of $\mathcal{I}_K$ on α , Π , and $\mathfrak{R}_\Pi(\alpha)$.
- 3 Let C be a circle in $\mathbb{C}$, and let $\widehat{C}$ be its stereographic image on Σ . If $\widehat{C}$ is a great circle, then (6.18) says that $\mathcal{I}_C$ stereographically induces reflection of Σ in $\widehat{C}$, but what transformation is induced if C is an *arbitrary* circle? Generalize the argument of figure [6.14] to show that $\mathcal{I}_C$ becomes *projection from the vertex v of the cone that touches Σ along $\widehat{C}$* . That is, if $w = \mathcal{I}_C(z)$ then $\widehat{w}$ is the second intersection point of Σ with the line in space that passes through the vertex v and the point $\widehat{z}$. Explain how (6.18) may be viewed as a limiting case of this more general result.
- 4 Use (3.42), p. 189 to show that if the Möbius transformation $M(z)$ has fixed points $\xi_{\pm}$, and the multiplier associated with ξ_+ is m , then

$$[M] = \begin{bmatrix} 1 & -\xi_+ \\ 1 & -\xi_- \end{bmatrix}^{-1} \begin{bmatrix} \sqrt{m} & 0 \\ 0 & 1/\sqrt{m} \end{bmatrix} \begin{bmatrix} 1 & -\xi_+ \\ 1 & -\xi_- \end{bmatrix}.$$

By putting $\xi_+ = \alpha$, $m = e^{-i\psi}$, and $\xi_- = -(1/\bar{\alpha})$, deduce (6.19), p. 328. [Hint: Remember that you are free to multiply a Möbius matrix by a constant.]

- 5 Show that the Möbius transformations (6.20) do indeed satisfy the differential equation (6.17).
- 6 (i) The *conjugate* $\bar{\mathbb{V}}$ of a quaternion $\mathbb{V} = v + \mathbf{V}$ is defined to be the conjugate transpose $\mathbb{V}^*$ of the corresponding matrix. Show that $\bar{\mathbb{V}} \equiv \mathbb{V}^* = v - \mathbf{V}$, and deduce that $\mathbb{V}$ is a *pure* quaternion (analogous to a purely imaginary complex number) if and only if $\bar{\mathbb{V}} = -\mathbb{V}$.
- (ii) The *length* $|\mathbb{V}|$ of $\mathbb{V}$ is defined (by analogy with complex numbers) by $|\mathbb{V}|^2 = \mathbb{V} \bar{\mathbb{V}}$. Show that $|\mathbb{V}|^2 = v^2 + |\mathbf{V}|^2 = |\bar{\mathbb{V}}|^2$.

- (iii) If $|\mathbf{V}| = 1$, then $\mathbf{V}$ is called a *unit quaternion*. Verify that $\mathbb{R}_{\mathbf{V}}^{\Psi}$ [see (6.28)] is a unit quaternion, and that $\overline{\mathbb{R}_{\mathbf{V}}^{\Psi}} = \left[\mathbb{R}_{\mathbf{V}}^{\Psi}\right]^* = \mathbb{R}_{\mathbf{V}}^{-\Psi}$.
- (iv) Show that $\overline{\mathbf{V} \mathbf{W}} = \overline{\mathbf{W}} \overline{\mathbf{V}}$ and deduce that $|\mathbf{V} \mathbf{W}| = |\mathbf{V}| |\mathbf{W}|$. Thus, for example, the product of two unit quaternions is another unit quaternion.
- (v) Show that $\mathbb{A}$ is a pure, unit quaternion if and only if $\mathbb{A}^2 = -1$.
- (vi) Show that any quaternion $\mathbb{Q}$ can be expressed as $\mathbb{Q} = |\mathbb{Q}| \mathbb{R}_{\mathbf{V}}^{\Psi}$ for some $\mathbf{v}$ and some ψ .
- (vii) Suppose we generalize the transformation (6.29) to $\mathbb{P} \mapsto \tilde{\mathbb{P}} = \mathbb{Q} \mathbb{P} \overline{\mathbb{Q}}$, where $\mathbb{Q}$ is an arbitrary quaternion. When interpreted in this way, deduce that $\mathbb{Q}$ represents a *dilative rotation of space*, and the product of two quaternions represents the composition of the corresponding dilative rotations. [This confirms the claim at the end of Chapter 1.]
- 7 [Do the previous exercise before this one.] The following proof of (6.29) is based on a paper of H. S. M. Coxeter (1946).
- (i) Use (6.27) to show that the pure quaternions $\mathbb{P}$ and $\mathbb{A}$ are orthogonal if and only if $\mathbb{P} \mathbb{A} + \mathbb{A} \mathbb{P} = 0$.
- (ii) If $\mathbb{A}$ has unit length, so that $\mathbb{A}^2 = -1$, deduce that the previous equation may be expressed as $\mathbb{P} = \mathbb{A} \mathbb{P} \mathbb{A}$.
- (iii) Now keep the pure, unit quaternion $\mathbb{A}$ fixed, but let $\mathbb{P}$ represent an *arbitrary* pure quaternion. Let $\Pi_{\mathbb{A}}$ denote the plane with normal vector $\mathbf{A}$ that passes through the origin, so that its equation is $\mathbf{P} \cdot \mathbf{A} = 0$. Now consider the transformation
- $$\mathbb{P} \mapsto \mathbb{P}' = \mathbb{A} \mathbb{P} \mathbb{A}. \quad (6.49)$$
- Show that (a) $\mathbb{P}'$ is automatically pure, and $|\mathbb{P}'| = |\mathbb{P}|$, so that (6.49) represents a motion of space; (b) every point on $\Pi_{\mathbb{A}}$ remains fixed; (c) every vector orthogonal to $\Pi_{\mathbb{A}}$ is reversed. Deduce that (6.49) represents reflection $\mathfrak{R}_{\Pi_{\mathbb{A}}}$ of space in the plane $\Pi_{\mathbb{A}}$.
- (iv) Deduce that if the angle from $\Pi_{\mathbb{A}}$ to a second plane $\Pi_{\mathbb{B}}$ is $(\psi/2)$, and the unit vector along the intersection of the planes is $\mathbf{V}$, then the rotation $\mathbb{R}_{\mathbf{V}}^{\Psi} = (\mathfrak{R}_{\Pi_{\mathbb{B}}} \circ \mathfrak{R}_{\Pi_{\mathbb{A}}})$ is given by
- $$\mathbb{P} \mapsto \tilde{\mathbb{P}} = (\mathbb{B} \mathbb{A}) \mathbb{P} (\mathbb{A} \mathbb{B}) = (-\mathbb{B} \mathbb{A}) \mathbb{P} (-\overline{\mathbb{B} \mathbb{A}}).$$
- (v) Use (6.27) to show that $-\mathbb{B} \mathbb{A} = \cos(\psi/2) + \mathbf{V} \sin(\psi/2)$, thereby simultaneously proving (6.29) and (6.28).
- 8 Here is another proof of (6.29). As in the text, we shall assume that $\mathbf{P}$ is a unit vector with its tip at the point $\hat{\mathbf{p}}$ on the unit sphere. If we represent the

stereographic images $\mathbf{p}$ and $\tilde{\mathbf{p}}$ of $\hat{\mathbf{p}}$ and $\hat{\tilde{\mathbf{p}}}$ by their homogeneous coordinate vectors $\mathbf{p}$ and $\tilde{\mathbf{p}}$ in $\mathbb{C}^2$, then we know that the rotation is represented as

$$\mathbf{p} \mapsto \tilde{\mathbf{p}} = \mathbb{R}_{\mathbf{v}}^{\Psi} \mathbf{p},$$

where $\mathbb{R}_{\mathbf{v}}^{\Psi}$ is being thought of as a 2×2 matrix.

(i) Show that in homogeneous coordinates, (3.21), p. 166, becomes

$$X + iY = \frac{2 \mathbf{p}_1 \overline{\mathbf{p}_2}}{|\mathbf{p}_1|^2 + |\mathbf{p}_2|^2} \quad \text{and} \quad Z = \frac{|\mathbf{p}_1|^2 - |\mathbf{p}_2|^2}{|\mathbf{p}_1|^2 + |\mathbf{p}_2|^2}.$$

(ii) To simplify this, recall that all multiples of $\mathbf{p}$ describe the same point $\mathbf{p}$ in $\mathbb{C}$. We can therefore choose the “length” of $\mathbf{p}$ to be $\sqrt{2}$:

$$\langle \mathbf{p}, \mathbf{p} \rangle \equiv |\mathbf{p}_1|^2 + |\mathbf{p}_2|^2 = 2.$$

With this choice, show that the above equations can be written as

$$\begin{bmatrix} 1 + Z & X + iY \\ X - iY & 1 - Z \end{bmatrix} = \begin{bmatrix} \mathbf{p}_1 \overline{\mathbf{p}_1} & \mathbf{p}_1 \overline{\mathbf{p}_2} \\ \mathbf{p}_2 \overline{\mathbf{p}_1} & \mathbf{p}_2 \overline{\mathbf{p}_2} \end{bmatrix} = \begin{bmatrix} \mathbf{p}_1 \\ \mathbf{p}_2 \end{bmatrix} \begin{bmatrix} \overline{\mathbf{p}_1} & \overline{\mathbf{p}_2} \end{bmatrix} = \mathbf{p} \mathbf{p}^*.$$

(iii) Verify that

$$\begin{bmatrix} 1 + Z & X + iY \\ X - iY & 1 - Z \end{bmatrix} = 1 - i\mathbb{P}.$$

(iv) Deduce that

$$1 - i\tilde{\mathbb{P}} = \mathbb{R}_{\mathbf{v}}^{\Psi} (1 - i\mathbb{P}) \left[\mathbb{R}_{\mathbf{v}}^{\Psi} \right]^* = 1 - i\mathbb{R}_{\mathbf{v}}^{\Psi} \mathbb{P} \mathbb{R}_{\mathbf{v}}^{-\Psi},$$

from which (6.29) follows immediately.

9 (i) Figure [6.40a] gave a two-step process for carrying a point z in the Beltrami–Poincaré disc to the corresponding point z' in the Klein model. Explain why the net mapping $z \mapsto z'$ of the disc to itself is the one shown in figure [a] below, where C is an arbitrary circle passing through z and orthogonal to the unit circle $\mathbb{U}$.

(ii) Figure [b] is a vertical cross section of [6.40a] through z and z' . Deduce that

$$\frac{|z'|}{a} = \frac{1}{b} \quad \text{and} \quad \frac{a}{|z|} = \frac{2}{b}.$$

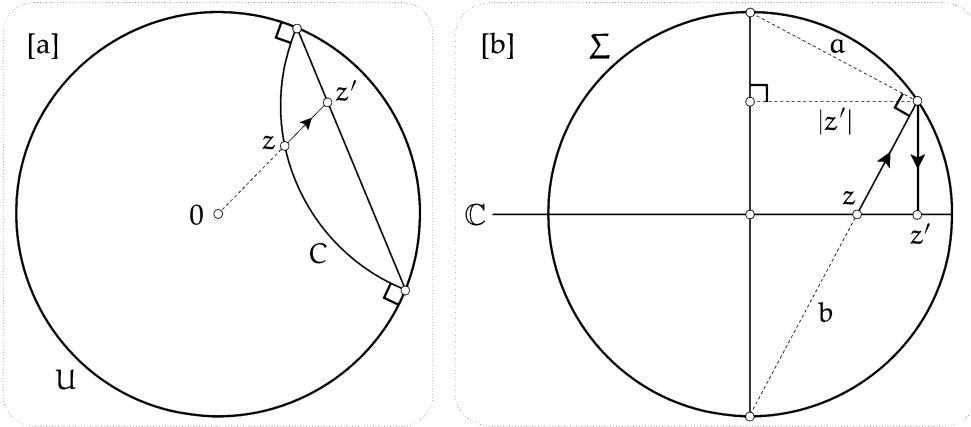
By multiplying these two equations, deduce that $z' = \frac{2z}{1+|z|^2}$.

[Thus we have a geometric explanation of the result (3.21), p. 166.]

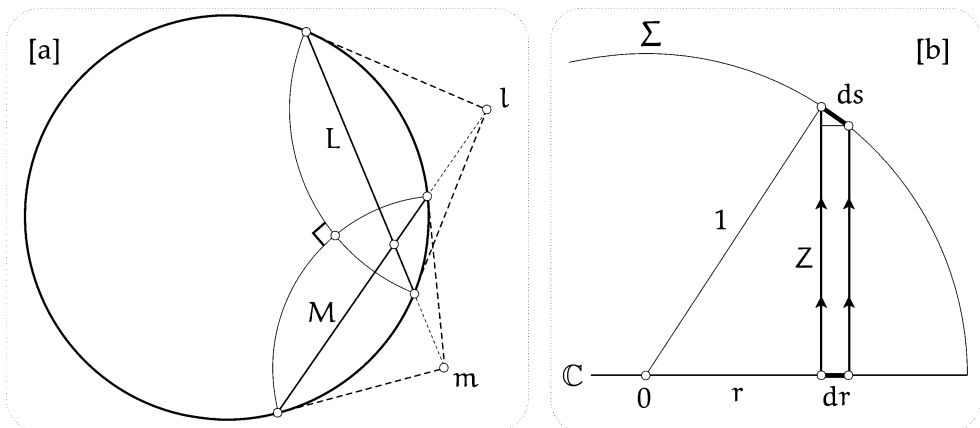
(iii) This formula can be derived directly from figure [a], without the assistance of the hemisphere. Redraw the figure with C chosen orthogonal to $0z$. Explain geometrically why the centre of C may be viewed as either $\mathcal{I}_{\mathbb{U}}(z')$, or as the midpoint of z and $\mathcal{I}_{\mathbb{U}}(z)$. Conclude that

$$1/\overline{z'} = \mathcal{I}_{\mathbb{U}}(z') = \frac{1}{2} [z + \mathcal{I}_{\mathbb{U}}(z)] = \frac{1}{2} [z + (1/\overline{z})],$$

from which the result follows immediately.



- 10 Think of the sphere as the surface of revolution generated by a semicircle. Construct a conformal map of the sphere by strict analogy with the construction of the map of the pseudosphere in [6.20]. Show that this is the Mercator map that you obtained in Ex. 14, p. 294.
- 11 (i) In the hyperbolic plane, show that the h-circumference of an h-circle of h-radius ρ is $2\pi \sinh \rho$. [Hint: Represent the h-circle as an origin-centred Euclidean circle in the Beltrami–Poincaré disc.]
- (ii) Let the inhabitants of the sphere of radius R draw a circle of (intrinsic) radius ρ . Use elementary geometry to show that the circle's circumference is $2\pi R \sin(\rho/R)$. Show that if we take the radius of the sphere to be $R = i$, then this becomes the formula in part (i)! [Compare this with Ex. 14.]
- 12 Let L and M be two intersecting chords of the unit circle, and let l and m be the intersection points of the pairs of tangents drawn at the ends of these chords. See figure [a] below. In the Beltrami–Klein model, show that L and M represent orthogonal h-lines if and only if l lies on M (produced) and m lies on L (produced).



13 Let $z = r e^{i\theta}$ denote a point in the Beltrami–Klein model of the hyperbolic plane.

- (i) On the hemisphere model lying above the Beltrami–Klein model, sketch the images under vertical upward projection of some circles $r = \text{const.}$ and some rays $\theta = \text{const.}$ Although angles are generally distorted in the Beltrami–Klein model, deduce that these circles and rays really are orthogonal, as they appear to be. Also, note that the Euclidean circles $r = \text{const.}$ are also h-circles.
- (ii) Figure [b] above shows a vertical cross section of the hemisphere model and Beltrami–Klein model taken through a ray $\theta = \text{const.}$ If the point z moves outward along this ray by dr , let ds denote the movement of its vertical projection on the hemisphere. Explain why the two shaded triangles are similar, and deduce that $ds = (dr/Z)$.
- (iii) Use the metric (6.47) of the hemisphere to conclude that the h-separation $d\hat{s}_r$ of the points in the Beltrami–Klein model with polar coordinates (r, θ) and $(r + dr, \theta)$ is given by

$$d\hat{s}_r = \frac{dr}{1 - r^2}.$$

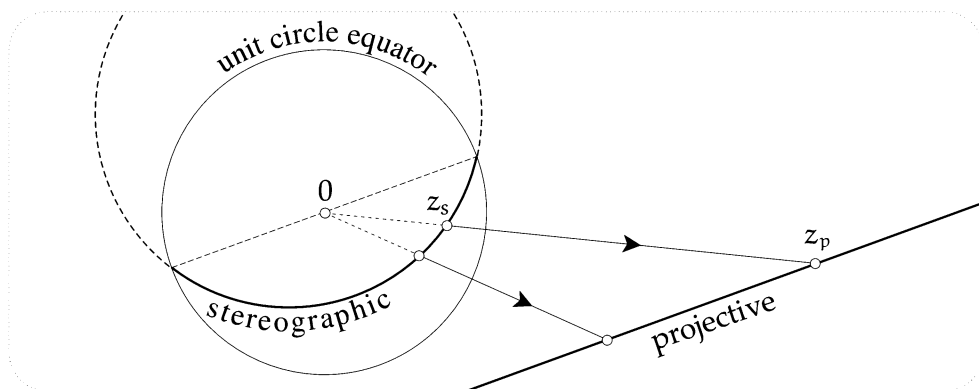
[Remarkably, this means that the formula for $\mathcal{H}\{0, z\}$ differs from the formula (6.45) in the Beltrami–Poincaré disc by a mere factor of two!]

- (iv) Use the same idea (of projecting onto the hemisphere) to show that the h-separation $d\hat{s}_\theta$ of the points (r, θ) and $(r, \theta + d\theta)$ is given by

$$d\hat{s}_\theta = \frac{r d\theta}{\sqrt{1 - r^2}}.$$

- (v) Deduce that the h-separation $d\hat{s}$ of the points (r, θ) and $(r + dr, \theta + d\theta)$ is given by the following formula of Beltrami (1868a):

$$d\hat{s}^2 = \frac{dr^2}{(1 - r^2)^2} + \frac{r^2 d\theta^2}{1 - r^2}. \quad (6.50)$$



- 14 (i) The figure above superimposes the stereographic and projective [see p. 323] images of a great circle on the unit sphere. Let z_s and z_p be the stereographic and projective images of the point whose spherical-polar coordinates are (ϕ, θ) . Referring to (3.22), p. 167, $z_s = \cot(\phi/2) e^{i\theta}$. Show that $z_p = [-\tan \phi] e^{i\theta}$, and deduce that

$$z_p = \frac{2z_s}{1 - |z_s|^2}.$$

Compare this with Ex. 9!

- (ii) Sketch the curves on the hemisphere that are centrally projected to the circles $|z_p| = \text{const.}$ and to the rays $\arg z_p = \text{const.}$ Although angles are generally distorted in the projective model, observe that these circles and rays really are orthogonal, as they appear to be.
- (iii) Now let the sphere have radius R , and write $z_p = r e^{i\theta}$ for the projective image of the point (ϕ, θ) . Thus $r = -R \tan \phi$. Show that if z_p moves a distance dr along the ray $\theta = \text{const.}$, then the corresponding point on the sphere moves a distance $d\hat{s}_r$ given by

$$d\hat{s}_r = \frac{dr}{1 + (r/R)^2}.$$

- (iv) Likewise, show that the separation $d\hat{s}_\theta$ of the points on the sphere corresponding to the points (r, θ) and $(r, \theta + d\theta)$ in the map is given by

$$d\hat{s}_\theta = \frac{r d\theta}{\sqrt{1 + (r/R)^2}}.$$

- (v) Deduce that the spherical separation $d\hat{s}$ corresponding to the points (r, θ) and $(r + dr, \theta + d\theta)$ is given by

$$d\hat{s}^2 = \frac{dr^2}{(1 + (r/R)^2)^2} + \frac{r^2 d\theta^2}{1 + (r/R)^2}.$$

- (vi) Here is a crazy idea, essentially due to Lambert; see Penrose (2005, § 2.6). Perhaps we can get a surface of constant *negative* curvature $\mathcal{K} = -(1/R^2) = +1/(iR)^2$ by allowing the radius R of the sphere to take on the *imaginary* value iR . Verify that if we substitute $R = i$ into the above formula, then it becomes the Beltrami metric (6.50) of the hyperbolic plane! [To make true sense of this idea, one must turn to Einstein's relativity theory; see Thurston (1997) and especially Penrose (2005).]

- 15 Take a stack of ten sheets of paper and staple them together, placing staples along three of the edges. Use a pair of compasses to draw the largest circle that will fit comfortably inside the top sheet. Pierce through all ten sheets in the centre of the circle. With heavy scissors, cut along the circle to obtain ten

identical discs, say of radius R . Repeat this whole process to double the number of discs to 20.

- (i) Cut a narrow sector out of the first disc, and tape the edges together to form a shallow cone. Repeat this process with the remaining discs, steadily increasing the angle of the sector each time, so that the cones get sharper and taller. Ensure that by the end of the process you are making very narrow cones, using only a quarter disc or less.
 - (ii) Stack these cones in the order you made them. Explain how it is that *you have created a model of a portion of a pseudosphere of radius R* . Create weird new (extrinsically asymmetric) surfaces of constant negative curvature by holding the tip of your structure and moving it from side to side!
 - (iii) Use the same idea to create a disc-like piece of “hyperbolic paper”, such as you would get if you could simply cut out a disc from your pseudosphere. Press it against the pseudosphere and verify that you can freely move it about and rotate it on the surface.
- 16 By holding a fairly short piece of string against the surface of the toy pseudosphere of the previous exercise, draw a segment of a typical geodesic. Extend this segment in both directions, one string-length at a time. Note the surprising way the geodesic only spirals a finite distance up the pseudosphere before spiralling down again.
- (i) Use the upper half-plane to verify mathematically that the tractrix generators are the *only* geodesics that extend all the way up to the top.
 - (ii) Let L be a typical geodesic, and let α be the angle between L and the tractrix generator at the point where L hits the rim $\sigma = 0$. Show that the maximum distance $\sigma_{\max}$ that L travels up the pseudosphere is given by $\sigma_{\max} = |\ln \sin \alpha|$.
- 17 Suppose we have a conformal map of a surface in the xy -plane, with the metric given by (6.15):

$$d\hat{s} = \Lambda ds = \Lambda(x, y) \sqrt{dx^2 + dy^2}.$$

An elegant result from differential geometry [see [Needham \(2021, §4.4\)](#)] states that the Gaussian curvature at any point on the surface is given by

$$\mathcal{K} = -\frac{1}{\Lambda^2} \Delta(\ln \Lambda),$$

where $\Delta \equiv (\partial_x^2 + \partial_y^2)$ is the Laplacian. Try this out on the metric (6.16) of the stereographic map of the sphere, and on the metrics (6.31) and (6.44) of the half-plane and disc models of the hyperbolic plane.

- 18 Use the Beltrami–Poincaré disc to rederive the formula $\tan(\Pi/2) = e^{-D}$ for the angle of parallelism. [Hint: Let one of the h-lines be a diameter.]
- 19 To derive the metric (6.44), consider the mapping (6.42) $z \mapsto w = D(z)$ from the upper half-plane to the Beltrami–Poincaré disc. An infinitesimal vector dz emanating from z is amplitwisted to an infinitesimal vector $dw = D'(z) dz$ emanating from w , and (by definition) the h-length $d\hat{s}$ of dw is the h-length of dz . Verify (6.44) by showing that

$$\frac{2|dw|}{1-|w|^2} = \frac{|dz|}{\text{Im } z} = d\hat{s}.$$

- 20 Consider the mapping $z \mapsto w = M_a^\phi(z)$ of the Beltrami–Poincaré disc to itself. Use the calculational approach of the previous exercise to show that $z \mapsto w$ is a hyperbolic motion, i.e., it preserves the metric:

$$\frac{2|dw|}{1-|w|^2} = d\hat{s} = \frac{2|dz|}{1-|z|^2}.$$

- 21 In the upper half-plane, the h-rotation $\mathcal{R}_i^\phi$ through angle ϕ about the point i is given by the following Möbius transformation:

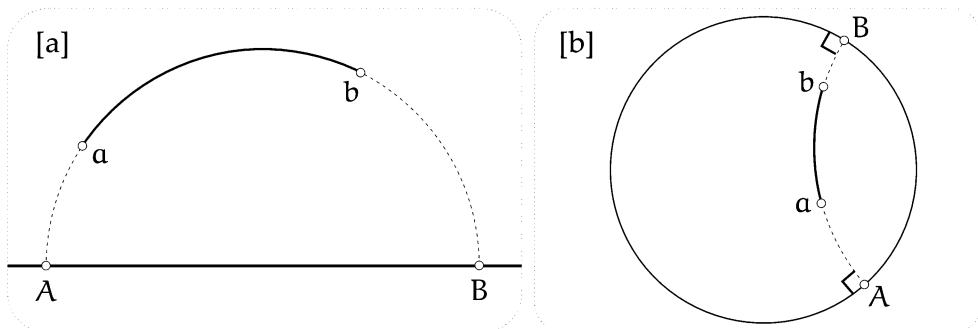
$$\mathcal{R}_i^\phi(z) = \frac{cz + s}{-sz + c}, \quad \text{where } c = \cos(\phi/2) \text{ and } s = \sin(\phi/2).$$

Prove this in three ways:

- Show that $\mathcal{R}_i^\phi(i) = i$ and $\{\mathcal{R}_i^\phi\}'(i) = e^{i\phi}$. Why does this prove the result?
- Use the formula for inversion [(3.4), p. 141] to calculate the composition $\mathcal{R}_i^\phi = (\mathcal{R}_B \circ \mathcal{R}_A)$, where A and B are h-lines through i , and the angle from A to B is $(\phi/2)$. [Hint: Take A to be the imaginary axis, and use a diagram to show that the semicircle B has centre $-(c/s)$ and radius $(1/s)$.]
- Describe and explain the geometrical effect of applying $(D \circ \mathcal{R}_i^\phi \circ D^{-1})$ to the Beltrami–Poincaré disc, where D is the mapping (6.42) from the upper half-plane to the Beltrami–Poincaré disc. Deduce that

$$(D \circ \mathcal{R}_i^\phi \circ D^{-1})(z) = e^{i\phi} z.$$

Re-express this equation in terms of products of Möbius matrices, and solve for the matrix $[\mathcal{R}_i^\phi]$.



- 22 (i) Referring to figure [a] above, show that the h-separation of two points in the upper half-plane may be expressed in terms of a cross-ratio as

$$\mathcal{H}\{a, b\} = \ln[a, B, b, A].$$

[Hint: Apply an h-rotation centred at a to bring b into a position vertically above a .]

- (ii) Show that the same formula applies to the Beltrami–Poincaré disc in figure [b].
- 23 (i) Let $\hat{a}$ and $\hat{z}$ be two point on Σ , and let a and z be their stereographic images in $\mathbb{C}$. If $S\{a, z\}$ is the distance (on the sphere) between $\hat{a}$ and $\hat{z}$, show that

$$S\{a, z\} = 2 \tan^{-1} \left| \frac{z - a}{\bar{a}z + 1} \right|.$$

[Hint: Use (6.20) to bring a to the origin.]

- (ii) Show that the h-distance formula (6.46) in the Beltrami–Poincaré disc can be re-expressed as

$$\mathcal{H}\{a, z\} = 2 \tanh^{-1} \left| \frac{z - a}{\bar{a}z - 1} \right|.$$

- 24 (i) Use a simple sketch to show that $z \mapsto f(z) = -\bar{z}$ is an opposite motion of the Beltrami–Poincaré upper half-plane.
- (ii) Let $\tilde{\mathcal{M}}$ be an arbitrary opposite motion. By considering $(f \circ \tilde{\mathcal{M}})$, show that

$$\tilde{\mathcal{M}}(z) = \frac{a\bar{z} + b}{c\bar{z} + d}, \quad \text{where } a, b, c, d \text{ are real, and } (ad - bc) > 0.$$

- (iii) Use this formula to show that $\tilde{\mathcal{M}}$ has two fixed points on the horizon (the real axis). If L is the h-line whose ends are these two points, explain why L is invariant under $\tilde{\mathcal{M}}$.
- (iv) Deduce that $\tilde{\mathcal{M}}$ is always a *glide reflection*: h-translation along L , followed by (or preceded by) h-reflection in L .
- 25 Given a point p not on the h-line L , draw an h-circle C of h-radius ρ centred at p . Draw the h-line orthogonal to L through p , cutting L in q . Draw an h-circle C' of h-radius ρ centred at q , and let q' be one of the intersection points with L . Through q' draw the h-line orthogonal to L , cutting C at a and b . Show that the h-lines joining p to a and b are the two asymptotic lines! [Hint: Take L to be a vertical line in the upper half-plane.] What happens if we perform this construction in the Euclidean plane?
- 26 Sketch a hyperbolic triangle Δ with vertices (in counterclockwise order) a , b , and c . Let ξ be an infinitesimal vector emanating from a and pointing along the edge ab . Carry ξ to b by h-translating it along this edge. Now carry it to c

along bc , and finally carry it home to a , along ca . In Euclidean geometry these three translations would simply cancel, and ξ would return home unaltered. Use your sketch to show that in hyperbolic geometry *the composition of these three h -translations is an h -rotation about vertex a through angle $\mathcal{E}(\Delta)$.*

[Suppose that Δ is instead a geodesic triangle on an *arbitrary* surface S of variable curvature. If an inhabitant of S wants to translate ξ along a “straight line” (a geodesic), all he has to do is keep its length constant, and keep the angle it makes with the line constant. This is called *parallel transport*, and it plays a central role in differential geometry; see [Needham \(2021, Act IV\)](#). The above argument still applies, and so when ξ is parallel transported round Δ , it returns home rotated through $\mathcal{E}(\Delta)$. By virtue of (6.6), this angle of rotation is the total amount of curvature inside Δ .]

- 27 Generalize the transformation from the upper half-plane to the Beltrami–Poincaré disc to obtain a model of hyperbolic space in the interior of the unit sphere. Describe the appearance of h -lines, h -planes, h -spheres, and horospheres in this model, and explain why an h -sphere is intrinsically the same as a Euclidean sphere of different radius.

